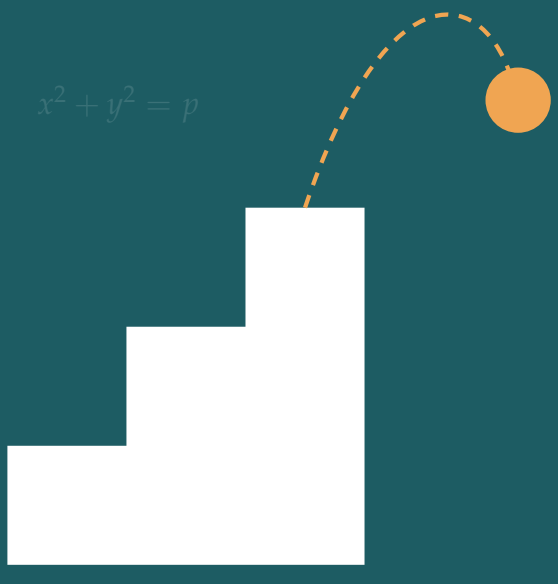
STEP PREPARATION · UNIVERSITY MATHEMATICS

# Beyond STEP

## STEP II AND III

2025 Worked Solutions



KEITH McNULTY

# Beyond STEP

Past STEP questions as a doorway
to university mathematics

**STEP II AND III**

2025 Worked Solutions

Keith McNulty

**Beyond STEP: Past STEP questions as a doorway to university mathematics**

**2025 Worked Solutions**

Worked solutions to all 24 questions of the 2025 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

# STEP II 2025, Question 1

Pure • Piecewise Functions

**STEP II 2025, Question 1**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

A function built from Min is just an ordinary function given by different formulas on different intervals. So the first job every time is to find where the two arguments cross, write the function piecewise, and then work interval by interval, keeping only the answers that land in the interval they came from. At the joins the function stays continuous but usually has a corner, and that corner is what makes the point in (iv) a maximum without the derivative being zero.

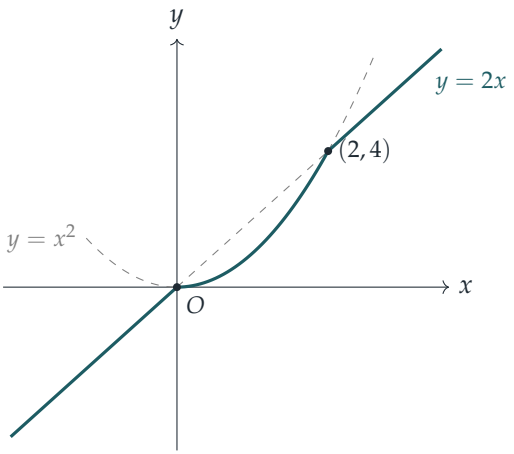
## Solution

### Part (i)

We have $x^2 \leq 2x \iff x(x-2) \leq 0 \iff 0 \leq x \leq 2$. So

$$\mathrm{Min}(x^2, 2x) = \begin{cases} x^2 & 0 \leq x \leq 2, \\ 2x & x < 0 \text{ or } x > 2. \end{cases}$$

The graph follows the line $y = 2x$ except between $x = 0$ and $x = 2$, where it dips onto the parabola, which lies below its chord there. Both straight pieces lie on the *same* line. There are corners at $(0, 0)$, where the parabola has gradient $0$ but the line has gradient $2$, and at $(2, 4)$, where the gradients are $4$ and $2$.




**Part (ii)**

Use the two pieces from (i).

- For $0 \leq x \leq 2$: $2x^2 = 5x - 3 \iff 2x^2 - 5x + 3 = 0 \iff (2x-3)(x-1) = 0$, so $x = 1$ or $x = \frac{3}{2}$. Both lie in $[0, 2]$, so both count.
- For $x < 0$ or $x > 2$: $4x = 5x - 3 \iff x = 3$, which satisfies $x > 2$, so it counts.

The solutions are $x = 1$, $x = \frac{3}{2}$ and $x = 3$.

**Part (iii)**

We also need the second function: $x^3 \leq 4x \iff x(x-2)(x+2) \leq 0 \iff x \leq -2$ or $0 \leq x \leq 2$. Putting the two together,

$$\text{Min}(x^2, 2x) + \text{Min}(x^3, 4x) = \begin{cases} x^3 + 2x & x \leq -2, \\ 6x & -2 < x < 0, \\ x^3 + x^2 & 0 \leq x \leq 2, \\ 6x & x > 2. \end{cases}$$

$m = 2$.

- $x \leq -2$: $x^3 + 2x = 2x$ gives $x = 0$, not in the interval.
- $-2 < x < 0$ and $x > 2$: $6x = 2x$ gives $x = 0$, in neither interval.
- $0 \leq x \leq 2$: $x^3 + x^2 - 2x = x(x+2)(x-1) = 0$ gives $x = 0$ or $x = 1$ (we reject $x = -2$).

So for $m = 2$ the solutions are $x = 0$ and $x = 1$.

? **Ask yourself** What happens to the equation $6x = 6x$?

It is true for *every* $x$, so when $m = 6$ whole intervals are solutions, not just isolated points.

$m = 6$**.**

- $x \le -2$: $x^3 + 2x = 6x \iff x(x-2)(x+2) = 0$, and only $x = -2$ is in the interval.
- $-2 < x < 0$ and $x > 2$: $6x = 6x$ holds identically, so every $x$ in these intervals is a solution.
- $0 \le x \le 2$: $x^3 + x^2 - 6x = x(x+3)(x-2) = 0$ gives $x = 0$ or $x = 2$ (we reject $x = -3$).

Combining, for $m = 6$ the solution set is

$$-2 \le x \le 0 \quad \text{or} \quad x \ge 2.$$

**Part (iv)**

Here $x^3 \le x^2 \iff x^2(x-1) \le 0 \iff x \le 1$ (the factor $x^2$ is never negative, and $x = 0$ gives equality). So

$$y = \begin{cases} 2x^3 - 5x & x \le 1, \\ 2x^2 - 5x & x > 1. \end{cases}$$

Both formulas give $y = -3$ at $x = 1$, so the curve is continuous there and passes through $(1, -3)$.

**The point** $(1, -3)$**.** It is not a stationary point: the gradient from the left is $6x^2 - 5 = 1$ at $x = 1$, and from the right it is $4x - 5 = -1$. More precisely,

- for $\sqrt{5/6} < x < 1$ we have $\mathrm{d}y/\mathrm{d}x = 6x^2 - 5 > 0$, so $y$ increases up to $-3$;
- for $1 < x < \frac{5}{4}$ we have $\mathrm{d}y/\mathrm{d}x = 4x - 5 < 0$, so $y$ decreases from $-3$.

So $y < -3$ at all points near $x = 1$ on either side, and $(1, -3)$ is a local maximum, a corner of the curve.

**The other turning points.** On $x < 1$, $\mathrm{d}y/\mathrm{d}x = 6x^2 - 5 = 0$ gives $x = \pm\sqrt{5/6} = \pm\frac{\sqrt{30}}{6}$, both less than 1. Using $2x^2 = \frac{5}{3}$,

$$y = x(2x^2 - 5) = -\tfrac{10}{3}x = \mp\tfrac{10}{3} \cdot \tfrac{\sqrt{30}}{6} = \mp\tfrac{5\sqrt{30}}{9}.$$

Since $\mathrm{d}^2y/\mathrm{d}x^2 = 12x$, the point with $x < 0$ is a maximum and the one with $x > 0$ a minimum. On $x > 1$, $4x - 5 = 0$ gives $x = \frac{5}{4}$, with $y = \frac{25}{8} - \frac{25}{4} = -\frac{25}{8}$, a minimum since the parabola opens upwards. So the other three are

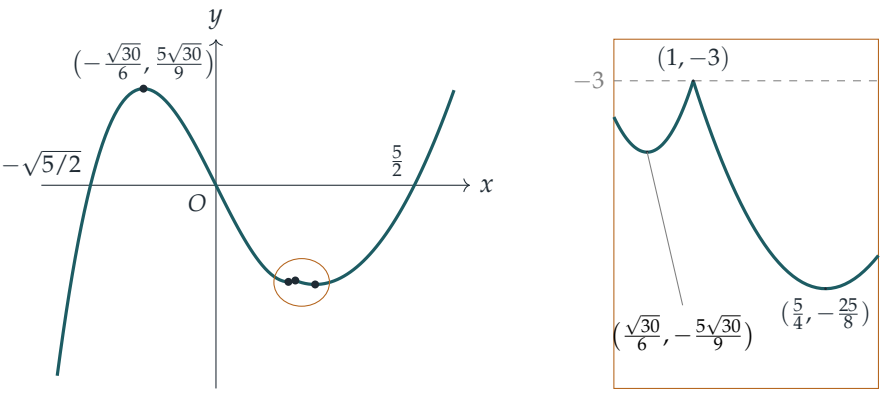
$$\text{maximum } \left(-\tfrac{\sqrt{30}}{6},\ \tfrac{5\sqrt{30}}{9}\right), \qquad \text{minima } \left(\tfrac{\sqrt{30}}{6},\ -\tfrac{5\sqrt{30}}{9}\right) \text{ and } \left(\tfrac{5}{4},\ -\tfrac{25}{8}\right).$$

**Sketch.** The cubic $2x^3 - 5x$ is odd, so the part of the curve with $-1 \leq x \leq 1$ has rotational symmetry about the origin. It crosses the axes where $x(2x^2 - 5) = 0$ with $x \leq 1$, giving $x = 0$ and $x = -\sqrt{5/2} \approx -1.58$, and where $x(2x - 5) = 0$ with $x > 1$, giving $x = \frac{5}{2}$. As $x \to -\infty$, $y \to -\infty$; as $x \to \infty$, $y \to \infty$.

The three points near $x = 1$ are very close together, so it is worth deciding their order. Squaring the positive numbers $\frac{5\sqrt{30}}{9}$ and $\frac{25}{8}$ gives $\frac{750}{81} \approx 9.26$ and $\frac{625}{64} \approx 9.77$, so

$$-\tfrac{25}{8} < -\tfrac{5\sqrt{30}}{9} < -3 :$$

the minimum at $x = \frac{5}{4}$ is the lower one, and both minima are below the corner. On a sensible scale this detail is invisible, so the inset magnifies it.




## Checks

- (ii): at $x = \frac{3}{2}$, $\mathrm{Min}(\frac{9}{4}, 3) = \frac{9}{4}$ and $2 \cdot \frac{9}{4} = \frac{9}{2} = 5 \cdot \frac{3}{2} - 3$; at $x = 3$, $\mathrm{Min}(9, 6) = 6$ and $12 = 15 - 3$.
- (iii), $m = 6$: try $x = -1$, where $\mathrm{Min}(1, -2) + \mathrm{Min}(-1, -4) = -2 - 4 = -6 = 6x$; and $x = 1$, not in the set, gives $1 + 1 = 2 \neq 6$.
- (iv): a computer scan of $y$ at steps of $10^{-5}$ on $[-3, 3]$ found exactly four turning points, at $x \approx -0.9129$ ($y \approx 3.0429$), $0.9129$ ($-3.0429$), $1$ ($-3$) and $1.25$ ($-3.125$), matching the values above.

## Common slips

- Keeping roots that fall outside the interval in which their formula applies, such as $x = -2$ in (iii) with $m = 2$.
- Solving $6x = 6x$ as if it gave a single value, instead of recognising that it holds on the whole interval.
- Smoothing the curve in (iv) at $(1, -3)$: it has a genuine corner there, with gradient $1$ on the left and $-1$ on the right.

## Beyond STEP

Functions like $\mathrm{Min}(f,g) = \frac{1}{2}\big(f+g-|f-g|\big)$ are always continuous when $f$ and $g$ are, but they need not be differentiable, and (iv) shows that a local maximum need not be a stationary point. In analysis courses this leads to the careful statement of the result that extrema of a *differentiable* function occur where the derivative vanishes, and to one-sided derivatives. Piecewise-defined functions of this kind are also central in optimisation, where the minimum of several linear functions gives the corners studied in linear programming. Abbott's *Understanding Analysis* treats continuity and differentiability with exactly this sort of example in mind.

WORKED SOLUTION

# STEP II 2025, Question 2

Pure • Complex Equations with Modulus

**STEP II 2025, Question 2**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Write $z = x + \mathrm{i}y$ and split the equation into real and imaginary parts. A modulus is always real, so it contributes nothing to the imaginary part, and the imaginary part of $z^2$ (or $z^3$) then factorises. That factorisation tells us which lines in the Argand diagram the roots must lie on, and on each line the problem collapses to a real equation in one variable.

## Solution

### Part (i)

**(a)** Let $z = x + \mathrm{i}y$ with $x, y$ real. Then $z^2 = x^2 - y^2 + 2xy\,\mathrm{i}$, while $|z + b|$ is a real number, and so is $a$. Comparing imaginary parts of $z^2 + |z + b| = a$ gives

$$2xy = 0,$$

so $x = 0$ (and $z$ is purely imaginary) or $y = 0$ (and $z$ is purely real). This is what we had to show.

**(b)** Suppose $z = \mathrm{i}y$ with $y$ real. Then $z^2 = -y^2$ and $|z + \frac{5}{2}| = \sqrt{y^2 + \frac{25}{4}}$, so we would need

$$\sqrt{y^2 + \tfrac{25}{4}} = y^2 + \tfrac{7}{2}.$$

Both sides are positive, so this is equivalent to the squared equation

$$y^2 + \tfrac{25}{4} = y^4 + 7y^2 + \tfrac{49}{4} \quad \Longleftrightarrow \quad y^4 + 6y^2 + 6 = 0.$$

The left-hand side is at least 6 for every real $y$, so there is no solution: the equation has no purely imaginary roots.

**(c)** Suppose $z = x$ is real, so we need $x^2 + |x + \frac{7}{2}| = \frac{5}{2}$. We split according to the sign of $x + \frac{7}{2}$.

- If $x \geq -\frac{7}{2}$: $x^2 + x + \frac{7}{2} = \frac{5}{2}$, that is $x^2 + x + 1 = 0$, whose discriminant is $1 - 4 = -3 < 0$. No real solutions.
- If $x < -\frac{7}{2}$: $x^2 - x - \frac{7}{2} = \frac{5}{2}$, that is $x^2 - x - 6 = (x-3)(x+2) = 0$. The roots 3 and $-2$ both fail the condition $x < -\frac{7}{2}$, so neither is a solution.

Hence there are no purely real roots.

**Ask yourself** Why is it not enough to solve $x^2 - x - 6 = 0$ and stop?

Each case only describes the original equation on its own range of $x$. A root of the case equation that lies outside that range solves a different equation (one with the wrong sign inside the modulus), so it must be thrown away.

**(d)** Now $\frac{1}{2} < b < \frac{3}{4}$; in particular $b > 0$.

*Imaginary root.* Put $z = \mathrm{i}y$ and write $Y = y^2 \geq 0$. The equation becomes $-Y + \sqrt{Y + b^2} = \frac{1}{2}$, that is

$$\sqrt{Y + b^2} = Y + \tfrac{1}{2}.$$

For $Y \geq 0$ the right-hand side is positive, so squaring gives an equivalent equation:

$$Y + b^2 = Y^2 + Y + \tfrac{1}{4} \quad \Longleftrightarrow \quad Y^2 = b^2 - \tfrac{1}{4}.$$

Since $b > \frac{1}{2}$, $b^2 - \frac{1}{4} > 0$, so $Y = \sqrt{b^2 - \frac{1}{4}}$ is a genuine positive value, and

$$z = \pm \mathrm{i} \left(b^2 - \tfrac{1}{4}\right)^{1/4}$$

are purely imaginary roots.

*Real root.* Put $z = x$ and look for a root with $x \geq -b$, where $|x + b| = x + b$. The equation becomes

$$x^2 + x + b - \tfrac{1}{2} = 0,$$

with discriminant $1 - 4(b - \frac{1}{2}) = 3 - 4b$, which is positive because $b < \frac{3}{4}$. Take the larger root

$$x_+ = \frac{-1 + \sqrt{3 - 4b}}{2}.$$

We must check that it lies in the range $x \geq -b$. Now $x_+ \geq -b$ is equivalent to $\sqrt{3 - 4b} \geq 1 - 2b$, and this is certainly true because $b > \frac{1}{2}$ makes the right-hand side negative while the left-hand side is positive. So $x_+$ is a purely real root.

Hence for $\frac{1}{2} < b < \frac{3}{4}$ the equation has at least one purely imaginary root and at least one purely real root, as required. (Each condition on $b$ was used exactly once: $b > \frac{1}{2}$ for the imaginary root and for $x_+ \geq -b$, and $b < \frac{3}{4}$ for the real discriminant.)

**Part (ii)**

Let $z = x + \mathrm{i}y$. Then

$$z^3 = x^3 - 3xy^2 + \left(3x^2y - y^3\right)\mathrm{i}, \qquad |z+2|^2 = (x+2)^2 + y^2.$$

The second is real, so the imaginary part of the equation is

$$y\left(3x^2 - y^2\right) = 0,$$

giving $y = 0$ or $y^2 = 3x^2$. (Geometrically, $z$ lies on the real axis or on one of the lines $y = \pm\sqrt{3}\,x$: exactly the directions whose argument is a multiple of $\frac{\pi}{3}$, which are the ones that make $z^3$ real.)

*Case* $y = 0$. The real part gives $x^3 + (x+2)^2 = 4$, that is

$$x^3 + x^2 + 4x = x\left(x^2 + x + 4\right) = 0.$$

The quadratic has discriminant $1 - 16 < 0$, so the only solution is $x = 0$, giving $z = 0$.

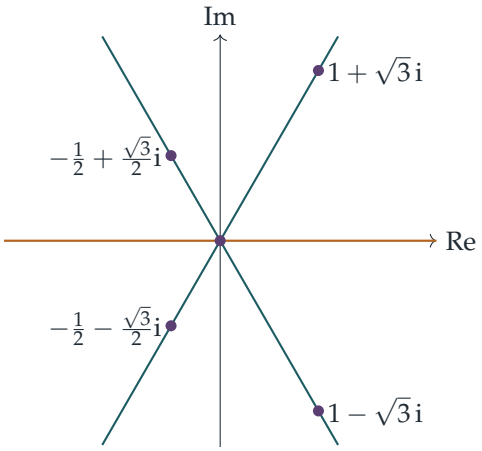
*Case* $y^2 = 3x^2$. Substituting $y^2 = 3x^2$ into the real part, $x^3 - 3xy^2 + (x+2)^2 + y^2 = 4$:

$$\begin{aligned} x^3 - 9x^3 + x^2 + 4x + 4 + 3x^2 &= 4 \\ -8x^3 + 4x^2 + 4x &= 0 \\ -4x\left(2x^2 - x - 1\right) &= 0 \\ -4x(2x+1)(x-1) &= 0. \end{aligned}$$

- $x = 0$ forces $y = 0$: this is $z = 0$ again.
- $x = 1$ gives $y^2 = 3$, so $z = 1 \pm \sqrt{3}\,\mathrm{i}$.
- $x = -\frac{1}{2}$ gives $y^2 = \frac{3}{4}$, so $z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\,\mathrm{i}$.

Every step was an equivalence (we only substituted and factorised), so all of these really are solutions. The equation therefore has exactly five roots:

$$z = 0, \qquad z = 1 \pm \sqrt{3}\,\mathrm{i}, \qquad z = -\tfrac{1}{2} \pm \tfrac{\sqrt{3}}{2}\,\mathrm{i}.$$



The roots lie on the three lines $y = 0$ and $y = \pm\sqrt{3}\,x$ allowed by the imaginary part.

## Checks

- In polar form, $1 + \sqrt{3}\,\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\pi/3}$, so $z^3 = 8\mathrm{e}^{\mathrm{i}\pi} = -8$, and $|3 + \sqrt{3}\,\mathrm{i}|^2 = 9 + 3 = 12$; indeed $-8 + 12 = 4$.
- $-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} = \mathrm{e}^{2\pi\mathrm{i}/3}$ is a cube root of unity, so $z^3 = 1$, and $|\frac{3}{2} + \frac{\sqrt{3}}{2}\mathrm{i}|^2 = \frac{9}{4} + \frac{3}{4} = 3$; indeed $1 + 3 = 4$.
- In (d), try $b = \frac{5}{8}$: then $Y = \sqrt{\frac{25}{64} - \frac{16}{64}} = \frac{3}{8}$, and $-\frac{3}{8} + \sqrt{\frac{3}{8} + \frac{25}{64}} = -\frac{3}{8} + \frac{7}{8} = \frac{1}{2}$. The real root is $x_+ = \frac{-1+\sqrt{1/2}}{2} \approx -0.146$, which is above $-\frac{5}{8}$ as needed.
- A computer search for roots of the part (ii) equation over the whole region $|x|, |y| \le 3$ found exactly the five roots above.

## Common slips

- Treating $|z + b|$ as if it had an imaginary part, or expanding $|z + 2|^2$ as $(z + 2)^2$. The modulus squared is $(x + 2)^2 + y^2$, a real number.
- Forgetting to check that a root of a case equation lies in the range where that case applies, as in (c) and (d).
- In (ii), dividing the imaginary part by $y$ and so losing the real root $z = 0$.

## Beyond STEP

Equations mixing $z$ with $|z|$ (or with $\bar{z}$) are not polynomial in $z$, so the fundamental theorem of algebra says nothing about how many roots they have: here a "cubic" has five. The right way to think of them is as a pair of real equations in two real unknowns, and you meet this viewpoint again in first-year complex analysis, where a function is differentiable in the complex sense only when it depends on $z$ alone and not on $\bar{z}$ (the Cauchy–Riemann equations). Needham's *Visual Complex Analysis* makes this geometric picture vivid.

WORKED SOLUTION

# STEP II 2025, Question 3

Pure • Comparing Powers

**STEP II 2025, Question 3**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Every comparison of the form $a^b$ against $b^a$ (with $a, b > 0$) becomes, after taking logarithms and dividing by $ab$, a comparison of $\frac{\ln a}{a}$ with $\frac{\ln b}{b}$. So everything comes down to the graph of $f(x) = \frac{\ln x}{x}$: it rises to a single peak at $x = \mathrm{e}$ and then falls. Two numbers on the same side of e are easy to compare; two on opposite sides need a stepping stone, such as the neat coincidence $f(2) = f(4)$.

## Solution

Throughout, write $f(x) = \dfrac{\ln x}{x}$ for $x > 0$. Since ln is increasing, for positive $a, b$

$$a^b > b^a \iff b \ln a > a \ln b \iff \frac{\ln a}{a} > \frac{\ln b}{b} \iff f(a) > f(b),$$

where we divided by $ab > 0$. We use this translation in every part.

### Part (i)

By the quotient rule,

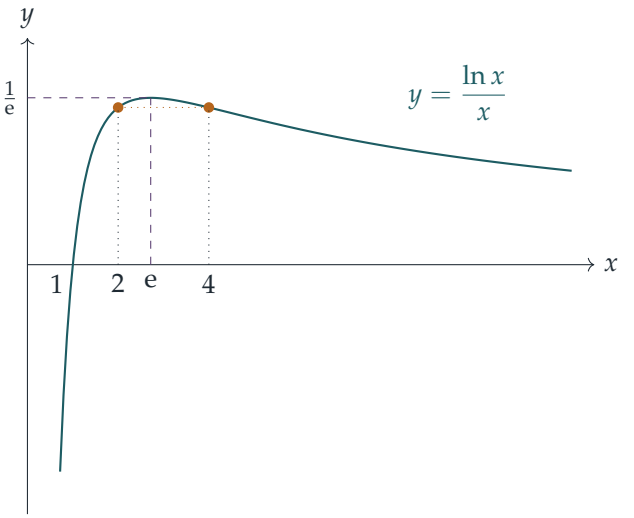
$$f'(x) = \frac{\frac{1}{x} \cdot x - \ln x}{x^2} = \frac{1 - \ln x}{x^2},$$

which is positive for $0 < x < \mathrm{e}$ and negative for $x > \mathrm{e}$. So $f$ is strictly increasing on $(0, \mathrm{e}]$ and strictly decreasing on $[\mathrm{e}, \infty)$, with a maximum value $f(\mathrm{e}) = 1/\mathrm{e}$.

Other features:

- $f(1) = 0$; $f < 0$ on $(0, 1)$ and $f > 0$ on $(1, \infty)$.
- As $x \to 0^+$, $\ln x \to -\infty$ while $1/x \to +\infty$, so $f(x) \to -\infty$: the $y$-axis is an asymptote.
- As $x \to \infty$, $f(x) \to 0^+$ because $\ln x$ grows more slowly than $x$: the $x$-axis is an asymptote.

- Differentiating again gives $f''(x) = \dfrac{2\ln x - 3}{x^3}$, so there is a point of inflection at $x = \mathrm{e}^{3/2} \approx 4.48$.




The amber points mark $f(2) = \frac{\ln 2}{2} = \frac{2\ln 2}{4} = f(4)$, which we shall use in part (iii).

**Part (ii)**

**(a)** By the translation, $3^\pi > \pi^3 \iff f(3) > f(\pi)$. Both 3 and $\pi$ exceed e, and $f$ is strictly decreasing on $[\mathrm{e}, \infty)$; since $3 < \pi$ we get $f(3) > f(\pi)$, so $3^\pi > \pi^3$.

**(b)** Here the inequality is $\left(\frac{9}{4}\right)^{\sqrt{5}} > \sqrt{5}^{9/4}$, which is equivalent to $f\left(\frac{9}{4}\right) > f(\sqrt{5})$. Now both numbers lie below e: indeed $\sqrt{5} < \frac{9}{4} = 2.25 < \mathrm{e}$, where $\sqrt{5} < \frac{9}{4}$ because $5 < \frac{81}{16}$. On $(0, \mathrm{e}]$ the function $f$ is strictly increasing, so the larger input gives the larger value: $f\left(\frac{9}{4}\right) > f(\sqrt{5})$, as required.

**Part (iii)**

With $a = x$ and $b = x + 2$, the translation gives

$$x^{x+2} > (x+2)^x \iff f(x) > f(x+2).$$

For $1 < x < 2$ we have $x \in (1, 2)$ and $x + 2 \in (3, 4)$, on opposite sides of e, so we need a stepping stone.

> **Ask yourself** How can we compare $f$ at a point left of e with $f$ at a point right of it?

Use the horizontal line through the two amber points. Since $1 < x < 2 < \mathrm{e}$ and $f$ is increasing there, $f(x) < f(2)$. Since $\mathrm{e} < x + 2 < 4$ and $f$ is decreasing there,

$f(x+2) > f(4)$. And $f(2) = f(4)$, so

$$f(x) < f(2) = f(4) < f(x+2).$$

Hence $f(x) < f(x+2)$, and so $(x+2)^x$ is the larger: $(x+2)^x > x^{x+2}$ for every $x$ in $(1,2)$.

**Part (iv)**

Rewrite both sides of the first inequality as powers of 3 and of $2\sqrt{2}$:

$$9^{\sqrt{2}} = \left(3^2\right)^{\sqrt{2}} = 3^{2\sqrt{2}}, \qquad \sqrt{2}^9 = \left(\sqrt{2}^3\right)^3 = \left(2\sqrt{2}\right)^3.$$

So the two inequalities have identical left-hand sides and identical right-hand sides; they are the same statement and hence equivalent.

By the translation, $3^{2\sqrt{2}} > (2\sqrt{2})^3 \iff f(3) > f(2\sqrt{2})$. Now $\mathrm{e}^2 < 8$ gives $\mathrm{e} < \sqrt{8} = 2\sqrt{2}$ (taking positive square roots), and $2\sqrt{2} < 3$ since $8 < 9$. So

$$\mathrm{e} < 2\sqrt{2} < 3,$$

and because $f$ is strictly decreasing on $[\mathrm{e}, \infty)$, $f(2\sqrt{2}) > f(3)$. Therefore $(2\sqrt{2})^3 > 3^{2\sqrt{2}}$, that is,

$$\sqrt{2}^9 > 9^{\sqrt{2}}: \quad \text{the larger is } \sqrt{2}^9.$$

**Part (v)**

Let $a = \sqrt[3]{3}$. We want to compare $8^a$ with $a^8$, that is, $f(8)$ with $f(a)$. Since $a < 2 < \mathrm{e} < 8$, the two inputs are again on opposite sides of the peak, and they turn out to be close ($f(a) \approx 0.254$, $f(8) \approx 0.260$), so we need something sharper than part (iii)'s trick.

**Ask yourself** Can we write $f(a)$ and $f(8)$ in terms of $f$ at simpler points?

The useful identity is $f(x^k) = \dfrac{k \ln x}{x^k}$. It gives

$$f(a) = \frac{\frac{1}{3}\ln 3}{a} = \frac{f(3)}{a}, \qquad f(8) = \frac{3\ln 2}{8} = \frac{3}{4}f(2), \qquad f(2\sqrt{2}) = \frac{\frac{3}{2}\ln 2}{2\sqrt{2}} = \frac{3}{2\sqrt{2}}f(2).$$

From part (iv), $f(3) < f(2\sqrt{2})$. Dividing by $a > 0$,

$$f(a) = \frac{f(3)}{a} < \frac{f(2\sqrt{2})}{a} = \frac{3f(2)}{2\sqrt{2}\,a}.$$

Finally, $a = 3^{1/3} > 2^{1/2} = \sqrt{2}$, because raising both to the sixth power gives $9 > 8$. Hence $2\sqrt{2}\,a > 2\sqrt{2} \cdot \sqrt{2} = 4$, and since $f(2) > 0$,

$$f(a) < \frac{3f(2)}{2\sqrt{2}\,a} < \frac{3f(2)}{4} = f(8).$$

So $f(8) > f(a)$, which by the translation means $8^a > a^8$: the larger is $8^{\sqrt[3]{3}}$.

## Checks

Numerically, $3^{\pi} \approx 31.54$ against $\pi^3 \approx 31.01$; $\left(\frac{9}{4}\right)^{\sqrt{5}} \approx 6.131$ against $\sqrt{5}^{9/4} \approx 6.114$; at $x = 1.5$, $1.5^{3.5} \approx 4.13$ against $3.5^{1.5} \approx 6.55$; $9^{\sqrt{2}} \approx 22.36$ against $\sqrt{2}^{9} \approx 22.63$; and $8^{\sqrt[3]{3}} \approx 20.07$ against $\sqrt[3]{3}^{8} \approx 18.72$. All agree with the conclusions above. The margins are small in (ii) and (iv), which is exactly why the graph argument is needed rather than rough estimation. A computer check of part (iii) at $10^5$ random points in $(1,2)$ found no exceptions.

## Common slips

- Comparing $f$ at two points on opposite sides of e as if $f$ were monotonic. In (iii) and (v) a stepping stone is essential.
- Forgetting that dividing $b \ln a > a \ln b$ by $ab$ keeps the direction only because $ab > 0$.
- In (ii)(b), not justifying which of $\frac{9}{4}$ and $\sqrt{5}$ is larger, or that both lie below e.

## Beyond STEP

The equation $a^b = b^a$ with $a \neq b$ is exactly $f(a) = f(b)$, and every horizontal line $y = c$ with $0 < c < 1/\mathrm{e}$ cuts the graph twice. Its solutions with $a < b$ can be parametrised as $a = (1 + \frac{1}{t})^t$, $b = (1 + \frac{1}{t})^{t+1}$ for $t > 0$; taking $t = 1$ gives $(2, 4)$, the only such pair of positive integers. As $t \to \infty$ both tend to e, one of the standard limits of first-year analysis. For the careful treatment of monotonicity and the mean value theorem behind arguments like these, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

# STEP II 2025, Question 4

Pure • Floor Function Identities

**STEP II 2025, Question 4**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Everything rests on one fact: $\lfloor x \rfloor = m$ means exactly that $m$ is an integer with $m \leq x < m+1$. Part (i) says that integer shifts pass straight through the floor. With that, $\mathrm{f}_n$ turns out to be periodic with period $\frac{1}{n}$, so we only need it on one short interval, where every term is easy to work out. Part (iii) is the case $n = 2$ in disguise, and it makes the infinite sum telescope.

## Solution

### Part (i)

Let $m = \lfloor x \rfloor$, so $m$ is an integer and $m \leq x < m+1$. Adding the integer $n$ gives

$$m + n \leq x + n < (m+n) + 1.$$

Since $m+n$ is an integer, it is the largest integer not exceeding $x+n$. So $\lfloor x+n \rfloor = m+n = \lfloor x \rfloor + n$, as required.

### Part (ii)(a)

Write $\mathrm{f}_n(x) = \sum_{k=0}^{n-1} \lfloor x + \frac{k}{n} \rfloor - \lfloor nx \rfloor$. Replacing $x$ by $x + \frac{1}{n}$ moves each term in the sum along by one place:

$$\mathrm{f}_n\left(x + \tfrac{1}{n}\right) = \sum_{k=1}^{n} \left\lfloor x + \tfrac{k}{n} \right\rfloor - \lfloor nx + 1 \rfloor.$$

The last term of the sum is $\lfloor x+1 \rfloor = \lfloor x \rfloor + 1$, and $\lfloor nx+1 \rfloor = \lfloor nx \rfloor + 1$, both by part (i) with $n = 1$. So

$$\mathrm{f}_n\left(x + \tfrac{1}{n}\right) = \lfloor x \rfloor + 1 + \sum_{k=1}^{n-1} \left\lfloor x + \tfrac{k}{n} \right\rfloor - \lfloor nx \rfloor - 1 = \mathrm{f}_n(x).$$

**Part (ii)(b)**

Let $0 \le t < \frac{1}{n}$. For $0 \le k \le n-1$,

$$0 \le t + \tfrac{k}{n} < \tfrac{1}{n} + \tfrac{n-1}{n} = 1,$$

so $\left\lfloor t + \frac{k}{n} \right\rfloor = 0$. Also $0 \le nt < 1$, so $\lfloor nt \rfloor = 0$. Every term is zero, and therefore $\mathrm{f}_n(t) = 0$.

**Part (ii)(c)**

**Ask yourself** Part (a) only lets us step *up* by $\frac{1}{n}$. How do we reach an $x$ that lies below $[0, \frac{1}{n})$?

Applying (a) at the point $x - \frac{1}{n}$ gives $\mathrm{f}_n(x) = \mathrm{f}_n\left(x - \frac{1}{n}\right)$, so we can step down as well as up. By induction, $\mathrm{f}_n\left(x + \frac{m}{n}\right) = \mathrm{f}_n(x)$ for every integer $m$, positive or negative.

Now take any real $x$ and let $m = \lfloor nx \rfloor$, so $m \le nx < m + 1$. Put $t = x - \frac{m}{n}$. Dividing by $n > 0$ gives $0 \le t < \frac{1}{n}$. Hence

$$\mathrm{f}_n(x) = \mathrm{f}_n\left(t + \tfrac{m}{n}\right) = \mathrm{f}_n(t) = 0$$

by (b). So $\mathrm{f}_n(x) = 0$ for all $x$, which is the identity

$$\lfloor x \rfloor + \left\lfloor x + \tfrac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \tfrac{n-1}{n} \right\rfloor = \lfloor nx \rfloor.$$

**Part (iii)(a)**

Take $n = 2$ in (ii)(c) and evaluate at $\frac{x}{2}$:

$$0 = \mathrm{f}_2\left(\tfrac{x}{2}\right) = \left\lfloor \tfrac{x}{2} \right\rfloor + \left\lfloor \tfrac{x}{2} + \tfrac{1}{2} \right\rfloor - \left\lfloor 2 \cdot \tfrac{x}{2} \right\rfloor,$$

which rearranges to $\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor$.

**Part (iii)(b)**

**Ask yourself** How can (iii)(a), which has a lone $+1$ in it, help with the term $\left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor$?

Divide the top and bottom by $2^k$. With $y = x/2^k$,

$$\left\lfloor \frac{x + 2^k}{2^{k+1}} \right\rfloor = \left\lfloor \frac{y+1}{2} \right\rfloor = \lfloor y \rfloor - \left\lfloor \frac{y}{2} \right\rfloor = \left\lfloor \frac{x}{2^k} \right\rfloor - \left\lfloor \frac{x}{2^{k+1}} \right\rfloor,$$

using (iii)(a) with $y$ in place of $x$. So the sum of the terms for $k = 0, 1, \ldots, K$ telescopes:

$$S_K = \sum_{k=0}^{K} \left\lfloor \frac{x + 2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \left\lfloor \frac{x}{2^{K+1}} \right\rfloor.$$

As $K \to \infty$, $x/2^{K+1} \to 0$, and the answer depends on which side of 0 it approaches from.

- If $x \geq 0$: once $2^{K+1} > x$ we have $0 \leq x/2^{K+1} < 1$, so the last floor is 0 and $S_K = \lfloor x \rfloor$ from then on.
- If $x < 0$: once $2^{K+1} \geq -x$ we have $-1 \leq x/2^{K+1} < 0$, so the last floor is $-1$ and $S_K = \lfloor x \rfloor + 1$ from then on.

In either case only finitely many terms are non-zero, so the infinite sum is

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{2^2} \right\rfloor + \cdots = \begin{cases} \lfloor x \rfloor, & x \geq 0, \\ \lfloor x \rfloor + 1, & x < 0. \end{cases}$$

For a non-negative integer $x$ the sum is simply $x$.

## Checks

Try $x = 7$. The terms are $\lfloor 4 \rfloor = 4$, $\lfloor \frac{9}{4} \rfloor = 2$, $\lfloor \frac{11}{8} \rfloor = 1$, $\lfloor \frac{15}{16} \rfloor = 0$, and every later term is 0. They add up to $7 = \lfloor 7 \rfloor$. Try $x = -\frac{5}{2}$: the terms are $\lfloor -\frac{3}{4} \rfloor = -1$, $\lfloor -\frac{1}{8} \rfloor = -1$, $\lfloor \frac{3}{16} \rfloor = 0$, and so on. The sum is $-2 = \lfloor -\frac{5}{2} \rfloor + 1$, which confirms that the negative case really does differ. For (ii), $n = 3$ and $x = 1.5$ give $1 + 1 + 2 = 4 = \lfloor 4.5 \rfloor$. A computer check on twenty thousand random rationals $x$ and values $1 \leq n \leq 30$ confirmed every identity in the question, as well as the telescoped partial sums.

## Common slips

- In (ii)(c), using (a) only to step upwards, which covers $x \geq 0$ but says nothing about negative $x$.
- In (iii)(b), assuming that $\lfloor x/2^{K+1} \rfloor \to 0$ for all $x$. For negative $x$ it settles at $-1$, not 0.
- Writing $\lfloor a + b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor$ when neither $a$ nor $b$ is an integer. Part (i) is precisely the case in which this is allowed.

## Beyond STEP

The identity in (ii)(c) is Hermite's identity. It belongs to a family of floor-function results that are central to elementary number theory, for example Legendre's formula $\sum_{k \geq 1} \lfloor N/p^k \rfloor$ for the power of a prime $p$ dividing $N!$. The sum in (iii)(b) counts something too: for a positive integer $x$, the $k$th term is the number of integers from 1 to $x$ that are divisible by $2^k$ but not by $2^{k+1}$, and every integer is counted exactly once. Davenport's *The Higher Arithmetic* is a good place to see these ideas at work.

WORKED SOLUTION

# STEP II 2025, Question 5

Pure • Integration by Substitution

**STEP II 2025, Question 5**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Every integral here runs from 0 to $\infty$, and a substitution of the form $x = u^{-m}$ maps $[0, \infty)$ onto itself, with the ends swapped. The minus sign from $\mathrm{d}x$ undoes the swap of limits, so we land on another integral over $[0, \infty)$. If the new integrand is minus the old one, the integral equals its own negative and so is zero. If it is a multiple of a different integrand of the same shape, we get a relation between two integrals that we never evaluate. In parts (iii) and (iv) we keep the exponent general and choose it so that the powers of $u$ cancel.

## Solution

Throughout, if $x = u^{-m}$ with $m > 0$, then $x \to 0$ as $u \to \infty$ and $x \to \infty$ as $u \to 0$, so

$$\int_0^\infty f(x)\,\mathrm{d}x = \int_\infty^0 f(u^{-m})\left(-mu^{-m-1}\right)\mathrm{d}u = \int_0^\infty f(u^{-m})\,mu^{-m-1}\,\mathrm{d}u.$$

We use this with $m = 1$ and $m = 2$, and then with general $m$.

### Part (i)

Call the integral $I$ and put $x = u^{-1}$, so $\mathrm{d}x \to u^{-2}\,\mathrm{d}u$ once the limits are put back in order. For $u > 0$,

$$x(x^3+1) = \frac{1}{u}\cdot\frac{1+u^3}{u^3} = \frac{u^3+1}{u^4}, \qquad \text{so} \qquad \sqrt{x(x^3+1)} = \frac{\sqrt{u^3+1}}{u^2},$$

and

$$\sqrt{x} - 1 = \frac{1}{\sqrt{u}} - 1 = \frac{1-\sqrt{u}}{\sqrt{u}}.$$

Hence

$$I = \int_0^\infty \frac{1-\sqrt{u}}{\sqrt{u}} \cdot \frac{u^2}{\sqrt{u^3+1}} \cdot \frac{1}{u^2}\,\mathrm{d}u = \int_0^\infty \frac{1-\sqrt{u}}{\sqrt{u(u^3+1)}}\,\mathrm{d}u$$
$$= -\int_0^\infty \frac{\sqrt{u}-1}{\sqrt{u(u^3+1)}}\,\mathrm{d}u = -I.$$

The last integral is $I$ again, since the name of the variable doesn't matter. So $2I = 0$, giving $I = 0$, as required.

Note that this does not say the integrand is zero. It is negative on $(0,1)$ and positive on $(1,\infty)$, and the substitution matches the two areas exactly.

#### Part (ii)

Now put $x = u^{-2}$, so $\mathrm{d}x \to 2u^{-3}\,\mathrm{d}u$ after reordering the limits. For $u > 0$,

$$x^3+1 = u^{-6}+1 = \frac{1+u^6}{u^6}, \qquad \sqrt{x^3+1} = \frac{\sqrt{u^6+1}}{u^3}.$$

Therefore

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^3+1}} = \int_0^\infty \frac{u^3}{\sqrt{u^6+1}} \cdot \frac{2}{u^3}\,\mathrm{d}u = 2\int_0^\infty \frac{\mathrm{d}u}{\sqrt{u^6+1}},$$

which is the required result once we rename $u$ as $x$.

#### Part (iii)

Part (i) is the case $s = 1$, $p = 3$, $r = \frac{1}{2}$, so we repeat its method with letters. Call the integral $J$ and put $x = u^{-1}$:

$$x^s(x^p+1) = u^{-s} \cdot \frac{1+u^p}{u^p} = \frac{u^p+1}{u^{s+p}}, \qquad \text{so} \qquad \sqrt{x^s(x^p+1)} = u^{-(s+p)/2}\sqrt{u^p+1},$$

and also $x^r - 1 = (1-u^r)/u^r$. So

$$J = \int_0^\infty \frac{(1-u^r)\,u^{-r}\,u^{(s+p)/2}\,u^{-2}}{\sqrt{u^p+1}}\,\mathrm{d}u = -\int_0^\infty \frac{(u^r-1)\,u^{(s+p)/2-r-2}}{\sqrt{u^p+1}}\,\mathrm{d}u.$$

Compare this with $J$ itself, written as

$$J = \int_0^\infty \frac{(u^r-1)\,u^{-s/2}}{\sqrt{u^p+1}}\,\mathrm{d}u.$$

**Ask yourself** Why is it enough to match the powers of $u$, rather than to evaluate anything?

If the two powers of $u$ agree, the substitution turns $J$ into $-J$, and then $J = 0$ exactly as in part (i). So we need

$$\frac{s+p}{2} - r - 2 = -\frac{s}{2} \quad \Longleftrightarrow \quad r = s + \frac{p}{2} - 2.$$

With this value of $r$ we get $J = -J$, so $J = 0$. (The question tells us that $p$ and $s$ are such that the integrals converge, and that is all the argument $J = -J \Rightarrow J = 0$ needs.) As a check, $s = 1$ and $p = 3$ give $r = 1 + \frac{3}{2} - 2 = \frac{1}{2}$, which is part (i).

#### Part (iv)

Part (ii) is the case $k = 2$, $p = 3$, $q = 6$, which came from $x = u^{-2}$. So we try $x = u^{-m}$ with $m > 0$, so that $\mathrm{d}x \to mu^{-m-1}\,\mathrm{d}u$: since $x^p + 1 = (1 + u^{mp})/u^{mp}$,

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^p + 1}} = m \int_0^\infty \frac{u^{mp/2}\, u^{-m-1}}{\sqrt{u^{mp} + 1}}\,\mathrm{d}u.$$

The stray power of $u$ disappears when

$$\frac{mp}{2} = m + 1 \quad \Longleftrightarrow \quad p = 2 + \frac{2}{m},$$

and then the right-hand side is $m \displaystyle\int_0^\infty \frac{\mathrm{d}u}{\sqrt{u^q + 1}}$ with $q = mp = 2m + 2$.

Now take $m = k$. For any $k > 0$, the choice

$$p = 2 + \frac{2}{k}, \qquad q = 2k + 2$$

gives

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^p + 1}} = k \int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^q + 1}}.$$

Both exponents exceed 2, because $k > 0$. That is exactly the condition for these integrals to converge, since the integrand behaves like $x^{-p/2}$ for large $x$. So the integrals in our relation exist. With $k = 2$ we recover $p = 3$, $q = 6$ from part (ii), and $k = 1$ gives the trivial case $p = q = 4$.

## Checks

- Numerically, $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^3 + 1}} \approx 2.8044$ and $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^6 + 1}} \approx 1.4022$, exactly half, as part (ii) says.
- The integral in part (i) evaluates to zero to machine precision, even though the negative contribution comes mostly from near $x = 0$ and the positive one is spread thinly over $(1, \infty)$.
- For random admissible $p$ and $s$, a computer check gave zero for the part (iii) integral with $r = s + \frac{p}{2} - 2$. Changing $r$ by 0.3 gave clearly non-zero values.
- For $k = \frac{1}{2}$ (so $p = 6$, $q = 3$) the relation in part (iv) is part (ii) read backwards, as it should be.

## Common slips

- Forgetting that the substitution swaps the limits. The minus sign from $\mathrm{d}x$ and the reversed limits cancel, and losing either one gives a wrong sign in part (i).
- Writing $\sqrt{u^{-s}} = u^{-s}/2$ instead of $u^{-s/2}$ when handling the general powers in part (iii).
- In part (iv), finding one pair $(p, q)$ for a particular $k$ rather than giving $p$ and $q$ as functions of every $k > 0$.

## Beyond STEP

These integrals can all be evaluated in closed form. Substituting $t = x^p/(1+x^p)$ turns $\int_0^\infty (1+x^p)^{-1/2}\,\mathrm{d}x$ into $\frac{1}{p}\,\mathrm{B}\left(\frac{1}{p}, \frac{1}{2} - \frac{1}{p}\right)$, where B is Euler's Beta function. Combined with the Gamma function, this gives the ratio in part (iv) explicitly. The trick of a symmetry $x \mapsto 1/x$ that sends an integral to minus itself appears again and again in analysis and in contour integration. For a careful treatment of improper integrals, including the convergence questions we were allowed to skip, see Abbott, *Understanding Analysis*. For many more STEP-style substitution tricks, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

# STEP II 2025, Question 6

Pure • Circles Touching a Parabola

**STEP II 2025, Question 6**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Every circle here is centred on the $y$-axis, so everything is symmetric in $x$ and "touching at two points" means touching at a mirror pair $(\pm x_0, y_0)$ with $x_0 \neq 0$. Tangency is easiest to capture by saying the normals agree: the radius to the point of contact must be along the normal to the other curve. Once parts (i) and (ii) each give a formula for $r^2$ in terms of the centre height $a$, part (iii) is about a quadratic in $a$ whose two roots are $a_1$ and $a_2$.

## Solution

### Part (i)

Suppose the circle touches the parabola at $P = (x_0, y_0)$, with $x_0 \neq 0$. (Two distinct points of contact cannot both lie on the axis, since the parabola meets $x = 0$ only at the origin; and by symmetry, if $(x_0, y_0)$ is a point of contact then so is $(-x_0, y_0)$.)

The normal to the circle at $P$ is along the radius, $(x_0,\ y_0 - a)$. Writing the parabola as $2ky - x^2 = 0$, its normal is along the gradient $(-2x_0,\ 2k)$. Tangency means these are parallel:

$$x_0 \cdot 2k - (y_0 - a)(-2x_0) = 0 \quad \Longrightarrow \quad 2x_0\,(k + y_0 - a) = 0.$$

Since $x_0 \neq 0$, we get $y_0 = a - k$. Then $P$ is on the parabola, so $x_0^2 = 2ky_0 = 2k(a-k)$, and $P$ is on the circle, so

$$r^2 = x_0^2 + (y_0 - a)^2 = 2k(a-k) + k^2 = k(2a-k),$$

as required.

Conversely, let $a > k > 0$ and $r^2 = k(2a-k)$. Put $y_0 = a - k > 0$ and $x_0 = \sqrt{2k(a-k)} > 0$. Both $(\pm x_0, y_0)$ lie on the parabola by construction, and on the circle because $x_0^2 + (y_0 - a)^2 = 2k(a-k) + k^2 = r^2$. At each of these points the normals $(\pm x_0, -k)$ and $(\mp 2x_0, 2k)$ are parallel, so the curves share a tangent there. Hence the circle touches the parabola at two distinct points, since $x_0 \neq 0$.

(As a cross-check, substituting $x^2 = 2ky$ into the circle gives $y^2 - 2(a-k)y + (a-k)^2 = 0$, that is $(y-(a-k))^2 = 0$: a repeated root, which is the algebraic signature of tangency.)

**Part (ii)**

The line $y = c + x$, or $x - y + c = 0$, is tangent exactly when its distance from the centre $(0, a)$ equals $r$:

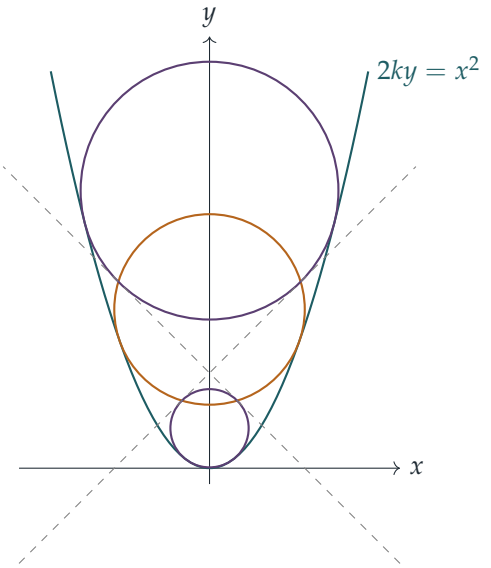
$$r = \frac{|0 - a + c|}{\sqrt{2}} \quad \Longrightarrow \quad r^2 = \frac{(a-c)^2}{2}.$$

The line $y = c - x$ is the mirror image in the $y$-axis, so it gives the same condition.

The point of contact is the foot of the perpendicular from $(0, a)$. The radius is perpendicular to the line, so it has direction $(1, -1)$; moving from $(0, a)$ by $t(1, -1)$ and landing on $y = c + x$ gives $a - t = c + t$, so $t = \frac{a-c}{2}$. The point of contact with $y = c + x$ is

$$\left(\tfrac{a-c}{2},\ \tfrac{a+c}{2}\right),$$

and by symmetry the point of contact with $y = c - x$ is $\left(-\tfrac{a-c}{2},\ \tfrac{a+c}{2}\right)$.




*The case $k = 1$, $c = 3$: the parabola, the two circles $C_1$, $C_2$ (plum) touching it and the lines $y = 3 \pm x$ (dashed), and the circle of part (iii)(b) (amber).*

**Part (iii)(a)**

Each circle touches the parabola at two points, so by part (i), $r_i^2 = k(2a_i - k)$; and each is tangent to the lines $y = c \pm x$, so by part (ii), $r_i^2 = \frac{1}{2}(a_i - c)^2$. Equating,

$$(a_i - c)^2 = 2k(2a_i - k) \quad \Longrightarrow \quad a_i^2 - 2(c + 2k)\, a_i + c^2 + 2k^2 = 0.$$

So $a_1$ and $a_2$ both satisfy the quadratic $a^2 - 2(c+2k)a + c^2 + 2k^2 = 0$. As $a_1 \neq a_2$, they are its two roots, and so

$$a_1 + a_2 = 2c + 4k, \qquad a_1 a_2 = c^2 + 2k^2.$$

Then

$$\begin{aligned} a_1^2 + a_2^2 &= (a_1 + a_2)^2 - 2a_1 a_2 \\ &= 4c^2 + 16kc + 16k^2 - 2c^2 - 4k^2 \\ &= 2c^2 + 16kc + 12k^2, \end{aligned}$$

which completes (a).

#### Part (iii)(b)

By part (ii), the points of tangency on $C_i$ are $\left(\pm\frac{a_i - c}{2},\ \frac{a_i + c}{2}\right)$. It is tidier to write $u_i = a_i - c$, so the points are $\left(\pm\frac{u_i}{2},\ c + \frac{u_i}{2}\right)$.

> **?** **Ask yourself** We need one circle through four points, but only have two unknowns $d$ and $p$. How can we pin them down without solving for $a_1$ and $a_2$ explicitly?

The four points are two mirror pairs, and any circle centred on the $y$-axis is symmetric, so it is enough to make the circle pass through $\left(\frac{u}{2}, c + \frac{u}{2}\right)$ for $u = u_1$ and $u = u_2$. This condition is

$$\frac{u^2}{4} + \left(c - d + \frac{u}{2}\right)^2 = p^2 \quad \Longleftrightarrow \quad u^2 + 2(c-d)\,u + 2(c-d)^2 - 2p^2 = 0. \qquad (*)$$

On the other hand, substituting $a = u + c$ into the quadratic from (a) gives

$$u^2 - 4k\,u + 2k^2 - 4kc = 0, \qquad (**)$$

whose roots are exactly $u_1$ and $u_2$, which are distinct because $a_1 \neq a_2$. The circle passes through all four points precisely when $u_1$ and $u_2$ are both roots of $(*)$. Two monic quadratics with the same two distinct roots are identical, so we compare coefficients with $(**)$:

$$2(c-d) = -4k \quad \Longrightarrow \quad d = c + 2k,$$

and then $2(c-d)^2 - 2p^2 = 8k^2 - 2p^2 = 2k^2 - 4kc$, so

$$p^2 = 3k^2 + 2kc = k(3k + 2c).$$

#### Part (iii)(c)

We use the converse in part (i). With $d = c + 2k$,

$$k(2d - k) = k(2c + 3k) = p^2,$$

so the circle has exactly the form $r^2 = k(2a - k)$, with $d$ in place of $a$. It remains to check that $d > k$.

**Ask yourself** Where does an inequality like $d > k$ come from, when nothing in (a) or (b) mentioned one?

It comes from the fact that $C_1$ and $C_2$ really do touch the parabola at two points. In part (i), touching at two points forced the points of contact to have $y_0 = a - k$ with $x_0^2 = 2k(a-k) > 0$ (as $x_0 \neq 0$), so $a_1 > k$ and $a_2 > k$. Adding these and using (a),

$$2c + 4k = a_1 + a_2 > 2k \quad \Longrightarrow \quad c > -k \quad \Longrightarrow \quad d = c + 2k > k.$$

So $d > k > 0$ and $p^2 = k(2d-k)$, and by the second half of part (i) the circle $x^2 + (y-d)^2 = p^2$ touches the parabola $2ky = x^2$ at two points, namely $\left(\pm\sqrt{2k(c+k)},\ c+k\right)$.

## Checks

Take $k = 1$, $c = 3$ (the picture above). The quadratic in (a) is $a^2 - 10a + 11 = 0$, so $a = 5 \pm \sqrt{14} \approx 8.742$ and $1.258$. Their sum is $10 = 2c + 4k$ and their sum of squares is $78 = 2 \cdot 9 + 16 \cdot 3 + 12$. Part (b) gives $d = 5$ and $p^2 = 9$. The circle $x^2 + (y-5)^2 = 9$ meets $2y = x^2$ where $2y + (y-5)^2 = 9$, that is $(y-4)^2 = 0$: a double root at $y = 4$, so it touches at $(\pm 2\sqrt{2}, 4)$, as (c) predicts. Each of the four points of tangency, for instance $\left(\frac{\sqrt{14}+2}{2}, \frac{\sqrt{14}+8}{2}\right)$, lies at distance 3 from $(0,5)$. All the results were also confirmed by computer for hundreds of random admissible $k$ and $c$.

## Common slips

- In (i), forgetting that the touching points have $x_0 \neq 0$. That is what allows us to divide by $x_0$, and it is also what gives $a > k$.
- In (iii)(a), using only $r^2 = k(2a-k)$ or only $r^2 = \frac{1}{2}(a-c)^2$. The quadratic for $a$ comes from needing both at once.
- In (iii)(c), checking $p^2 = k(2d-k)$ but not $d > k$. Without it the "circle" might touch only at the origin, or not at all.

## Beyond STEP

By part (i), the circles centred on the $y$-axis that touch the parabola twice are $x^2 + y^2 - k^2 - 2a(y-k) = 0$: an equation *linear* in the parameter $a$. Such a family is called a pencil (or coaxal system) of circles, and $C_1$, $C_2$ and the circle in (b) all belong to it. Matching normals is the first step towards the idea of *contact* between curves; the circle that touches a curve to second order is its circle of curvature, met in a first course on the differential geometry of curves. Siklos, *Advanced Problems in Mathematics*, has many more STEP-style coordinate geometry problems in this spirit.

WORKED SOLUTION

# STEP II 2025, Question 7

Pure • Nonlinear Differential Equations

**STEP II 2025, Question 7**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

The right-hand side $2x\dfrac{\mathrm{d}x}{\mathrm{d}t}$ is exactly the derivative of $x^2$ with respect to $t$, so the equation integrates at once to a first-order equation $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2 + C$. The starting conditions fix $C$, and the three parts are the three possible signs of $C$: zero, positive and negative. Each first-order equation then separates, and the behaviour of the particle can be read off from the formula, or even straight from the sign of $x^2 + C$.

## Solution

### Integrating once

Since $\dfrac{\mathrm{d}}{\mathrm{d}t}\left(x^2\right) = 2x\dfrac{\mathrm{d}x}{\mathrm{d}t}$, the equation says

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(x^2\right), \qquad \text{so} \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = x^2 + C$$

for a constant $C$. At $t = 0$ we have $x = a$, so $C$ is the initial velocity minus $a^2$. In the three parts $C = 0$, $C = p^2$ and $C = -q^2$ respectively.

### Part (i)

Here $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2$. The velocity is positive whenever $x \neq 0$, so $x$ increases from $a > 0$ and never reaches 0; we may therefore divide by $x^2$ and separate:

$$\int_a^x \frac{\mathrm{d}u}{u^2} = t \quad \Longrightarrow \quad \frac{1}{a} - \frac{1}{x} = t \quad \Longrightarrow \quad x = \frac{a}{1 - at}.$$

As $t$ increases from 0 towards $\frac{1}{a}$, the particle moves in the positive direction ever faster (its velocity is $x^2$), and $x \to \infty$ as $t \to \frac{1}{a}$. The particle escapes to infinity in the finite time $\frac{1}{a}$, and the solution does not exist beyond that time.

**Part (ii)**

Now $\dfrac{\mathrm{d}x}{\mathrm{d}t} = x^2 + p^2$, which is always positive, so we can separate:

$$\int_a^x \frac{\mathrm{d}u}{u^2+p^2} = t \quad \Longrightarrow \quad \frac{1}{p}\left(\arctan\frac{x}{p} - \arctan\frac{a}{p}\right) = t.$$

Writing $\alpha = \arctan\dfrac{a}{p}$, which lies in $(0, \frac{\pi}{2})$ because $a, p > 0$,

$$x = p\tan(pt+\alpha).$$

This holds while $pt + \alpha < \frac{\pi}{2}$. The particle again moves in the positive direction with increasing speed, and $x \to \infty$ as $pt + \alpha \to \frac{\pi}{2}$, that is, as

$$t \to \frac{1}{p}\left(\frac{\pi}{2} - \arctan\frac{a}{p}\right) = \frac{1}{p}\arctan\frac{p}{a},$$

using $\arctan\theta + \arctan\frac{1}{\theta} = \frac{\pi}{2}$ for $\theta > 0$. So it escapes to infinity in a finite time, sooner than in (i), as you'd expect from the larger initial velocity.

**Part (iii)**

Now

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x^2 - q^2 = (x-q)(x+q).$$

The velocity is zero at $x = \pm q$, so these two positions are equilibria. The sign of $x^2 - q^2$ already tells us the story: the particle moves right when $x > q$, left when $-q < x < q$. So the answer depends on how $a$ compares with $q$.

Separate using partial fractions, $\dfrac{1}{x^2-q^2} = \dfrac{1}{2q}\left(\dfrac{1}{x-q} - \dfrac{1}{x+q}\right)$:

$$\frac{1}{2q}\ln\left|\frac{x-q}{x+q}\right| = t + \text{constant}.$$

Exponentiating and fixing the constant by $x = a$ at $t = 0$ suggests

$$\frac{x-q}{x+q} = Ke^{2qt}, \qquad K = \frac{a-q}{a+q}, \qquad \text{i.e.} \qquad x = q\,\frac{1+Ke^{2qt}}{1-Ke^{2qt}}.$$

To be sure the signs are right in every case, check this directly. Put $E = Ke^{2qt}$, so $\frac{\mathrm{d}E}{\mathrm{d}t} = 2qE$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} = q\cdot\frac{2\,\frac{\mathrm{d}E}{\mathrm{d}t}}{(1-E)^2} = \frac{4q^2E}{(1-E)^2}, \qquad x^2 - q^2 = q^2\,\frac{(1+E)^2-(1-E)^2}{(1-E)^2} = \frac{4q^2E}{(1-E)^2},$$

which agree. At $t = 0$, $1 + K = \frac{2a}{a+q}$ and $1 - K = \frac{2q}{a+q}$, so $x = q\cdot\frac{a}{q} = a$. So this formula is the solution, wherever $1 - E \neq 0$. Note that $-1 < K < 1$ since $a > 0$.

**Case $a > q$.** Then $0 < K < 1$ and $E = Ke^{2qt}$ increases from $K$. The particle starts with positive velocity $a^2 - q^2$ and moves in the positive direction, faster and faster. The denominator $1 - E$ reaches 0 when $e^{2qt} = \frac{1}{K}$, so

$$x \to \infty \quad \text{as} \quad t \to \frac{1}{2q} \ln \frac{a+q}{a-q}.$$

As in (i) and (ii), the particle escapes to infinity in finite time.

**Case $a = q$.** Then $K = 0$ and $x = q = a$ for all $t$. The initial velocity is 0, and so is the acceleration $2x\frac{\mathrm{d}x}{\mathrm{d}t}$: the particle stays at rest at $x = a$.

**Case $0 < a < q$.** Then $-1 < K < 0$, so $E < 0$ and the denominator $1 - E > 1$ never vanishes: the solution exists for all $t \geq 0$.

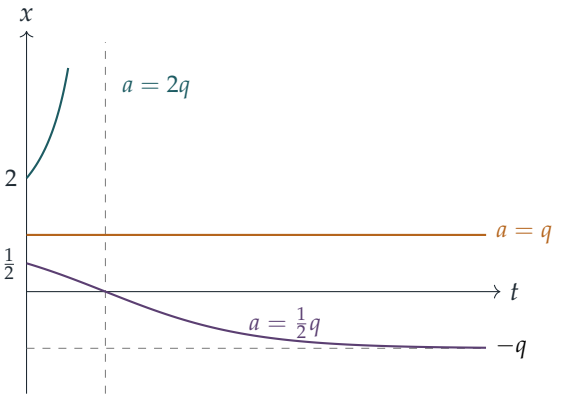
**Ask yourself** Does the particle stop at the origin, or go through it, and where does it end up?

The initial velocity $a^2 - q^2$ is negative, so the particle moves in the negative direction. It is at the origin when $E = -1$, i.e. at $t = \frac{1}{2q} \ln \frac{q+a}{q-a}$, and it passes straight through, since the velocity there is $-q^2 \neq 0$. As $t \to \infty$, $E \to -\infty$ and

$$x = q\,\frac{1+E}{1-E} \to -q.$$

Throughout, $|x| < q$ (because $\frac{1+E}{1-E} \in (-1, 1)$ for $E < 0$), so the velocity $x^2 - q^2$ stays negative: the particle moves steadily to the left and approaches $x = -q$ without ever reaching it. Its speed $q^2 - x^2$ is greatest, $q^2$, as it passes the origin, and then dies away to zero.

In hyperbolic form these solutions are $x = -q \coth\left(qt - \operatorname{arcoth} \frac{a}{q}\right)$ for $a > q$ and $x = -q \tanh\left(qt - \operatorname{artanh} \frac{a}{q}\right)$ for $a < q$. The sketch shows the three cases with $q = 1$.




For $a = 2q$ and $a = \frac{1}{2}q$ the special time $\frac{1}{2q} \ln 3$ (dashed vertical line) is the blow-up time and the time of crossing the origin respectively.

## Checks

- Letting $p \to 0$ in (ii): $p\tan(pt + \arctan\frac{a}{p}) \to \frac{a}{1-at}$ and $\frac{1}{p}\arctan\frac{p}{a} \to \frac{1}{a}$, matching (i). The same happens as $q \to 0$ in (iii) with $a > q$.
- Differentiate $x = \frac{a}{1-at}$: $\dot{x} = \frac{a^2}{(1-at)^2} = x^2$ and $\ddot{x} = \frac{2a^3}{(1-at)^3} = 2x\dot{x}$.
- Stepping the original second-order equation forward numerically for, say, $a = 0.5$, $q = 1.4$ gives $x \approx -1.0814$ at $t = 1$ and $x \approx -1.3987$ at $t = 3$, agreeing with the formula, and the particle is at the origin at $t \approx 0.2669$, as predicted.

## Common slips

- Not noticing that $2x\frac{\mathrm{d}x}{\mathrm{d}t}$ is an exact derivative, and trying a substitution such as $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ with $v\frac{\mathrm{d}v}{\mathrm{d}x}$, which works but takes longer.
- Dropping the modulus in the logarithm in (iii) and so getting the wrong sign of $K$ when $a < q$.
- Missing the equilibrium case $a = q$, or saying the particle "stops at $-q$" when it only approaches it.

## Beyond STEP

The finite-time blow-up seen here is a hallmark of nonlinear equations: $\dot{x} = x^2$ has a perfectly smooth right-hand side, yet its solutions only exist for a limited time, something that never happens for linear equations. Part (iii) is a first look at phase-line analysis: the equilibria $x = q$ (unstable) and $x = -q$ (stable) organise all the motion, and you can predict it from the sign of $x^2 - q^2$ before solving anything. Boyce and DiPrima's *Elementary Differential Equations* treats both ideas, including existence and uniqueness of solutions.

WORKED SOLUTION

# STEP II 2025, Question 8

Pure • Balanced Partitions of Integers

**STEP II 2025, Question 8**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Shifting every element of a set by the same amount $c$ changes its sum and its sum of squares in a controlled way: the change depends only on how many elements there are, their sum and (for squares) nothing more. So if two sets of the same size already agree in their sums and sums of squares, shifting one copy of each and swapping them over builds a bigger balanced partition. Part (ii) is the algebra behind this, and parts (iii) and (iv) are two uses of it.

## Solution

### Part (i)

The total of $1, 2, \ldots, 8$ is 36 and the total of their squares is 204, so each half needs sum 18 and sum of squares 102.

Put 8 in $B$. The other three elements of $B$ must have sum 10 and squares summing to $102 - 64 = 38$. The triples of distinct numbers from 1 to 7 with sum 10 are

$$\{1,2,7\},\ \{1,3,6\},\ \{1,4,5\},\ \{2,3,5\},$$

with sums of squares 54, 46, 42 and 38. Only $\{2,3,5\}$ works, so

$$A = \{1,4,6,7\}, \qquad B = \{2,3,5,8\}.$$

Check: $1+4+6+7 = 18 = 2+3+5+8$ and $1+16+36+49 = 102 = 4+9+25+64$. (The search also shows this is the only balanced partition into fours, apart from swapping $A$ and $B$.)

### Part (ii)

Expand each cube with the binomial theorem:

$$\sum_{k=1}^{m} (c+b_k)^3 = mc^3 + 3c^2 \sum_{k=1}^{m} b_k + 3c \sum_{k=1}^{m} b_k^2 + \sum_{k=1}^{m} b_k^3,$$

and similarly with $a_k$ in place of $b_k$. Hence

$$\left(\sum a_k^3 + \sum (c+b_k)^3\right) - \left(\sum b_k^3 + \sum (c+a_k)^3\right)$$
$$= 3c^2\left(\sum b_k - \sum a_k\right) + 3c\left(\sum b_k^2 - \sum a_k^2\right),$$

because the $mc^3$ terms cancel, as do the two copies of $\sum a_k^3$ and of $\sum b_k^3$. Both brackets are zero by the hypotheses, so the two sides are equal for every real $c$, as required.

Notice that the same expansion one power lower gives

$$\sum a_k^2 + \sum (c+b_k)^2 - \sum b_k^2 - \sum (c+a_k)^2 = 2c\left(\sum b_k - \sum a_k\right) = 0,$$

and with first powers $\sum a_k + \sum (c+b_k) = \sum b_k + \sum (c+a_k)$, since both equal $mc + \sum a_k + \sum b_k$.

#### Part (iii)

Take the balanced partition from (i), with $a_k$ running through $1, 4, 6, 7$ and $b_k$ through $2, 3, 5, 8$, and take $c = 8$. Then $c + a_k$ runs through $9, 12, 14, 15$ and $c + b_k$ through $10, 11, 13, 16$. Set

$$A = \{1, 4, 6, 7\} \cup \{10, 11, 13, 16\}, \qquad B = \{2, 3, 5, 8\} \cup \{9, 12, 14, 15\}.$$

These are disjoint, each has 8 elements, and together they are $\{1, 2, \ldots, 16\}$, because $\{1, \ldots, 8\}$ and $\{9, \ldots, 16\}$ are each split between them.

> **Ask yourself** Why put the shifted copy of $B$ into $A$, rather than the shifted copy of $A$?

Because that is exactly the pattern in (ii): the left-hand side there is $\sum a_k^3 + \sum (c+b_k)^3$, which is the sum of cubes over our $A$, and the right-hand side is the sum of cubes over our $B$. So (ii) with $c = 8$ gives equal sums of cubes. The two extra identities at the end of (ii) give equal sums and equal sums of squares, so the partition is balanced too.

Directly: both halves have sum 68, sum of squares 748 and sum of cubes 9248.

#### Part (iv)

Write the elements of $S$ as $(n+k)^2$ for $k = 0, 1, \ldots, 11$. For any set $K$ of values of $k$,

$$\sum_{k \in K} (n+k)^2 = |K|\, n^2 + 2n \sum_{k \in K} k + \sum_{k \in K} k^2.$$

So if we split $\{0, 1, \ldots, 11\}$ into two sets that have the same *size*, the same sum and the same sum of squares, the corresponding squares will have equal totals for every $n$.

The given partition of $\{1, \ldots, 11\}$ has equal sums (33 each) and equal sums of squares (253 each), but its parts have sizes 6 and 5. Adding $k = 0$ to the smaller part fixes this: it increases the size by one and adds nothing to the sum or the sum of squares. So we use

$$\{1, 3, 4, 5, 9, 11\} \quad \text{and} \quad \{0, 2, 6, 7, 8, 10\},$$

which have size 6, sum 33 and sum of squares 253 each. Therefore

$$C = \{(n+1)^2, (n+3)^2, (n+4)^2, (n+5)^2, (n+9)^2, (n+11)^2\},$$
$$D = \{n^2, (n+2)^2, (n+6)^2, (n+7)^2, (n+8)^2, (n+10)^2\}$$

both have element sum $6n^2 + 66n + 253$. The twelve squares are distinct since $n > 0$, so $C$ and $D$ are disjoint and their union is $S$.

## Checks

- With $n = 1$ in (iv), $C = \{4, 16, 25, 36, 100, 144\}$ and $D = \{1, 9, 49, 64, 81, 121\}$, both summing to $325 = 6 + 66 + 253$.
- In (iii), the sums 68 and sums of squares 748 are half of 136 and 1496, the totals for 1 to 16; the sums of cubes are 9248, half of $136^2 = 18496$.
- A computer search over all ways of choosing four of $1, \ldots, 8$ confirms the partition in (i) is unique up to swapping, and the construction in (iv) was checked for $n$ up to 499.

## Common slips

- In (iv), using the given $A$ and $B$ directly: the sizes 6 and 5 differ, so the $n^2$ terms do not cancel. The missing element $n^2$ (that is, $k = 0$) must go with $B$.
- In (iii), shifting the wrong half: putting $\{9, 12, 14, 15\}$ (the shifted $a_k$) with $\{1, 4, 6, 7\}$ still balances sums and squares, but the cube sums then differ by $2\left(\sum b_k^3 - \sum a_k^3\right) = 2(672 - 624) = 96$.

## Beyond STEP

This is the Prouhet–Tarry–Escott problem: split a set of integers into two parts whose sums of $j$th powers agree for $j = 1, \ldots, k$. Repeating the shift-and-swap of (iii) with $c = 16, 32, \ldots$ produces the Thue–Morse partition of $\{0, 1, \ldots, 2^{k+1} - 1\}$, which balances all powers up to $k$; the proof is the binomial expansion of (ii) done by induction. Davenport's *The Higher Arithmetic* is a good companion for this kind of elementary number theory.

WORKED SOLUTION

# STEP II 2025, Question 9

Mechanics • Collisions and Projectiles

**STEP II 2025, Question 9**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

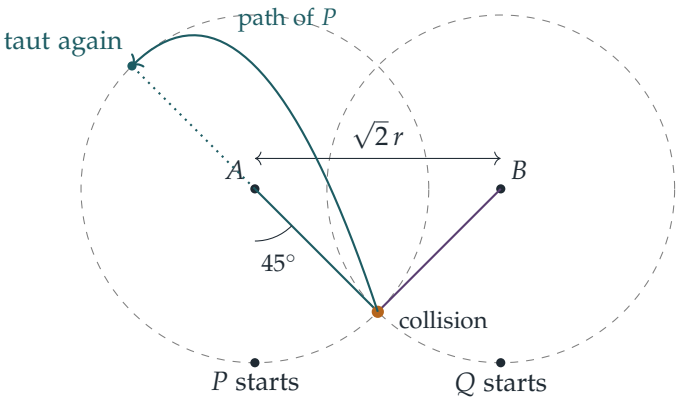
www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Symmetry does most of the work. The particles must meet on the perpendicular bisector of $AB$, which fixes where they are and how fast they move when they collide. Just after the collision each string goes slack, so each particle is an ordinary projectile until its distance from the point of suspension is back up to $r$. Writing that distance condition out gives a quartic in $t$. A triple root has a physical meaning: the particle reaches the circle travelling along it, so the string tightens without a jerk.

## Solution

Model: $P$ and $Q$ are particles of mass $m$ (small, smooth spheres for the collision), the strings are light and inextensible, and there is no air resistance. Put $P$ on the left, hanging from $A$, and take axes through $A$ with $x$ horizontal (towards $B$) and $y$ vertically up.




The figure shows $P$'s flight after the collision in the special case of part (iv), where it rejoins its circle at the point 45° above the horizontal on the far side.

**Part (i)**

The set-up is symmetric about the perpendicular bisector of $AB$, so the particles meet on it. Each has then moved horizontally $\frac{1}{2}\sqrt{2}\,r = r\sin 45^\circ$ from its lowest point, so each string makes $45^\circ$ with the downward vertical. Each particle has risen by $r - r\cos 45^\circ = r\left(1 - \frac{1}{\sqrt{2}}\right)$. The string is taut up to this point and its tension does no work, so by conservation of energy the speed $v$ just before impact satisfies

$$v^2 = u^2 - 2gr\left(1 - \tfrac{1}{\sqrt{2}}\right) = u^2 - gr(2 - \sqrt{2}).$$

(We need $u^2 > gr(2-\sqrt{2})$ for the particles to get this far.)

Each particle is moving along its circle, at $45^\circ$ to the horizontal, so

$$\text{velocity of } P = \left(\tfrac{v}{\sqrt{2}},\ \tfrac{v}{\sqrt{2}}\right), \qquad \text{velocity of } Q = \left(-\tfrac{v}{\sqrt{2}},\ \tfrac{v}{\sqrt{2}}\right).$$

The particles meet at the same height, so the line of centres is horizontal and the impulse between them is horizontal. Vertical components are therefore unchanged. Horizontally, the momenta are equal and opposite, so the total is zero both before and after. Newton's law of restitution then makes each particle rebound with $e$ times its approach speed. Immediately after the collision,

$$\text{horizontal speed} = \frac{ev}{\sqrt{2}} = \tfrac{1}{2}ev\sqrt{2}, \qquad \text{vertical component} = \tfrac{1}{2}\sqrt{2}\,v \text{ (upwards)},$$

with $P$ now moving away from $Q$, to the left.

**Ask yourself** Could the strings exert an impulse during the collision as well?

No. A string can only pull, and only along itself. Just after the collision, $P$ is at $\left(\frac{r}{\sqrt{2}}, -\frac{r}{\sqrt{2}}\right)$ relative to $A$ with velocity $\left(-\frac{ev}{\sqrt{2}}, \frac{v}{\sqrt{2}}\right)$. Its component along $\overrightarrow{AP}$ is $-\frac{1}{2}(1+e)v < 0$, so $P$ is moving towards $A$. The string goes slack straight away, and an impulsive tension is never needed.

**Part (ii)**

Measure $t$ from the collision. Until its string is taut again, $P$ is a projectile, so relative to $A$

$$x = \frac{r}{\sqrt{2}} - \frac{ev}{\sqrt{2}}\,t, \qquad y = -\frac{r}{\sqrt{2}} + \frac{v}{\sqrt{2}}\,t - \tfrac{1}{2}gt^2.$$

The string is taut exactly when $x^2 + y^2 = r^2$. Multiplying by 2, so that $\sqrt{2}\,x$ and $\sqrt{2}\,y$ appear, gives

$$(r - evt)^2 + \left(-r + vt - \tfrac{1}{2}\sqrt{2}\,gt^2\right)^2 = 2r^2.$$

This holds at $t = 0$, when the string is still taut at the instant of collision, so the time we want is a non-zero root. By symmetry $Q$ does the same at the same moment. The two particles move apart horizontally, so they cannot collide again first.

**Part (iii)**

Expanding and cancelling the $2r^2$:

$$\tfrac{1}{2}g^2t^4 - \sqrt{2}\,gv\,t^3 + \big((1+e^2)v^2 + \sqrt{2}\,rg\big)t^2 - 2(1+e)rv\,t = 0.$$

Dividing by $t \neq 0$ leaves a cubic. Now

$$\frac{z}{c} = \frac{vt}{r}\cdot\frac{rg}{\sqrt{2}\,v^2} = \frac{gt}{\sqrt{2}\,v}, \qquad \text{so} \qquad t = \frac{\sqrt{2}\,v}{g}\cdot\frac{z}{c}.$$

Substitute this, and divide the whole cubic by $\sqrt{2}\,v^3/g$. Writing $w = z/c$, the terms become

$$\begin{aligned}
\tfrac{1}{2}g^2t^3 &\longrightarrow w^3,\\
-\sqrt{2}\,gv\,t^2 &\longrightarrow -2w^2,\\
\big((1+e^2)v^2 + \sqrt{2}\,rg\big)t &\longrightarrow \Big(1+e^2+\frac{\sqrt{2}\,rg}{v^2}\Big)w = \Big(\frac{2}{c}+1+e^2\Big)w,\\
-2(1+e)rv &\longrightarrow -\frac{\sqrt{2}\,rg}{v^2}(1+e) = -\frac{2}{c}(1+e),
\end{aligned}$$

using $\dfrac{2}{c} = \dfrac{2rg}{\sqrt{2}\,v^2} = \dfrac{\sqrt{2}\,rg}{v^2}$. This is the required equation

$$\Big(\frac{z}{c}\Big)^3 - 2\Big(\frac{z}{c}\Big)^2 + \Big(\frac{2}{c}+1+e^2\Big)\Big(\frac{z}{c}\Big) - \frac{2}{c}(1+e) = 0.$$

**Part (iv)**

If the cubic in $w$ has a triple root $\alpha$, it must be $(w-\alpha)^3 = w^3 - 3\alpha w^2 + 3\alpha^2 w - \alpha^3$. Comparing coefficients,

$$3\alpha = 2, \qquad 3\alpha^2 = \frac{2}{c}+1+e^2, \qquad \alpha^3 = \frac{2}{c}(1+e).$$

So $\alpha = \frac{2}{3}$. Write $k = 2/c$. Then $k = \frac{4}{3} - 1 - e^2 = \frac{1}{3} - e^2$, and the last equation becomes

$$\Big(\tfrac{1}{3} - e^2\Big)(1+e) = \tfrac{8}{27} \iff 27e^3 + 27e^2 - 9e - 1 = 0 \iff (3e-1)(9e^2+12e+1) = 0.$$

The quadratic factor has roots $\frac{-2\pm\sqrt{3}}{3}$, both negative, so they cannot be coefficients of restitution. Hence $e = \frac{1}{3}$. Then $k = \frac{1}{3} - \frac{1}{9} = \frac{2}{9}$, so $c = 9$, and

$$v^2 = \frac{c\,rg}{\sqrt{2}} = \frac{9rg}{\sqrt{2}} = \tfrac{9}{2}\sqrt{2}\,rg,$$

as required.

? **Ask yourself** What does a repeated root say about the motion?

Let $D(t) = x^2 + y^2 - r^2$. Up to a positive factor, $D$ equals $w(w-\alpha)^3$. This is negative for $0 < w < \alpha$, so $P$ stays strictly inside its circle until the string tightens. Because $\alpha$ is a repeated root, $D' = 0$ there too. But $D' = 2(x\dot{x} + y\dot{y})$, so the component of velocity along the string is zero at that moment: the particle arrives moving along the circle. An impulsive tension would act along the string and would only be needed to remove a component of velocity along it. Here there is no such component, so there is no impulse, the velocity does not change, and no kinetic energy is lost.

#### Part (v)

Energy is lost only in the collision. Between the collision and the bottom of the circle, only gravity does work: during flight there is no tension, the tightening is jerk-free, and afterwards the tension is perpendicular to the motion. Just after the collision, the speed squared is

$$\frac{e^2v^2}{2} + \frac{v^2}{2} = \frac{(1+e^2)v^2}{2},$$

and the particle is $r\left(1 - \frac{1}{\sqrt{2}}\right)$ above its lowest point. Therefore

$$\begin{aligned} U^2 &= \frac{(1+e^2)v^2}{2} + gr\left(2-\sqrt{2}\right) = \frac{1}{2}\cdot\frac{10}{9}\cdot\frac{9rg}{\sqrt{2}} + gr\left(2-\sqrt{2}\right) \\ &= \frac{5\sqrt{2}}{2}rg + (2-\sqrt{2})rg = \left(2+\tfrac{3}{2}\sqrt{2}\right)rg = \frac{4+3\sqrt{2}}{2}rg. \end{aligned}$$

We should also check that the string stays taut on the way down. With $w = \frac{2}{3}$, $t = \frac{2\sqrt{2}v}{3g}$. Substituting gives position $\left(-\frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}}\right)$ and velocity $\left(-\frac{v}{3\sqrt{2}}, -\frac{v}{3\sqrt{2}}\right)$. So $P$ rejoins its circle 45° above the horizontal, on the side away from $B$, with speed squared $\frac{v^2}{9} = \frac{rg}{\sqrt{2}}$. Resolving towards $A$ there,

$$T + mg\sin 45^\circ = \frac{m}{r}\cdot\frac{rg}{\sqrt{2}} \quad\Longrightarrow\quad T = 0,$$

so the string becomes taut with zero tension. As $P$ swings down, its speed increases and the inward component of gravity decreases, so $T > 0$ from then on. The particle follows the circle down to its lowest point, and the same value of $U^2$ comes from energy measured from this point: $\frac{rg}{\sqrt{2}} + 2gr\left(1+\frac{1}{\sqrt{2}}\right) = \left(2+\frac{3}{2}\sqrt{2}\right)rg$.

## Checks

- With $r = 1$ and $g = 9.81$, a step-by-step computer simulation of the flight and the circular swing gave $U^2 \approx 40.430$. This agrees with $\frac{1}{2}(4+3\sqrt{2})g \approx 40.430$, and the tension stayed non-negative throughout the swing.
- The two routes to $U^2$ (from the collision point, and from the point where the string becomes taut) agree.
- If $e = 1/3$ and $c = 9$, then $w^3 - 2w^2 + \frac{4}{3}w - \frac{8}{27} = \left(w - \frac{2}{3}\right)^3$ exactly.

## Common slips

- Taking the height risen as $r\cos 45^\circ$ (the depth below the point of suspension) rather than $r - r\cos 45^\circ$.
- Applying restitution to the whole velocity. Only the component along the line of centres (here horizontal) is affected.
- Forgetting to rule out the negative roots of the cubic in $e$.

## Beyond STEP

The repeated-root condition says that two curves (the projectile's parabola and the circle) touch to high order: they meet, share a tangent, and here even share curvature, which is why the tension is exactly zero when the string becomes taut. Order of contact and osculating circles appear in first-year courses on curves and differential geometry. Impulsive constraints, and when they dissipate energy, are a standard topic in dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see these ideas developed.

WORKED SOLUTION

# STEP II 2025, Question 10

Mechanics • Statics of a Rigid Rod

**STEP II 2025, Question 10**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

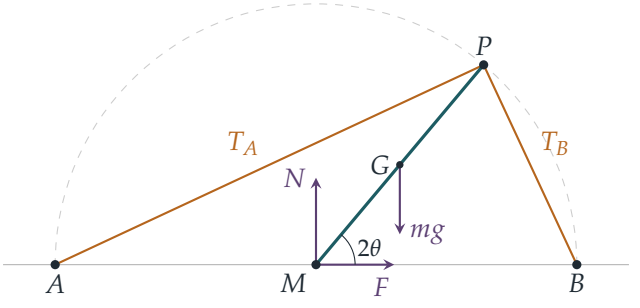
## The key idea

Because $MA = MB = MP = a$, the point $P$ lies on the circle with diameter $AB$, and that one observation drives everything: it gives the right angle at $P$, the string lengths in terms of $\theta$, and the angles each string makes with the rod. With Hooke's law the tension is $\lambda(\text{length} - \ell)/\ell$, which splits naturally into a "$\lambda \times \text{length}/\ell$" part and a constant "$-\lambda$" part; that split is exactly the decomposition asked for in (i). After that, moments about well-chosen points remove the forces we don't want.

## Solution

Model the rod as a uniform rigid rod with its weight $mg$ acting at its midpoint $G$, and the strings as light elastic strings obeying Hooke's law. Put $M$ at the origin with $B = (a, 0)$, $A = (-a, 0)$, so that

$$P = (a\cos 2\theta,\ a\sin 2\theta).$$




(The direction of the friction $F$ in the diagram is justified in part (iii).)

**Part (i)**

Since $MA = MB = MP = a$, the point $P$ lies on the circle with centre $M$ and diameter $AB$. By the angle-in-a-semicircle theorem, $\angle APB = 90°$.

**String lengths.** Triangle $MPA$ is isosceles with $MP = MA$ and apex angle $\angle PMA = 180° - 2\theta$, so its base angles are $\angle MPA = \angle MAP = \theta$. Similarly triangle $MPB$ has apex angle $2\theta$, so $\angle MPB = 90° - \theta$. In the right-angled triangle $APB$ with hypotenuse $AB = 2a$ and $\angle PAB = \theta$,

$$PA = 2a\cos\theta, \qquad PB = 2a\sin\theta.$$

Since $\theta < 45°$, $PB < PA$; both strings are in tension, so $2a\sin\theta > \ell$.

**Splitting the force.** Let $\mathbf{u}_A$ and $\mathbf{u}_B$ be the unit vectors from $P$ towards $A$ and $B$. By Hooke's law the tension in string $PA$ is $\lambda(PA - \ell)/\ell$, so the force it exerts on the rod at $P$ is

$$\frac{\lambda(PA-\ell)}{\ell}\,\mathbf{u}_A = \frac{\lambda}{\ell}\,PA\,\mathbf{u}_A - \lambda\mathbf{u}_A = \frac{\lambda}{\ell}\,\overrightarrow{PA} - \lambda\mathbf{u}_A,$$

and likewise for $PB$. Adding the two,

$$\text{total string force} = \frac{\lambda}{\ell}\left(\overrightarrow{PA} + \overrightarrow{PB}\right) - \lambda(\mathbf{u}_A + \mathbf{u}_B).$$

Now $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PM}$ because $M$ is the midpoint of $AB$, and $|\overrightarrow{PM}| = a$. So the first term is a force of magnitude $2a\lambda/\ell$ along the rod (towards $M$).

For the second term, $\mathbf{u}_A$ and $\mathbf{u}_B$ are perpendicular unit vectors (the angle at $P$ is a right angle), so $\mathbf{u}_A + \mathbf{u}_B$ is the diagonal of a unit square: it has length $\sqrt{2}$ and points along the bisector of $\angle APB$. Hence $-\lambda(\mathbf{u}_A + \mathbf{u}_B)$ is a force of magnitude $\sqrt{2}\,\lambda$ along the bisector of $\angle APB$ (pointing away from $AB$). This is the required decomposition.

**Part (ii)**

Take moments about $M$; the friction and normal reaction there have no moment. Measure anticlockwise as positive (so $B$ is to the right).

**Ask yourself** How far is $M$ from the line of each string?

The line $PA$ makes angle $\angle MPA = \theta$ with $PM$, and $PM = a$, so the perpendicular distance from $M$ to line $PA$ is $a\sin\theta$. Likewise the distance from $M$ to line $PB$ is $a\sin(90° - \theta) = a\cos\theta$. The weight acts at horizontal distance $\frac{1}{2}a\cos 2\theta$ to the right of $M$.

The string $PA$ pulls $P$ towards $A$ (anticlockwise about $M$); $PB$ and the weight turn the rod clockwise. Writing $T_A$, $T_B$ for the tensions,

$$T_A\,a\sin\theta \;=\; T_B\,a\cos\theta + mg\cdot\tfrac{1}{2}a\cos 2\theta.$$

With $T_A = \lambda(2a\cos\theta - \ell)/\ell$ and $T_B = \lambda(2a\sin\theta - \ell)/\ell$,

$$\begin{aligned} T_A\sin\theta - T_B\cos\theta &= \frac{\lambda}{\ell}\left(2a\cos\theta\sin\theta - \ell\sin\theta - 2a\sin\theta\cos\theta + \ell\cos\theta\right) \\ &= \lambda(\cos\theta - \sin\theta). \end{aligned}$$

(This is just the moment of the $\sqrt{2}\,\lambda$ bisector force, since the force along the rod has no moment about $M$.) So the moment equation becomes

$$\lambda(\cos\theta - \sin\theta) = \tfrac{1}{2}mg(\cos^2\theta - \sin^2\theta) = \tfrac{1}{2}mg(\cos\theta - \sin\theta)(\cos\theta + \sin\theta).$$

Since $0 < \theta < 45°$, $\cos\theta - \sin\theta > 0$, so we may divide by it:

$$\cos\theta + \sin\theta = \frac{2\lambda}{mg}.$$

**The range of $\lambda$.** Write $\cos\theta + \sin\theta = \sqrt{2}\sin(\theta + 45°)$. The rod is inclined to the ground, so $0 < \theta < 45°$, giving $45° < \theta + 45° < 90°$, where sin is strictly increasing from $\frac{1}{\sqrt{2}}$ to 1. Hence

$$1 < \cos\theta + \sin\theta < \sqrt{2} \quad\Longrightarrow\quad \tfrac{1}{2}mg < \lambda < \tfrac{1}{2}\sqrt{2}\,mg,$$

which is therefore a necessary condition for such an equilibrium.

#### Part (iii)

**Ask yourself** Which point makes both string forces disappear from a moment equation?

Both strings act at $P$, so take moments about $P$: then only the weight and the two ground forces at $M$ appear. Relative to $P$, the point $M$ is at horizontal distance $a\cos 2\theta$ (to the left) and vertical distance $a\sin 2\theta$ (below); $G$ is at half those distances.

Suppose the friction $F$ acts towards $B$ (we confirm this below). About $P$:

- $N$ (upwards at $M$, to the left of $P$) has clockwise moment $N\,a\cos 2\theta$;
- $F$ (towards $B$ at $M$, below $P$) has anticlockwise moment $F\,a\sin 2\theta$;
- $mg$ (downwards at $G$, to the left of $P$) has anticlockwise moment $mg\cdot\frac{1}{2}a\cos 2\theta$.

For equilibrium,

$$Na\cos 2\theta = Fa\sin 2\theta + \tfrac{1}{2}mg\,a\cos 2\theta.$$

Since $2\theta < 90°$, $\cos 2\theta > 0$, and dividing by $a\cos 2\theta$ gives

$$N - F\tan 2\theta = \tfrac{1}{2}mg,$$

as required.

**Direction of friction.** Resolving horizontally, $F$ balances the horizontal part of the string force. From (i), $\mathbf{u}_A = (-\cos\theta, -\sin\theta)$ and $\mathbf{u}_B = (\sin\theta, -\cos\theta)$,

and $\overrightarrow{PM} = (-a\cos 2\theta, -a\sin 2\theta)$, so the string force has horizontal component $-\frac{2a\lambda}{\ell}\cos 2\theta + \lambda(\cos\theta - \sin\theta)$. Hence the friction towards $B$ is

$$F = \frac{2a\lambda}{\ell}\cos 2\theta - \lambda(\cos\theta - \sin\theta) = \lambda(\cos\theta - \sin\theta)\left(\frac{2a(\cos\theta + \sin\theta)}{\ell} - 1\right),$$

which is positive because $2a\sin\theta > \ell$ (both strings taut). So friction does act towards $B$, and $F$ is its magnitude.

## Checks

- The limits in (ii) make sense: as $\theta \to 0$ the rod lies nearly flat and $\lambda \to \frac{1}{2}mg$; as $\theta \to 45^\circ$ the rod becomes vertical and $\lambda \to mg/\sqrt{2}$.
- Numerically, with $a = 1$, $mg = 10$, $\theta = 20^\circ$, $\ell = 0.5$: then $\lambda = 5(\cos 20^\circ + \sin 20^\circ) \approx 6.408$, the tensions are $T_A \approx 17.68$ and $T_B \approx 2.36$, and resolving gives $N \approx 18.26$, $F \approx 15.81$, so $N - F\tan 40^\circ \approx 5.00 = \frac{1}{2}mg$. Moments about $M$ also balance. A computer check over thousands of random configurations confirmed every result to rounding error.

## Common slips

- Taking the tension as $\lambda \times$ extension rather than $\lambda \times$ extension$/\ell$.
- Dividing by $\cos\theta - \sin\theta$ or $\cos 2\theta$ without noting that $\theta < 45^\circ$ makes them positive.
- Taking moments about $M$ in (iii): it removes $N$ and $F$, which are exactly the forces we want. Moments about $P$ remove the strings instead.

## Beyond STEP

The split in (i) is a neat instance of a general fact: a set of forces acting on a rigid body is equivalent to a single force plus a couple, and you are free to group forces however makes the moments easiest. The "$-\lambda$ per string" piece here behaves like a pair of constant-tension strings, whose resultant is fixed by geometry alone. Statics of rigid bodies, virtual work and stability of equilibria are developed in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

# STEP II 2025, Question 11

Probability and Statistics • Expectation and Binomial Sums

**STEP II 2025, Question 11**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Both games end at the first $2k$, so each is a sum over the number of rolls $r$. For Ali the sum is geometric-like, and part (i) is exactly the tool for the expected number of rolls. For Zen, a winning game is fixed by *which* numbers appear before the $2k$, because once you know the set, the increasing order is forced. Counting these sets brings in binomial coefficients, so the sum is the one from part (iii).

## Solution

### Part (i)

For $|x| < 1$ the geometric series gives

$$\sum_{r=0}^{\infty} x^r = \frac{1}{1-x}.$$

A power series can be differentiated term by term inside its interval of convergence, so differentiating both sides with respect to $x$ gives

$$\sum_{r=1}^{\infty} r x^{r-1} = \frac{1}{(1-x)^2}, \qquad |x| < 1,$$

as required.

### Part (ii)

On any one roll there are three kinds of outcome:

$$\mathrm{P}(\text{a } 2k) = \frac{1}{2k}, \qquad \mathrm{P}(\text{even, not } 2k) = \frac{k-1}{2k}, \qquad \mathrm{P}(\text{odd}) = \frac{1}{2}.$$

Let $R$ be the number of rolls. Ali wins with $R = r$ exactly when rolls $1, \ldots, r-1$ are even numbers other than $2k$ and roll $r$ is $2k$. The rolls are independent, so with $x = \dfrac{k-1}{2k}$,

$$\mathrm{P}(\text{Ali wins and } R = r) = \frac{1}{2k}\, x^{r-1}.$$

**(a)** Summing the geometric series (note $0 \le x < 1$):

$$\mathrm{P}(\text{Ali wins}) = \frac{1}{2k} \cdot \frac{1}{1-x} = \frac{1}{2k} \cdot \frac{2k}{k+1} = \frac{1}{k+1}.$$

There is a quick way to see this. Ignore the even numbers other than $2k$, since they don't affect the result. The game is decided by whichever comes first, an odd number or the $2k$. There are $k$ odd faces and one face showing $2k$, all equally likely, so the chance that the $2k$ comes first is $\frac{1}{k+1}$.

**(b)** Ali earns £$r$ if he wins with $R = r$ and nothing otherwise, so by part (i)

$$\mathrm{E}(\text{Ali's earnings}) = \sum_{r=1}^{\infty} r \cdot \frac{1}{2k}\, x^{r-1} = \frac{1}{2k} \cdot \frac{1}{(1-x)^2} = \frac{1}{2k}\left(\frac{2k}{k+1}\right)^2 = \frac{2k}{(k+1)^2} \text{ pounds.}$$

#### Part (iii)

Call the expression $S$. Its general term is $(r+1)\binom{n}{r}x^r$ for $r = 0, 1, \ldots, n$. The factor $r+1$ suggests differentiating $x^{r+1}$:

$$S = \sum_{r=0}^{n} (r+1)\binom{n}{r} x^r = \frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_{r=0}^{n} \binom{n}{r} x^{r+1}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\Big(x(1+x)^n\Big),$$

using the binomial theorem. By the product rule,

$$S = (1+x)^n + nx(1+x)^{n-1} = (1+x)^{n-1}\big(1 + (n+1)x\big).$$

#### Part (iv)

Suppose Zen wins with exactly $r$ rolls. The last roll is $2k$, and the first $r-1$ rolls are strictly increasing numbers from $\{1, 2, \ldots, 2k-1\}$. (They must all be less than $2k$ anyway, since the game stops at the first $2k$.)

> **Ask yourself** How many winning sequences of length $r$ are there?

A strictly increasing sequence is fixed once we know which numbers appear in it, because there is only one way to put a set of distinct numbers in increasing order. So the winning sequences of length $r$ correspond to the $(r-1)$-element subsets of $\{1, \ldots, 2k-1\}$. There are $\binom{2k-1}{r-1}$ of them, and $1 \le r \le 2k$.

Each particular sequence of $r$ rolls has probability $\left(\frac{1}{2k}\right)^r$. Hence

$$\mathrm{E}(\text{Zen's earnings}) = \sum_{r=1}^{2k} r\binom{2k-1}{r-1}\left(\frac{1}{2k}\right)^r = \frac{1}{2k}\sum_{j=0}^{2k-1}(j+1)\binom{2k-1}{j}\left(\frac{1}{2k}\right)^j,$$

where $j = r - 1$. This is the sum in part (iii) with $n = 2k - 1$ and $x = \frac{1}{2k}$, and then $1 + (n+1)x = 1 + 2k \cdot \frac{1}{2k} = 2$. Therefore

$$\text{E(Zen's earnings)} = \frac{1}{2k}\left(1 + \frac{1}{2k}\right)^{2k-2} \cdot 2 = \frac{1}{k}\left(1 + \frac{1}{2k}\right)^{2k-2} \text{ pounds.}$$

**Part (v)**

Divide Zen's expected earnings by Ali's:

$$\rho = \frac{1}{k}\left(1 + \frac{1}{2k}\right)^{2k-2} \cdot \frac{(k+1)^2}{2k} = \frac{(k+1)^2}{2k^2} \cdot \frac{\left(1 + \frac{1}{2k}\right)^{2k}}{\left(1 + \frac{1}{2k}\right)^2}.$$

Now let $k$ be large. Then

- $\dfrac{(k+1)^2}{k^2} = \left(1 + \dfrac{1}{k}\right)^2 \approx 1$;
- $\left(1 + \dfrac{1}{2k}\right)^2 \approx 1$;
- $\left(1 + \dfrac{1}{2k}\right)^{2k} \approx \mathrm{e}$, by the given approximation with $n = 2k$.

So

$$\rho \approx \frac{\mathrm{e}}{2} \approx \frac{2.71828}{2} = 1.359\ldots,$$

and for large $k$ Zen's expected earnings are about 1.359 times Ali's, which is a little over 35% more, as required.

## Checks

- For $k = 1$ the die is a coin with faces $1, 2$. Ali wins only if the first roll is 2, which has probability $\frac{1}{2} = \frac{1}{k+1}$, and then earns £1, giving expected earnings $\frac{1}{2} = \frac{2k}{(k+1)^2}$. Zen wins with the sequences 2 (probability $\frac{1}{2}$, earning £1) and $1, 2$ (probability $\frac{1}{4}$, earning £2), so her expected earnings are $\frac{1}{2} + \frac{1}{2} = 1$. The formula agrees: with $k = 1$ the exponent $2k - 2$ is 0, so it gives $\frac{1}{1} \cdot 1 = 1$.
- For $k = 3$, Ali's expected earnings are $\frac{3}{8}$ and Zen's are $\frac{1}{3}\left(\frac{7}{6}\right)^4 = \frac{2401}{3888} \approx 0.618$. Both agree with a direct enumeration of all winning sequences and with a computer simulation of the games.
- The ratio $\rho$ is about 1.456 at $k = 10$, 1.369 at $k = 100$ and 1.3592 at $k = 10^4$. It is decreasing towards $\mathrm{e}/2 = 1.35914\ldots$.

## Common slips

- In part (iv), counting ordered sequences with a factorial instead of subsets. There is only one increasing arrangement of each set.
- Forgetting that the earlier rolls in Zen's game can't include $2k$. This makes the range $\{1, \ldots, 2k-1\}$, so $n = 2k-1$, not $2k$.
- In part (ii), using $\frac{k}{2k}$ instead of $\frac{k-1}{2k}$ as the probability of continuing. The roll $2k$ is even, but it ends the game.

## Beyond STEP

Part (iii) is a small example of *generating functions*: you encode a sequence as the coefficients of a power series and then use calculus on the series. Probability generating functions take this further. If $G(s) = \mathrm{E}(s^X)$, then $\mathrm{E}(X) = G'(1)$, and that is exactly the differentiation trick used twice here. The limit $\left(1 + \frac{1}{n}\right)^n \to \mathrm{e}$ turns up all over probability, for instance in the Poisson approximation to the binomial. Grimmett and Welsh, *Probability: An Introduction*, has a good chapter on generating functions.

WORKED SOLUTION

# STEP II 2025, Question 12

Probability and Statistics • Poisson Distribution

**STEP II 2025, Question 12**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

For a Poisson distribution, consecutive probabilities have the simple ratio $p_r/p_{r-1} = \lambda/r$, so every difference $d_r$ can be written as a known probability times a simple rational factor in $r$. The sign of $d_r$ tells us where $p_r$ rises and falls; going one level further, the sign of the *second* difference $d_{r+1} - d_r$ tells us where $d_r$ rises and falls, and that sign is controlled by a quadratic in $r$. The awkward-looking number $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}}$ is just (one more than) a root of that quadratic.

## Solution

Throughout, $p_r = \mathrm{e}^{-\lambda}\lambda^r/r!$, so for $r \geq 1$

$$p_r = \frac{\lambda}{r}\,p_{r-1}, \qquad p_{r-1} = \frac{r}{\lambda}\,p_r,$$

and every $p_r$ is strictly positive.

### Part (i)

For $r \geq 1$,

$$d_r = p_r - p_{r-1} = p_{r-1}\left(\frac{\lambda}{r} - 1\right) = \frac{p_{r-1}}{r}\,(\lambda - r).$$

Since $p_{r-1}/r > 0$, the sign of $d_r$ is the sign of $\lambda - r$, and $d_r$ is never zero because $\lambda$ is not an integer. Let $m = \lfloor \lambda \rfloor$, the largest integer below $\lambda$, so that $\lambda - 1 < m < \lambda$ (both inequalities strict because $\lambda \notin \mathbb{Z}$). Then

- for $1 \leq r \leq m$ we have $r < \lambda$, so $d_r > 0$ and $p_0 < p_1 < \cdots < p_m$;
- for $r \geq m+1$ we have $r > \lambda$, so $d_r < 0$ and $p_m > p_{m+1} > p_{m+2} > \cdots$.

(If $\lambda < 1$ then $m = 0$ and the first list is empty.) Hence $p_r < p_m$ for every $r \neq m$: $m$ is the unique most likely value, and $\lambda - 1 < m < \lambda$.

**Part (ii)**

Now look at how $d_r$ itself changes. For $r \geq 1$, put $e_r = d_{r+1} - d_r = p_{r+1} - 2p_r + p_{r-1}$. Writing everything in terms of $p_r$,

$$e_r = p_r\left(\frac{\lambda}{r+1} - 2 + \frac{r}{\lambda}\right) = \frac{p_r}{\lambda(r+1)}\Big(\lambda^2 - 2\lambda(r+1) + r(r+1)\Big).$$

The prefactor is positive, so the sign of $e_r$ is the sign of

$$Q(r) = r^2 + (1-2\lambda)r + \lambda^2 - 2\lambda.$$

This quadratic has discriminant $(1-2\lambda)^2 - 4(\lambda^2 - 2\lambda) = 1 + 4\lambda$, so its roots are

$$\alpha = \lambda - \tfrac{1}{2} - \sqrt{\lambda + \tfrac{1}{4}}, \qquad \beta = \lambda - \tfrac{1}{2} + \sqrt{\lambda + \tfrac{1}{4}},$$

and $Q(r) < 0$ exactly when $\alpha < r < \beta$. Thus

$$d_{r+1} < d_r \iff \alpha < r < \beta, \qquad d_{r+1} > d_r \iff r < \alpha \text{ or } r > \beta.$$

Note that $\beta + 1 = \lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}}$, which is not an integer, so $\beta$ is not an integer, and $k = \lfloor \beta + 1 \rfloor = \lfloor \beta \rfloor + 1$ is exactly the integer with

$$k < \lambda + \tfrac{1}{2} + \sqrt{\lambda + \tfrac{1}{4}} < k + 1.$$

**Ask yourself** Where should we look for the minimum of $d_r$ at all, given that $d_r$ also increases when $r < \alpha$?

From part (i), $d_r > 0$ for $r < \lambda$ and $d_r < 0$ for $r > \lambda$, so the minimum must be among the negative terms, $r \geq m+1$. Also $\beta > \lambda$ (since $\sqrt{\lambda + \frac{1}{4}} > \frac{1}{2}$) and $\alpha < \lambda$, so every $r \geq m+1$ satisfies $r > \alpha$, and $k > \beta > \lambda$ gives $k \geq m+1$. On the integers $r \geq m+1$ we therefore have:

- for $m+1 \leq r \leq k-1$: $r < \beta$, so $e_r < 0$, giving $d_{m+1} > d_{m+2} > \cdots > d_k$;
- for $r \geq k$: $r > \beta$, so $e_r > 0$, giving $d_k < d_{k+1} < d_{k+2} < \cdots$.

(The first list is empty when $k = m+1$.) So $d_k$ is strictly smaller than every other negative $d_r$, and smaller than all the positive ones too. The minimum of $d_r$ occurs at $r = k$, as required. In words: $p_r$ falls most steeply between $r = k-1$ and $r = k$.

**Part (iii)**

First, if $\lambda < 1$ then every $d_r$ is negative by part (i). But $d_r \to 0$ as $r \to \infty$ (because $p_r \to 0$), so for any $r$ there is a later term larger than $d_r$: no maximum is attained at all, and certainly not at $r = 1$. Hence we need $\lambda > 1$ ($\lambda = 1$ is excluded since $\lambda$ is not an integer).

Now take $\lambda > 1$, so $d_1 > 0$. The positive terms are $d_1, \ldots, d_m$, and any maximum lies among them.

- If the maximum is at $r = 1$, then $d_2 < d_1$, i.e. $e_1 < 0$.
- Conversely, if $e_1 < 0$ then $\alpha < 1 < \beta$, so every integer $r$ with $1 \leq r < \beta$ has $e_r < 0$. Since $m < \lambda < \beta$, this gives $d_1 > d_2 > \cdots > d_m$, and all later terms are negative. So the maximum is at $r = 1$.

So, for $\lambda > 1$, the maximum is at $r = 1$ exactly when $e_1 < 0$, i.e. when $Q(1) < 0$:

$$Q(1) = 1 + 1 - 2\lambda + \lambda^2 - 2\lambda = \lambda^2 - 4\lambda + 2 < 0 \iff 2 - \sqrt{2} < \lambda < 2 + \sqrt{2}.$$

Combining with $\lambda > 1$ (and $2 - \sqrt{2} < 1$), the condition is

$$1 < \lambda < 2 + \sqrt{2}.$$

**Part (iv)**

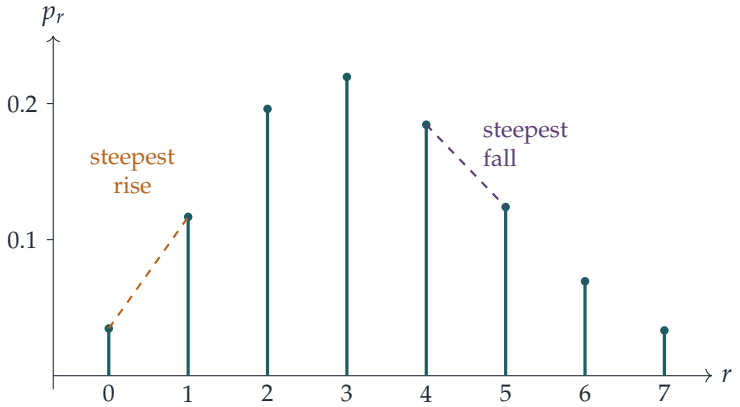
With $\lambda = 3.36$ we have $\lambda + \frac{1}{4} = 3.61 = 1.9^2$, so:

- the mode is $m = 3$, by part (i);
- $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} = 3.86 + 1.9 = 5.76$, so $k = 5$: the steepest fall is from $p_4$ to $p_5$;
- $1 < 3.36 < 2 + \sqrt{2} \approx 3.41$, so the steepest rise is from $p_0$ to $p_1$ (only just: $\alpha = 0.96$ is close to 1, so $d_1$ and $d_2$ are nearly equal).

Using $p_r = \frac{3.36}{r} p_{r-1}$ and $p_0 = \mathrm{e}^{-3.36} \approx 0.035$ gives the approximate values

$$p_0, \ldots, p_7 \approx 0.035,\ 0.117,\ 0.196,\ 0.220,\ 0.185,\ 0.124,\ 0.069,\ 0.033.$$

The sketch should show a rise that is steep from the start and then flattens to a peak at $r = 3$, followed by a fall that gets steeper, is steepest between $r = 4$ and $r = 5$, and then eases off.




## Checks

- Computing $p_r$ and $d_r$ for 20,000 random values of $\lambda \in (0, 40)$ confirms that the mode is $\lfloor \lambda \rfloor$, that the minimum of $d_r$ is at $\lfloor \lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} \rfloor$, and that (for $\lambda > 1$)

the maximum of $d_r$ is at $r = 1$ exactly when $\lambda < 2 + \sqrt{2} \approx 3.414$.

- For $\lambda = 3.36$: $d_1 \approx 0.0820$ and $d_2 \approx 0.0794$, so $d_1$ is (narrowly) the largest; $d_4, d_5, d_6 \approx -0.035, -0.061, -0.055$, so $d_5$ is the smallest, as predicted.
- Small $\lambda$: as $\lambda \to 0$, $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} \to 1$, so $k = 1$. Indeed then nearly all the probability is at 0, and the biggest drop is the first one, $p_1 - p_0$.

## Common slips

- Forgetting that $d_r$ also *increases* for $r < \alpha$, and so not explaining why that region cannot contain the minimum (it is where $d_r > 0$).
- In part (iii), checking only $d_2 < d_1$ and missing the case $\lambda < 1$, where every $d_r$ is negative and the sequence never attains a maximum.
- Writing $\lambda - 1 \leq m \leq \lambda$: the inequalities are strict precisely because $\lambda$ is not an integer.

## Beyond STEP

The differences $d_r$ and $e_r$ are discrete analogues of the first and second derivatives, and $k$ marks a discrete "point of inflection" of the Poisson distribution. For large $\lambda$ the Poisson distribution is close to a normal distribution with mean $\lambda$ and standard deviation $\sqrt{\lambda}$, whose inflection points are at $\lambda \pm \sqrt{\lambda}$; compare $\alpha, \beta \approx \lambda \pm \sqrt{\lambda}$ here. The ratio trick $p_r / p_{r-1}$ is the standard way to find modes of discrete distributions, and it is explored further in Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

# STEP III 2025, Question 1

Pure • Beta Function Integrals

**STEP III 2025, Question 1**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Everything here is a substitution or the identity $x + (1 - x) = 1$. Reflecting $x \mapsto 1 - x$ swaps the roles of $p$ and $q$; splitting the integrand with $x + (1 - x)$ relates neighbouring values; $x = \sin^2\theta$ turns the integral into a trigonometric one; and $x = t/(1+t)$ stretches $[0, 1)$ onto $[0, \infty)$. The final integral is then a chain of the earlier results, ending at an integral of $\sin^4\theta$ that we can do by hand.

## Solution

### Part (i)

Substitute $u = 1 - x$, so $\mathrm{d}x = -\mathrm{d}u$, and $x = 0, 1$ correspond to $u = 1, 0$:

$$\mathrm{b}(p, q) = \int_1^0 (1-u)^{p-1} u^{q-1} \, (-\mathrm{d}u) = \int_0^1 u^{q-1}(1-u)^{p-1} \, \mathrm{d}u = \mathrm{b}(q, p).$$

### Part (ii)

Adding the two integrals on the right-hand side,

$$\begin{aligned} \mathrm{b}(p+1, q) + \mathrm{b}(p, q+1) &= \int_0^1 x^{p-1}(1-x)^{q-1}\big[x + (1-x)\big] \, \mathrm{d}x \\ &= \int_0^1 x^{p-1}(1-x)^{q-1} \, \mathrm{d}x = \mathrm{b}(p, q), \end{aligned}$$

which rearranges to $\mathrm{b}(p+1, q) = \mathrm{b}(p, q) - \mathrm{b}(p, q+1)$.

Now put $q = p$. By part (i), $\mathrm{b}(p, p+1) = \mathrm{b}(p+1, p)$, so

$$\mathrm{b}(p+1, p) = \mathrm{b}(p, p) - \mathrm{b}(p+1, p) \quad \Longrightarrow \quad \mathrm{b}(p+1, p) = \tfrac{1}{2}\,\mathrm{b}(p, p).$$

**Part (iii)**

Let $x = \sin^2\theta$, so $\mathrm{d}x = 2\sin\theta\cos\theta\,\mathrm{d}\theta$ and $1 - x = \cos^2\theta$. As $\theta$ runs from 0 to $\frac{1}{2}\pi$, $x$ increases from 0 to 1, and both $\sin\theta$ and $\cos\theta$ are non-negative, so $x^{p-1} = (\sin\theta)^{2p-2}$ and $(1-x)^{q-1} = (\cos\theta)^{2q-2}$ with no sign trouble. Hence

$$\mathrm{b}(p,q) = \int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-2}(\cos\theta)^{2q-2}\cdot 2\sin\theta\cos\theta\,\mathrm{d}\theta = 2\int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-1}(\cos\theta)^{2q-1}\,\mathrm{d}\theta.$$

For the second result, take $q = p$ and use $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$:

$$\mathrm{b}(p,p) = 2\int_0^{\frac{1}{2}\pi} \left(\tfrac{1}{2}\sin 2\theta\right)^{2p-1}\mathrm{d}\theta = 2^{2-2p}\int_0^{\frac{1}{2}\pi} (\sin 2\theta)^{2p-1}\,\mathrm{d}\theta = 2^{1-2p}\int_0^{\pi} (\sin\phi)^{2p-1}\,\mathrm{d}\phi,$$

where $\phi = 2\theta$.

> **Ask yourself** How do we get back to an integral over $[0, \frac{1}{2}\pi]$ that looks like $\mathrm{b}(p,\cdot)$?

The graph of $\sin\phi$ on $[0,\pi]$ is symmetric about $\phi = \frac{1}{2}\pi$ (substitute $\phi \mapsto \pi - \phi$), so the integral over $[0,\pi]$ is twice the integral over $[0, \frac{1}{2}\pi]$. Also, the trigonometric form with $q = \frac{1}{2}$ has $(\cos\theta)^{2q-1} = (\cos\theta)^0 = 1$, giving

$$\mathrm{b}\left(p, \tfrac{1}{2}\right) = 2\int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-1}\,\mathrm{d}\theta.$$

Therefore

$$\mathrm{b}(p,p) = 2^{1-2p}\cdot 2\int_0^{\frac{1}{2}\pi} (\sin\phi)^{2p-1}\,\mathrm{d}\phi = 2^{1-2p}\,\mathrm{b}\left(p,\tfrac{1}{2}\right) = \frac{1}{2^{2p-1}}\,\mathrm{b}\left(p,\tfrac{1}{2}\right),$$

as required.

**Part (iv)**

We want $x$ to go from 0 to 1 as $t$ goes from 0 to $\infty$, and $1 - x$ to be simple; $x = \dfrac{t}{1+t}$ does both, with

$$1 - x = \frac{1}{1+t}, \qquad \mathrm{d}x = \frac{(1+t)-t}{(1+t)^2}\,\mathrm{d}t = \frac{\mathrm{d}t}{(1+t)^2}.$$

As $t$ increases from 0 to $\infty$, $x$ increases from 0 towards 1. So

$$\mathrm{b}(p,q) = \int_0^{\infty} \frac{t^{p-1}}{(1+t)^{p-1}}\cdot\frac{1}{(1+t)^{q-1}}\cdot\frac{1}{(1+t)^2}\,\mathrm{d}t = \int_0^{\infty}\frac{t^{p-1}}{(1+t)^{p+q}}\,\mathrm{d}t.$$

**Part (v)**

Match the integrand with part (iv): $p-1=\frac{3}{2}$ and $p+q=6$, so $p=\frac{5}{2}$, $q=\frac{7}{2}$. The integral is $\mathrm{b}\left(\frac{5}{2},\frac{7}{2}\right)$, and we reduce it step by step:

$$\begin{aligned}\mathrm{b}\left(\tfrac{5}{2},\tfrac{7}{2}\right) &= \mathrm{b}\left(\tfrac{7}{2},\tfrac{5}{2}\right) && \text{by (i)}\\ &= \tfrac{1}{2}\,\mathrm{b}\left(\tfrac{5}{2},\tfrac{5}{2}\right) && \text{by (ii) with } p=\tfrac{5}{2}\\ &= \tfrac{1}{2}\cdot\frac{1}{2^4}\,\mathrm{b}\left(\tfrac{5}{2},\tfrac{1}{2}\right) && \text{by (iii) with } p=\tfrac{5}{2}.\end{aligned}$$

Finally, from the trigonometric form with $p=\frac{5}{2}$, $q=\frac{1}{2}$,

$$\mathrm{b}\left(\tfrac{5}{2},\tfrac{1}{2}\right)=2\int_0^{\frac{1}{2}\pi}\sin^4\theta\,\mathrm{d}\theta.$$

Using $\sin^2\theta=\frac{1}{2}(1-\cos 2\theta)$ twice,

$$\sin^4\theta=\tfrac{1}{4}\left(1-2\cos 2\theta+\cos^2 2\theta\right)=\tfrac{1}{8}\left(3-4\cos 2\theta+\cos 4\theta\right).$$

The cosine terms integrate to zero over $[0,\frac{1}{2}\pi]$ (their antiderivatives $\sin 2\theta$ and $\sin 4\theta$ vanish at both ends), so $\int_0^{\frac{1}{2}\pi}\sin^4\theta\,\mathrm{d}\theta=\frac{3}{8}\cdot\frac{1}{2}\pi=\frac{3\pi}{16}$ and $\mathrm{b}\left(\frac{5}{2},\frac{1}{2}\right)=\frac{3\pi}{8}$. Hence

$$\int_0^\infty\frac{t^{\frac{3}{2}}}{(1+t)^6}\,\mathrm{d}t=\frac{1}{32}\cdot\frac{3\pi}{8}=\frac{3\pi}{256}.$$

## Checks

- With $p=q=1$ the definition gives $\mathrm{b}(1,1)=1$, and (ii) gives $\mathrm{b}(2,1)=\frac{1}{2}$, which matches $\int_0^1 x\,\mathrm{d}x$ directly.
- Also with $p=1$, part (iii) says $\mathrm{b}(1,1)=\frac{1}{2}\,\mathrm{b}(1,\frac{1}{2})=\frac{1}{2}\int_0^1(1-x)^{-1/2}\,\mathrm{d}x=\frac{1}{2}\cdot 2=1$. Consistent.
- Numerically, $3\pi/256\approx 0.036816$, and a direct numerical integration of $t^{3/2}/(1+t)^6$ over $[0,\infty)$ by computer agrees to many decimal places. As a sanity check on size: the integrand peaks at $t=\frac{1}{3}$ with value about 0.034, and its hump has width of order 1, so a value of a few hundredths is plausible.

## Common slips

- In (iv), forgetting the factor $(1+t)^{-2}$ from $\mathrm{d}x$, which gives the wrong power $p+q-2$.
- In (iii), integrating $(\sin 2\theta)^{2p-1}$ only over $[0,\frac{1}{2}\pi]$ in $\theta$ but forgetting that $\phi=2\theta$ doubles the range and halves $\mathrm{d}\theta$.
- In (v), matching $p$ to the exponent $\frac{3}{2}$ rather than to $\frac{3}{2}+1$.

## Beyond STEP

The function $\mathrm{b}(p,q)$ is Euler's beta function, usually written $B(p,q)$. It is linked to the gamma function by $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$, and the result of part (iii) is a form of Legendre's duplication formula for $\Gamma$; with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ you can recover $3\pi/256$ in one line. The beta function also appears in probability as the normalising constant of the beta distribution. For more on such integrals, see Siklos, *Advanced Problems in Mathematics*, or Blitzstein and Hwang, *Introduction to Probability*, for the beta distribution.

WORKED SOLUTION

# STEP III 2025, Question 2

Pure • Iterated Functions and Periodicity

**STEP III 2025, Question 2**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 8.8, *Period Three Implies Chaos*: https://beyondstep.keithmcnulty.org/ch08/period-three.html

## The key idea

Every function in this question is built from straight lines, so the whole question is about keeping track of which piece you are on. Work out $\mathrm{f}(\mathrm{f}(x))$ and $\mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$ piece by piece, using the sign of the inner function to remove each modulus. Then read the sequence questions backwards: a solution of $\mathrm{f}^n(x) = x$ is a starting value that comes back to itself after $n$ steps, and every number less than 7 has exactly two preimages under f, which are negatives of each other.

## Solution

Throughout, write $\mathrm{f}^2(x) = \mathrm{f}(\mathrm{f}(x))$ and $\mathrm{f}^3(x) = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$. All three functions are even, because f depends only on $|x|$, so it is enough to work for $x \geq 0$ and reflect in the $y$-axis.

### Part (i)

**(a)** The graph of $y = \mathrm{f}(x)$ is an inverted V with vertex $(0, 7)$, gradients $\mp 2$, crossing the $x$-axis at $x = \pm\frac{7}{2}$.

For $\mathrm{f}^2$ we need the sign of $\mathrm{f}(x)$. If $0 \leq x \leq \frac{7}{2}$ then $\mathrm{f}(x) = 7 - 2x \geq 0$, so

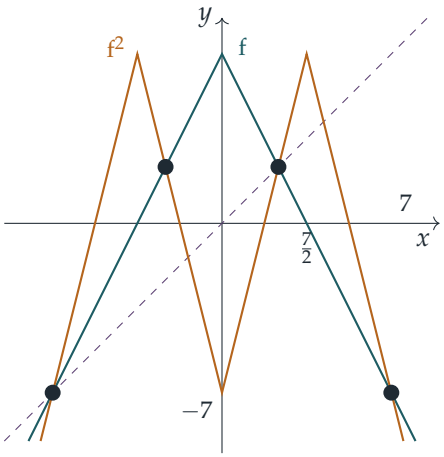
$$\mathrm{f}^2(x) = 7 - 2(7 - 2x) = 4x - 7.$$

If $x \geq \frac{7}{2}$ then $\mathrm{f}(x) \leq 0$, so $|\mathrm{f}(x)| = 2x - 7$ and

$$\mathrm{f}^2(x) = 7 - 2(2x - 7) = 21 - 4x.$$

So $y = \mathrm{f}^2(x)$ is a W shape: value $-7$ at $x = 0$, peaks of height 7 at $x = \pm\frac{7}{2}$ (exactly where f vanishes), gradients $\pm 4$, and $x$-intercepts at $\pm\frac{7}{4}$ and $\pm\frac{21}{4}$. Solving $7 - 2x = 4x - 7$ and

$7 - 2x = 21 - 4x$ shows that the two graphs cross at $(\pm\frac{7}{3}, \frac{7}{3})$ and $(\pm 7, -7)$.




(Teal: $y = \mathrm{f}(x)$; amber: $y = \mathrm{f}^2(x)$; dashed: $y = x$.)

**(b)** We solve on each of the four pieces and keep only roots in the right range.

- $0 \le x \le \frac{7}{2}$: $4x - 7 = x$ gives $x = \frac{7}{3}$, which is in range.
- $x \ge \frac{7}{2}$: $21 - 4x = x$ gives $x = \frac{21}{5} = 4.2$, in range.
- $-\frac{7}{2} \le x \le 0$: here $\mathrm{f}^2(x) = -4x - 7$, and $-4x - 7 = x$ gives $x = -\frac{7}{5}$, in range.
- $x \le -\frac{7}{2}$: here $\mathrm{f}^2(x) = 21 + 4x$, and $21 + 4x = x$ gives $x = -7$, in range.

So $\mathrm{f}(\mathrm{f}(x)) = x$ has exactly four solutions:

$$x = -7, \quad -\tfrac{7}{5}, \quad \tfrac{7}{3}, \quad \tfrac{21}{5}.$$

**(c)** The sequence has period 2 exactly when $u_2 = u_0$, that is $\mathrm{f}^2(a) = a$, but $u_1 \ne u_0$, that is $\mathrm{f}(a) \ne a$.

**?** **Ask yourself** Do all four answers to (b) give sequences of period 2?

No. Two of them are fixed points of f itself: $\mathrm{f}(\frac{7}{3}) = 7 - \frac{14}{3} = \frac{7}{3}$ and $\mathrm{f}(-7) = 7 - 14 = -7$. These give constant sequences, which have period 1. The other two swap:

$$\mathrm{f}\left(-\tfrac{7}{5}\right) = 7 - \tfrac{14}{5} = \tfrac{21}{5}, \qquad \mathrm{f}\left(\tfrac{21}{5}\right) = 7 - \tfrac{42}{5} = -\tfrac{7}{5}.$$

So the sequence has period 2 exactly when $a = -\frac{7}{5}$ or $a = \frac{21}{5}$.

**(d)** With $a = \frac{28}{5}$,

$$u_1 = 7 - \tfrac{56}{5} = -\tfrac{21}{5}, \qquad u_2 = 7 - \tfrac{42}{5} = -\tfrac{7}{5}, \qquad u_3 = \mathrm{f}\left(-\tfrac{7}{5}\right) = \tfrac{21}{5}.$$

From $u_2$ on, the sequence is the sequence of part (c) starting at $-\frac{7}{5}$, so $u_2, u_3, u_4, \ldots$ is $-\frac{7}{5}, \frac{21}{5}, -\frac{7}{5}, \ldots$ and has period 2. Finally $u_0 = \frac{28}{5}$ and $u_1 = -\frac{21}{5}$ are both different from

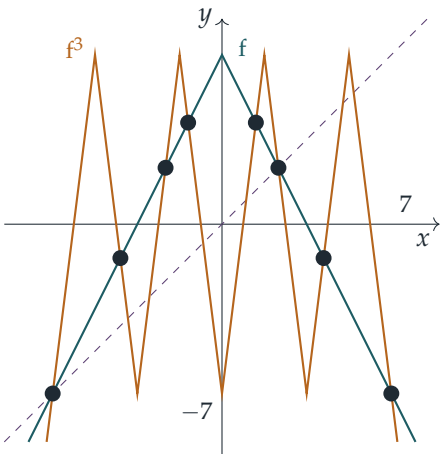
each of $u_2 = -\frac{7}{5}$ and $u_3 = \frac{21}{5}$. (We have to check all four pairs, not just $u_0 \neq u_2$ and $u_1 \neq u_3$.) This is what was required.

**Part (ii)**

**(a)** Now apply f to $\mathrm{f}^2(x)$, splitting each piece of $\mathrm{f}^2$ where it changes sign, at $x = \frac{7}{4}$ and $x = \frac{21}{4}$. For $x \geq 0$:

$$\mathrm{f}^3(x) = 7 - 2\left|\mathrm{f}^2(x)\right| = \begin{cases} 7 + 2(4x - 7) = 8x - 7, & 0 \leq x \leq \frac{7}{4}, \\ 7 - 2(4x - 7) = 21 - 8x, & \frac{7}{4} \leq x \leq \frac{7}{2}, \\ 7 - 2(21 - 4x) = 8x - 35, & \frac{7}{2} \leq x \leq \frac{21}{4}, \\ 7 + 2(21 - 4x) = 49 - 8x, & x \geq \frac{21}{4}. \end{cases}$$

So the graph zig-zags between $-7$ and $7$ with gradients $\pm 8$: troughs at $(0, -7)$ and $(\pm\frac{7}{2}, -7)$, peaks at $(\pm\frac{7}{4}, 7)$ and $(\pm\frac{21}{4}, 7)$, then falls away for $|x| > \frac{21}{4}$. Solving $7 - 2x$ against each piece, it crosses $y = \mathrm{f}(x)$ at $(\pm\frac{7}{5}, \frac{21}{5})$, $(\pm\frac{7}{3}, \frac{7}{3})$, $(\pm\frac{21}{5}, -\frac{7}{5})$ and $(\pm 7, -7)$.




(Teal: $y = \mathrm{f}(x)$; amber: $y = \mathrm{f}^3(x)$; dashed: $y = x$.)

**(b)** Starting from $a = 1$: $u_1 = 7 - 2 = 5$, $u_2 = 7 - 10 = -3$, $u_3 = 7 - 6 = 1$. Starting from $a = -\frac{7}{9}$:

$$u_1 = 7 - \tfrac{14}{9} = \tfrac{49}{9}, \qquad u_2 = 7 - \tfrac{98}{9} = -\tfrac{35}{9}, \qquad u_3 = 7 - \tfrac{70}{9} = -\tfrac{7}{9}.$$

So both sequences return to their start after three steps: they are the 3-cycles $1 \to 5 \to -3 \to 1$ and $-\frac{7}{9} \to \frac{49}{9} \to -\frac{35}{9} \to -\frac{7}{9}$.

Every term of a 3-cycle satisfies $\mathrm{f}^3(x) = x$, not just the starting value, so this gives six solutions $1$, $5$, $-3$, $-\frac{7}{9}$, $\frac{49}{9}$, $-\frac{35}{9}$. Also any fixed point of f satisfies $\mathrm{f}^3(x) = x$, which adds $\frac{7}{3}$ and $-7$ from part (i). That makes eight distinct solutions.

**Ask yourself** How do we know there are no more?

The formula in (ii)(a) shows that $f^3$ is made of eight straight pieces (four for $x \geq 0$ and their mirror images), each with gradient $\pm 8$. Since no piece is parallel to $y = x$, each piece meets the line $y = x$ at most once, so there are at most eight solutions. We have found eight, so the complete solution set of $f(f(f(x))) = x$ is

$$x = -7, \quad -\tfrac{35}{9}, \quad -3, \quad -\tfrac{7}{9}, \quad 1, \quad \tfrac{7}{3}, \quad 5, \quad \tfrac{49}{9}.$$

(The sketch agrees: the dashed line crosses each of the eight amber segments once.)

**(c)** We copy the idea of (i)(d) and work backwards from a 3-cycle. Since $f(y) = c$ means $|y| = \frac{1}{2}(7 - c)$, every $c < 7$ has exactly two preimages, $\pm\frac{1}{2}(7 - c)$. Take $u_3 = 1$ from the cycle $1 \to 5 \to -3 \to 1$. The preimage of 1 inside the cycle is $-3$, so to leave the cycle we choose the other preimage, $u_2 = 3$, which is not in the cycle. Then

$$f(u_1) = 3 \Rightarrow |u_1| = 2, \quad \text{take } u_1 = 2; \qquad f(u_0) = 2 \Rightarrow |u_0| = \tfrac{5}{2}, \quad \text{take } u_0 = \tfrac{5}{2}.$$

So $a = \frac{5}{2}$ works. Checking forwards:

$$\tfrac{5}{2},\ 2,\ 3,\ 1,\ 5,\ -3,\ 1,\ 5,\ -3,\ \ldots$$

From $u_3$ on, the sequence runs round the cycle $1, 5, -3$ and has period 3, while $u_0, u_1, u_2 = \frac{5}{2}, 2, 3$ are all different from 1, 5 and $-3$.

(Other choices also work: for example $a = -\frac{5}{2}$ or $a = \pm\frac{9}{2}$, which lead to $u_1 = -2$, or $a = \frac{3}{2}$, which enters the cycle at $u_3 = 5$ via $u_2 = -1$.)

## Checks

- Each claimed solution can be iterated by hand: for instance $\frac{49}{9} \to -\frac{35}{9} \to -\frac{7}{9} \to \frac{49}{9}$, and $\frac{28}{5} \to -\frac{21}{5} \to -\frac{7}{5} \to \frac{21}{5} \to -\frac{7}{5}$.
- A computer scan of $f^3(x) - x$ over $-30 \leq x \leq 30$ finds sign changes only near $-7$, $-3.889$, $-3$, $-0.778$, 1, 2.333, 5 and 5.444, matching the eight solutions; the same scan for $f^2(x) - x$ finds only $-7$, $-1.4$, 2.333 and 4.2.
- Counting check: $f^n$ has $2^n$ linear pieces of gradient $\pm 2^n$, so $f^n(x) = x$ has $2^n$ solutions. For $n = 2$ that is $4 = 2 + 2$ (two fixed points and one 2-cycle); for $n = 3$ it is $8 = 2 + 3 + 3$ (two fixed points and two 3-cycles).

## Common slips

- Giving all four solutions of (i)(b) in (i)(c): the fixed points give period 1.
- In (ii)(b), listing only 1 and $-\frac{7}{9}$, or forgetting that the fixed points $\frac{7}{3}$ and $-7$ also satisfy $f^3(x) = x$.
- Misplacing where the graphs cross in the sketches; finding the crossing points algebraically fixes this.

## Beyond STEP

This is a first taste of discrete dynamical systems. Since f is a tent map with slope 2, $f^n$ has $2^n$ pieces, so there are periodic points of every period, and the map is a standard example of chaotic behaviour. Sharkovskii's theorem says that for a continuous map of the line, having a point of period 3 forces points of every period, the famous slogan "period three implies chaos". Strogatz, *Nonlinear Dynamics and Chaos*, gives a readable introduction to iterated maps.

WORKED SOLUTION

# STEP III 2025, Question 3

Pure • Inequalities of Means

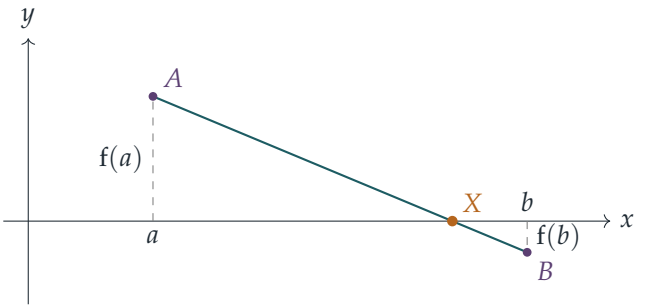
**STEP III 2025, Question 3**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

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## The key idea

Because $A$ sits above the axis and $B$ below it, the crossing point $X$ divides $AB$ in the ratio of the two heights $\mathrm{f}(a) : \mathrm{f}(b)$. So $m$ is a *weighted average* of $a$ and $b$, and everything that follows is about how the weights change as f changes. Once $m$ is written as a function of the single ratio $\mathrm{f}(a)/\mathrm{f}(b)$, comparing two choices of f becomes comparing two numbers.




## Solution

### Part (i)

The triangles formed by $A$, its foot $(a, 0)$ and $X$, and by $B$, its foot $(b, 0)$ and $X$, are similar (vertically opposite angles at $X$, right angles at the feet). Hence

$$\frac{m-a}{b-m} = \frac{\mathrm{f}(a)}{\mathrm{f}(b)} \quad \Longrightarrow \quad m\,\mathrm{f}(b) - a\,\mathrm{f}(b) = b\,\mathrm{f}(a) - m\,\mathrm{f}(a),$$

so

$$\boxed{m = \frac{a\,\mathrm{f}(b) + b\,\mathrm{f}(a)}{\mathrm{f}(a) + \mathrm{f}(b)}}.$$

(The denominator is positive because $\mathrm{f} > 0$; the same answer comes from the line's equation $y - \mathrm{f}(a) = -\frac{\mathrm{f}(a)+\mathrm{f}(b)}{b-a}(x-a)$ with $y = 0$.)

It is worth rewriting this with $r = \mathrm{f}(a)/\mathrm{f}(b) > 0$, dividing top and bottom by $\mathrm{f}(b)$:

$$m = \frac{a+br}{1+r} = b - \frac{b-a}{1+r}. \qquad (*)$$

**Part (ii)**

With $\mathrm{f}(x) = \sqrt{x}$,

$$m = \frac{a\sqrt{b}+b\sqrt{a}}{\sqrt{a}+\sqrt{b}} = \frac{\sqrt{a}\sqrt{b}\,(\sqrt{a}+\sqrt{b})}{\sqrt{a}+\sqrt{b}} = \sqrt{ab},$$

as required.

For the second target, the numerator $a^{n+1}+b^{n+1}$ should come from $a\,\mathrm{f}(b)+b\,\mathrm{f}(a)$ after clearing fractions, which suggests $\mathrm{f}(b)$ should behave like $b^{-n}$. Try $\mathrm{f}(x) = x^{-n}$, which is positive for $x > 0$:

$$m = \frac{ab^{-n}+ba^{-n}}{a^{-n}+b^{-n}} = \frac{a^{n+1}+b^{n+1}}{b^n+a^n},$$

multiplying top and bottom by $a^n b^n$. So $\mathrm{f}(x) = x^{-n}$ works (as does any positive constant multiple of it, since $m$ only depends on the ratio $\mathrm{f}(a)/\mathrm{f}(b)$). Note that $n = -\frac{1}{2}$ recovers the first result.

**Part (iii)**

Let $r_1 = \mathrm{g}_1(a)/\mathrm{g}_1(b)$ and $r_2 = \mathrm{g}_2(a)/\mathrm{g}_2(b)$, both positive. By $(*)$, $M_i = b - \dfrac{b-a}{1+r_i}$. Since $b-a > 0$, the expression $b - \frac{b-a}{1+r}$ is strictly increasing in $r$ for $r > 0$, so it is enough to show $r_1 > r_2$.

> **Ask yourself** How does a condition on the *function* $\mathrm{g}_1/\mathrm{g}_2$ turn into a condition on $r_1$ and $r_2$?

Write $h = \mathrm{g}_1/\mathrm{g}_2$. Then

$$\frac{r_1}{r_2} = \frac{\mathrm{g}_1(a)\,\mathrm{g}_2(b)}{\mathrm{g}_1(b)\,\mathrm{g}_2(a)} = \frac{h(a)}{h(b)} > 1,$$

because $h$ is decreasing and $a < b$ give $h(a) > h(b) > 0$. As $r_2 > 0$, this gives $r_1 > r_2$, and hence $M_1 > M_2$.

**Hence.** We need three functions whose values of $m$ are the three means.

- $\mathrm{f}(x) = 1$ (constant) gives $m = \dfrac{a+b}{2}$.
- $\mathrm{f}(x) = \sqrt{x}$ gives $m = \sqrt{ab}$, by part (ii).

▶ $\mathrm{f}(x) = x$ gives $m = \dfrac{ab + ba}{a + b} = \dfrac{2ab}{a + b}$.

Now apply the result just proved twice. With $\mathrm{g}_1 = 1$, $\mathrm{g}_2 = \sqrt{x}$, the ratio $\mathrm{g}_1/\mathrm{g}_2 = x^{-1/2}$ is decreasing for $x > 0$, so $\frac{a+b}{2} > \sqrt{ab}$. With $\mathrm{g}_1 = \sqrt{x}$, $\mathrm{g}_2 = x$, the ratio is again $x^{-1/2}$, decreasing, so $\sqrt{ab} > \frac{2ab}{a+b}$. Together,

$$\frac{a + b}{2} > \sqrt{ab} > \frac{2ab}{a + b}.$$

**Part (iv)**

The curve passes through $A$ and $B$, so

$$\mathrm{f}(a) = p(c - a)^3, \qquad -\mathrm{f}(b) = p(c - b)^3, \text{ i.e. } \mathrm{f}(b) = p(b - c)^3.$$

Since $\mathrm{f}(b) > 0$ we have $p \neq 0$ and $b \neq c$, so we may divide:

$$\frac{\mathrm{f}(a)}{\mathrm{f}(b)} = \left(\frac{c - a}{b - c}\right)^3.$$

Every real number has exactly one real cube root, so taking cube roots is reversible and

$$\frac{c - a}{b - c} = \left(\frac{\mathrm{f}(a)}{\mathrm{f}(b)}\right)^{1/3},$$

as required. Writing $k$ for the right-hand side ($k > 0$), $c - a = k(b - c)$, so $c(1 + k) = a + kb$ and

$$c = \frac{a + kb}{1 + k} = \frac{a\,\mathrm{f}(b)^{1/3} + b\,\mathrm{f}(a)^{1/3}}{\mathrm{f}(a)^{1/3} + \mathrm{f}(b)^{1/3}},$$

multiplying top and bottom by $\mathrm{f}(b)^{1/3}$.

? **Ask yourself** Where have we seen this formula before?

It is exactly the formula of part (i) with f replaced by $\mathrm{f}^{1/3}$, which is also defined and positive for $x > 0$. So $c$ is the value of $m$ for the function $\mathrm{f}^{1/3}$. Apply part (iii) with $\mathrm{g}_1 = \mathrm{f}$ and $\mathrm{g}_2 = \mathrm{f}^{1/3}$:

$$\frac{\mathrm{g}_1(x)}{\mathrm{g}_2(x)} = \mathrm{f}(x)^{2/3}.$$

If f is decreasing and positive, then so is $\mathrm{f}^{2/3}$, since $t \mapsto t^{2/3}$ is increasing for $t > 0$. Part (iii) therefore gives $M_1 > M_2$, where $M_1 = m$ (for f itself) and $M_2 = c$. Hence $c < m$.

## Checks

▶ Take $a = 1$, $b = 4$. The three means are $2.5 > 2 > 1.6$, and $\mathrm{f} = \sqrt{x}$ gives $m = (1 \cdot 2 + 4 \cdot 1)/3 = 2 = \sqrt{4}$.

- With $\mathrm{f}(x) = 1/x$, $a = 1$, $b = 4$: $m = \frac{1/4+4}{5/4} = 3.4$, while $k = 4^{1/3} \approx 1.587$ gives $c \approx 2.84 < 3.4$, as part (iv) predicts. Both $m$ and $c$ lie between $a$ and $b$, as weighted averages must.
- Setting $n = 0$ in part (ii) gives $\mathrm{f} = 1$ and $m = \frac{a+b}{2}$; setting $n = -1$ gives $\mathrm{f} = x$ and $m = \frac{2ab}{a+b}$, consistent with part (iii).

These, and random tests of every result over many choices of $a$, $b$ and f, were confirmed by computer.

## Common slips

- Forgetting that $B$ is at height $-\mathrm{f}(b)$, which gives a denominator $\mathrm{f}(b) - \mathrm{f}(a)$ and a point that may not lie between $a$ and $b$.
- In part (iii), trying to compare $M_1$ and $M_2$ by cross-multiplying four function values at once instead of reducing to the single ratio $r$.
- In part (iv), taking $\mathrm{g}_1 = \mathrm{f}^{1/3}$, $\mathrm{g}_2 = \mathrm{f}$ the wrong way round: their ratio $\mathrm{f}^{-2/3}$ is *increasing*, so part (iii) does not apply in that order.

## Beyond STEP

Every $m$ here is a weighted mean $\lambda a + (1-\lambda)b$, and part (iii) is a neat way of ordering the whole family of power means $\left(a^{n+1} + b^{n+1}\right) / \left(a^n + b^n\right)$ (the Lehmer means), which increase with $n$. Inequalities between means, and the convexity arguments behind them, are a staple of first-year analysis. Abbott's *Understanding Analysis* and Siklos's *Advanced Problems in Mathematics* are good places to see more of this style of argument.

WORKED SOLUTION

# STEP III 2025, Question 4

Pure • Transformations of Curves

**STEP III 2025, Question 4**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

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## The key idea

Suppose a map $T$ of the plane has an inverse. Then the curve $f\left(T^{-1}(x,y)\right)=0$ is the image under $T$ of the curve $f(x,y)=0$, because a point lies on the new curve exactly when $T^{-1}$ sends it to the old one. Part (i) shows this for a rotation. Part (ii) does the same for a reflection, where $T^{-1}=T$. In part (iii) we spot that the substitution $u=x-2y$ turns the equation into a horizontal shear of $y=\sin x$, and a parametrisation then gives the tangents with no implicit-differentiation headaches.

## Solution

### Part (i)

Multiplying out the matrix product gives

$$x_2=\frac{x_1-y_1}{\sqrt{2}},\qquad y_2=\frac{x_1+y_1}{\sqrt{2}}.$$

Substitute $(x,y)=(x_2,y_2)$ into the two brackets in the equation of $G_2$:

$$\frac{x_2}{\sqrt{2}}+\frac{y_2}{\sqrt{2}}=\frac{(x_1-y_1)+(x_1+y_1)}{2}=x_1,$$
$$-\frac{x_2}{\sqrt{2}}+\frac{y_2}{\sqrt{2}}=\frac{-(x_1-y_1)+(x_1+y_1)}{2}=y_1.$$

So the left-hand side of the equation of $G_2$ at $(x_2,y_2)$ is $\frac{x_1^2}{9}+\frac{y_1^2}{4}$. This equals 1 because $(x_1,y_1)$ is on $G_1$. Hence $(x_2,y_2)$ lies on $G_2$.

For the rotation, note that $\frac{1}{\sqrt{2}}=\cos 45^\circ=\sin 45^\circ$, so the matrix is

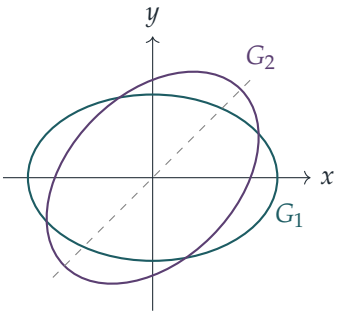
$$R=\begin{pmatrix}\cos 45^\circ & -\sin 45^\circ\\ \sin 45^\circ & \cos 45^\circ\end{pmatrix},$$

the anticlockwise rotation through $45^\circ$ about the origin. We have just shown that $R$ maps every point of $G_1$ onto $G_2$. To show that $G_2$ is *exactly* the rotated curve, we also

need every point of $G_2$ to come from $G_1$. Take any $(x, y)$ on $G_2$ and put

$$x_1 = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}, \qquad y_1 = -\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}},$$

which is $R^{-1}(x, y)$ (rotation back through 45°). The equation of $G_2$ says precisely that $\frac{x_1^2}{9} + \frac{y_1^2}{4} = 1$, so $(x_1, y_1) \in G_1$, and $R(x_1, y_1) = (x, y)$. Therefore $G_2 = R(G_1)$: the ellipse $G_2$ is $G_1$ rotated anticlockwise through 45° about the origin.




**Part (ii)(a)**

Call the matrix $M$. A point $(x, y)$ is invariant when $M\binom{x}{y} = \binom{x}{y}$, that is,

$$-0.6x + 0.8y = x, \qquad 0.8x + 0.6y = y.$$

The first gives $0.8y = 1.6x$, so $y = 2x$, and the second gives $0.8x = 0.4y$, which is the same thing. So the line of invariant points (the mirror line) is

$$y = 2x.$$

**Part (ii)(b)**

Write $(X, Y) = M(x, y) = (-0.6x + 0.8y,\ 0.8x + 0.6y)$. The second equation says exactly

$$Y = 2^X.$$

So $(x, y)$ is on the second graph if and only if $M(x, y)$ is on $y = 2^x$. A reflection undoes itself ($M^2 = I$, as you can check by multiplying out), so $(x, y) = M(X, Y)$. This is the part (i) argument again: the second graph is the reflection of $y = 2^x$ in the line $y = 2x$.

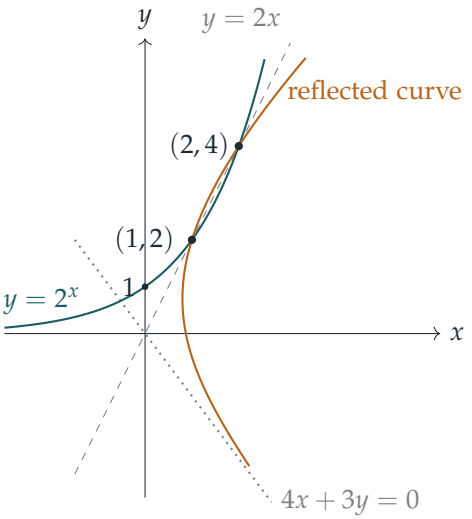
? **Ask yourself** Where does the asymptote of the reflected curve go, and which way does the curve run off to infinity?

Reflect the features of $y = 2^x$ one at a time.

- *Asymptote.* As $x \to -\infty$, $y = 2^x$ approaches the $x$-axis. The direction $(1, 0)$ reflects to $M(1, 0) = (-0.6, 0.8)$, so the $x$-axis reflects to the line $4x + 3y = 0$, i.e. $y = -\frac{4}{3}x$. Equivalently, $Y = 0.8x + 0.6y = 2^X > 0$, so the reflected curve lies strictly on the side

$4x + 3y > 0$ and approaches $4x + 3y = 0$ as $X \to -\infty$. Far out, the points of $y = 2^x$ are about $(X, 0)$ with $X$ large and negative, and these reflect to about $(-0.6X, 0.8X)$. So the reflected curve approaches its asymptote in the fourth quadrant, with $x \to +\infty$ and $y \to -\infty$.

- *Points on the mirror.* The reflection fixes every point on $y = 2x$. Now $2^x = 2x$ at $x = 1$ and $x = 2$ (and only there, since $2^x - 2x$ is convex). So both curves pass through $(1, 2)$ and $(2, 4)$, and they cross there. Searching by computer along $y = 2^x$ finds no other common points.
- *Other points.* $(0, 1)$ reflects to $(0.8, 0.6)$.
- *Leftmost point.* The reflected curve is $\left(-0.6t + 0.8 \cdot 2^t,\ 0.8t + 0.6 \cdot 2^t\right)$. Its $x$-coordinate is least where $0.8 \cdot 2^t \ln 2 = 0.6$, at about $(0.80, 0.74)$. There the tangent is vertical, so the reflected curve is not the graph of a function of $x$.
- *Growth.* As $X \to +\infty$ the points $(X, 2^X)$ reflect to about $2^X(0.8, 0.6)$. So the curve heads up and to the right, roughly in the direction of slope $\frac{3}{4}$.




**Part (iii)**

The graph $y = \sin x$ for $0 \le x \le 2\pi$ is the familiar single wave.

For $y = \sin(x - 2y)$, put $u = x - 2y$. A point $(x, y)$ is on the curve if and only if $y = \sin u$, and then $x = u + 2y$. So the curve is exactly

$$x = u + 2\sin u, \qquad y = \sin u.$$

This is $y = \sin x$ with every point pushed sideways by twice its height: a horizontal shear $(u, v) \mapsto (u + 2v, v)$.

*Which $u$ give $0 \le x \le 2\pi$?* Let $x(u) = u + 2\sin u$. Then $x(-u) = -x(u)$ and $x(u + 2\pi) = x(u) + 2\pi$. Since $x'(u) = 1 + 2\cos u$, the function $x(u)$ increases on $[0, \frac{2\pi}{3}]$ from 0 to

$\frac{2\pi}{3} + \sqrt{3} \approx 3.83$, then decreases on $[\frac{2\pi}{3}, \pi]$ down to $\pi$. So it lies in $(0, 2\pi)$ for $0 < u \leq \pi$, and on $[\pi, 2\pi]$ the symmetry $x(2\pi - u) = 2\pi - x(u)$ keeps it in $(0, 2\pi]$. For $u > 2\pi$ we get $x(u) = x(u - 2\pi) + 2\pi > 2\pi$ by induction, and for $u < 0$ oddness gives $x(u) < 0$. So the required piece of the curve is traced exactly once as $u$ runs from 0 to $2\pi$, from $(0, 0)$ to $(2\pi, 0)$.

*Tangents.* Along the curve

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 1 + 2\cos u, \qquad \frac{\mathrm{d}y}{\mathrm{d}u} = \cos u, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos u}{1 + 2\cos u}.$$

These two derivatives are never zero together, since $\cos u = 0$ makes $1 + 2\cos u = 1$.

- Horizontal tangents occur where $\cos u = 0$, i.e. $u = \frac{\pi}{2}, \frac{3\pi}{2}$:

$$\left(\tfrac{\pi}{2} + 2,\ 1\right) \quad \text{and} \quad \left(\tfrac{3\pi}{2} - 2,\ -1\right).$$

- Vertical tangents occur where $\cos u = -\frac{1}{2}$, i.e. $u = \frac{2\pi}{3}, \frac{4\pi}{3}$:

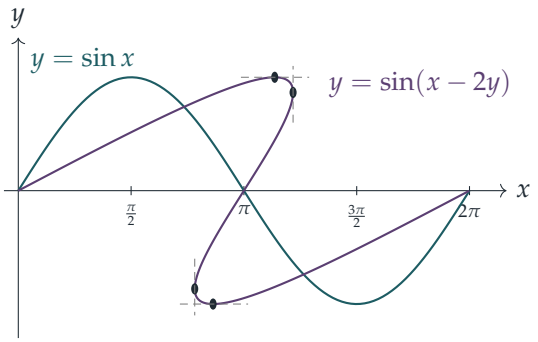
$$\left(\tfrac{2\pi}{3} + \sqrt{3},\ \tfrac{\sqrt{3}}{2}\right) \quad \text{and} \quad \left(\tfrac{4\pi}{3} - \sqrt{3},\ -\tfrac{\sqrt{3}}{2}\right).$$

Numerically these are about $(3.57, 1)$, $(2.71, -1)$, $(3.83, 0.87)$ and $(2.46, -0.87)$.

**Ask yourself** The horizontal tangent at the top is at $x \approx 3.57$, but the curve still reaches $x \approx 3.83$ after that. What does the curve actually look like?

Follow $u$ from 0 to $2\pi$. The curve leaves the origin with gradient $\frac{1}{3}$ (put $u = 0$), which is shallower than $y = \sin x$. It rises to its maximum at $(\frac{\pi}{2} + 2, 1)$, then moves right and down to the vertical tangent at $(\frac{2\pi}{3} + \sqrt{3}, \frac{\sqrt{3}}{2})$. Then it bends back to the left, passing through $(\pi, 0)$ with gradient $\frac{-1}{-1} = 1$. It reaches the second vertical tangent at $(\frac{4\pi}{3} - \sqrt{3}, -\frac{\sqrt{3}}{2})$, turns right through the minimum $(\frac{3\pi}{2} - 2, -1)$, and rises to $(2\pi, 0)$. The curve overhangs like a breaking wave. Between $x \approx 2.46$ and $x \approx 3.83$ there are three values of $y$ for each $x$. The whole picture has half-turn symmetry about $(\pi, 0)$.

For interest, the two graphs meet where $\sin x = \sin(x - 2\sin x)$. Either $\sin x = 0$, giving $x = 0, \pi, 2\pi$, or $x - \sin x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$, giving $x \approx 2.31$ and $x \approx 3.97$.



## Checks

- Part (i): take the vertex $(3,0)$ of $G_1$. It maps to $(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}})$, and both brackets in the equation of $G_2$ give 3 and 0, so the left-hand side is 1.
- Part (ii): $M\binom{1}{2} = \binom{-0.6+1.6}{0.8+1.2} = \binom{1}{2}$, as an invariant point must. The point $(1,2)$ also satisfies the second equation: $0.8 + 1.2 = 2 = 2^{-0.6+1.6}$.
- Part (iii): at $(\frac{\pi}{2}+2, 1)$ we get $x - 2y = \frac{\pi}{2}$, and $\sin\frac{\pi}{2} = 1 = y$. Similarly $(\frac{2\pi}{3} + \sqrt{3}, \frac{\sqrt{3}}{2})$ gives $x - 2y = \frac{2\pi}{3}$, and $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$. Implicit differentiation gives $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos(x-2y)}{1+2\cos(x-2y)}$. Its numerator is 0 at the first point and its denominator is 0 at the second.

## Common slips

- Showing only that $G_1$ maps *into* $G_2$. To say $G_2$ *is* the rotated curve you also need every point of $G_2$ to come from $G_1$, which is the inverse map.
- Treating $y = \sin(x - 2y)$ as a function of $x$, so the curve gets drawn as a single wave without its overhang. The vertical tangents are exactly where it folds back.
- Forgetting that the reflected exponential has an oblique asymptote, $4x + 3y = 0$, and lies wholly on one side of it.

## Beyond STEP

The principle behind this question is that applying an invertible map $T$ to a curve $f(x,y) = 0$ gives the curve $f\big(T^{-1}(x,y)\big) = 0$. In a first-year linear algebra course this becomes the study of how quadratic forms change under a change of basis. Rotating axes to remove the $xy$ term from a conic is the two-dimensional case of diagonalising a symmetric matrix. Strang's *Introduction to Linear Algebra* covers rotations, reflections and their eigenvectors (the invariant line in (ii)(a) is an eigenvector with eigenvalue 1) very readably.

WORKED SOLUTION

# STEP III 2025, Question 5

Pure • Vectors in Three Dimensions

**STEP III 2025, Question 5**

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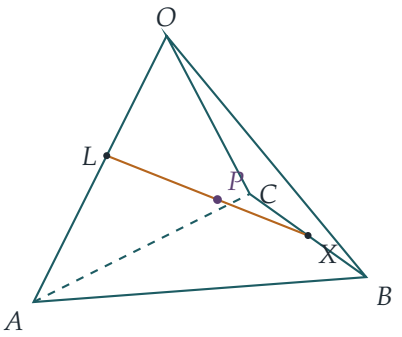
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## The key idea

Because $O$ is above the plane of the non-collinear points $A$, $B$, $C$, the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are linearly independent, so we may *compare coefficients* in any vector equation written in terms of them. Every point in the question is then found by writing it with unknown parameters and matching coefficients. Heights above the plane are just multiples of the coefficients, which makes "horizontal" easy to handle.

## Solution

Since $O$ is not in the plane $ABC$, no one of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is a combination of the other two (if, say, $\mathbf{c} = p\mathbf{a} + q\mathbf{b}$ then $O$, $A$, $B$, $C$ would be coplanar). So $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = x'\mathbf{a} + y'\mathbf{b} + z'\mathbf{c}$ forces $x = x'$, $y = y'$, $z = z'$. Note also that $k > 0$.




**Part (i)**

Write $\overrightarrow{OL} = \lambda\mathbf{a}$ and $\overrightarrow{OX} = (1-t)\mathbf{b} + t\mathbf{c}$. Since $P$ lies on $LX$, for some $s$

$$\overrightarrow{OP} = (1-s)\lambda\,\mathbf{a} + s(1-t)\,\mathbf{b} + st\,\mathbf{c}.$$

Comparing with $\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}$:

$$(1-s)\lambda=\alpha, \qquad s(1-t)=\beta, \qquad st=\gamma.$$

Adding the last two gives $s=\beta+\gamma$, and then $t=\gamma/(\beta+\gamma)$, so

$$\overrightarrow{OX}=\frac{\beta\mathbf{b}+\gamma\mathbf{c}}{\beta+\gamma}.$$

Finally $1-s=1-\beta-\gamma=k+\alpha$, which is positive, so

$$\overrightarrow{OL}=\frac{\alpha}{1-s}\,\mathbf{a}=\frac{\alpha}{k+\alpha}\,\mathbf{a},$$

as required. (The coefficients were forced, so $L$ and $X$ are unique; and since $0<s<1$ and $0<t<1$, $P$ lies strictly between $L$ and $X$, with $X$ strictly between $B$ and $C$.)

**Part (ii)**

The same argument with the letters cycled ($\mathbf{a}\to\mathbf{b}\to\mathbf{c}\to\mathbf{a}$, $\alpha\to\beta\to\gamma\to\alpha$) gives

$$\overrightarrow{OM}=\frac{\beta}{k+\beta}\mathbf{b}, \quad \overrightarrow{ON}=\frac{\gamma}{k+\gamma}\mathbf{c}, \quad \overrightarrow{OY}=\frac{\gamma\mathbf{c}+\alpha\mathbf{a}}{\gamma+\alpha}, \quad \overrightarrow{OZ}=\frac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta}.$$

**Ask yourself** How can we tell whether a plane through three given points is horizontal?

It is horizontal exactly when the three points are at the same height. Let $O$ be at height $h>0$ above the plane $ABC$. Then $A$, $B$, $C$ are each a vertical distance $h$ below $O$, so the point $\mu\mathbf{a}$ on $OA$ is a distance $\mu h$ below $O$, and similarly on $OB$ and $OC$. So $L$, $M$, $N$ (which are not collinear, lying on three different edges away from $O$) are at depths

$$\frac{\alpha h}{k+\alpha}, \qquad \frac{\beta h}{k+\beta}, \qquad \frac{\gamma h}{k+\gamma}$$

below $O$. Now

$$\frac{\alpha}{k+\alpha}=\frac{\beta}{k+\beta} \iff \alpha k+\alpha\beta=\beta k+\alpha\beta \iff k(\alpha-\beta)=0 \iff \alpha=\beta,$$

using $k>0$. Similarly for the other pairs, so

$$\text{plane } LMN \text{ is horizontal} \iff \alpha=\beta=\gamma.$$

Next, where does $OP$ meet the plane $ABC$? A point $x\mathbf{a}+y\mathbf{b}+z\mathbf{c}$ lies in that plane exactly when $x+y+z=1$ (it is then $\mathbf{a}+y(\mathbf{b}-\mathbf{a})+z(\mathbf{c}-\mathbf{a})$, and every point of the plane has this form). The point $\mu\overrightarrow{OP}$ has coefficient sum $\mu(\alpha+\beta+\gamma)$, so the line $OP$ meets the plane at

$$\overrightarrow{OG}=\frac{\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}}{\alpha+\beta+\gamma}.$$

By comparing coefficients, this equals $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ if and only if

$$\frac{\alpha}{\alpha+\beta+\gamma}=\frac{\beta}{\alpha+\beta+\gamma}=\frac{\gamma}{\alpha+\beta+\gamma}=\frac{1}{3},$$

that is, if and only if $\alpha=\beta=\gamma$. Both conditions are equivalent to $\alpha=\beta=\gamma$, so the plane $LMN$ is horizontal if and only if $OP$ meets the plane $ABC$ at the centroid $G$ of triangle $ABC$. This completes the "show that".

In this case $\overrightarrow{OX}=\frac{1}{2}(\mathbf{b}+\mathbf{c})$, and likewise for $Y$ and $Z$: the points $X$, $Y$, $Z$ are the **midpoints** of $BC$, $CA$ and $AB$ respectively.

#### Part (iii)

We can write

$$\overrightarrow{OP}=k\,\mathbf{0}+\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c},\qquad k+\alpha+\beta+\gamma=1,$$

with all four weights $k,\alpha,\beta,\gamma$ positive, precisely because $\alpha+\beta+\gamma<1$. So $P$ is a weighted average of $O$, $A$, $B$, $C$ with strictly positive weights, which means $P$ lies strictly **inside** the tetrahedron $OABC$.

Put another way: positivity of $\alpha$, $\beta$, $\gamma$ alone places $P$ inside the infinite three-sided cone at $O$ spanned by the edges $OA$, $OB$, $OC$. The depth of $P$ below $O$ is $(\alpha+\beta+\gamma)h$, so the condition $\alpha+\beta+\gamma<1$ says that $P$ is above the plane $ABC$, on the same side as $O$. (If the sum were 1, $P$ would be on the face $ABC$; if greater than 1, beyond it.)

## Checks

- Take $\alpha=\beta=\gamma=\frac{1}{4}$, so $k=\frac{1}{4}$: then $\overrightarrow{OL}=\frac{1}{2}\mathbf{a}$, $\overrightarrow{OM}=\frac{1}{2}\mathbf{b}$, $\overrightarrow{ON}=\frac{1}{2}\mathbf{c}$, and $LMN$ is clearly the horizontal mid-section, as part (ii) predicts.
- As $k\to 0$ (so $P$ approaches the face $ABC$), $\overrightarrow{OL}\to\mathbf{a}$: $L$ moves down to $A$, which makes sense since $L$, $P$, $X$ then all lie in the plane $ABC$.
- A computer check with 2000 random tetrahedra and random $\alpha,\beta,\gamma$ confirmed that $P$ lies on $LX$ (residual about $10^{-15}$), and that $LMN$ is horizontal exactly when $G$ is the centroid.

## Common slips

- Comparing coefficients without saying why it is allowed: it needs $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ to be independent, which is where "$O$ lies above the plane" is used.
- Forgetting that $k>0$ is needed to cancel in $k(\alpha-\beta)=0$.
- Proving only one direction of the "if and only if".

## Beyond STEP

The numbers $(k, \alpha, \beta, \gamma)$ are the *barycentric coordinates* of $P$ with respect to the tetrahedron $OABC$: a point is inside exactly when all of them are positive, and on a face when one of them is zero. They are the natural coordinates for affine geometry and turn up again in linear algebra (affine combinations and convex hulls) and in computer graphics, where triangles are shaded by interpolating with barycentric weights. Strang's *Introduction to Linear Algebra* is a good place to see linear independence and coordinates used this way.

WORKED SOLUTION

# STEP III 2025, Question 6

Pure • Symmetric Functions of Roots

**STEP III 2025, Question 6**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Conditions on $a+b+c$, $a^2+b^2+c^2$ and $a^3+b^3+c^3$ are really conditions on the elementary symmetric functions $a+b+c$, $ab+bc+ca$ and $abc$. Once you translate, the numbers become the roots of a very simple polynomial (in (i)), or a classic factorisation does all the work (in (ii) and (iii)). The final part is the same algebra run in reverse.

## Solution

**Part (i)**

Squaring $a+b+c=0$ gives

$$0=(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)=2(ab+bc+ca),$$

so $ab+bc+ca=0$ as well.

*Not all real.* If $a$, $b$ and $c$ were all real, then $a^2$, $b^2$, $c^2$ would be non-negative reals with sum 0, forcing $a=b=c=0$. This contradicts the numbers being non-zero, so they cannot all be real.

*Equal moduli.* Write $p=abc$, which is non-zero. Since $a+b+c=0$ and $ab+bc+ca=0$,

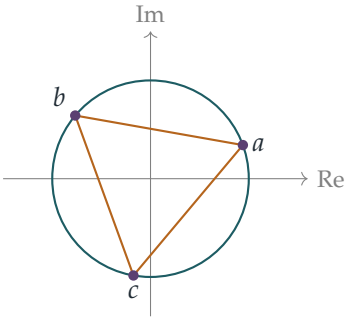
$$(z-a)(z-b)(z-c)=z^3-(a+b+c)z^2+(ab+bc+ca)z-abc=z^3-p.$$

So each of $a$, $b$, $c$ is a root of $z^3=p$, that is,

$$a^3=b^3=c^3=p.$$

Taking moduli, $|a|^3=|b|^3=|c|^3=|p|$, and since moduli are non-negative real numbers, $|a|=|b|=|c|=|p|^{1/3}$. So $a$, $b$ and $c$ all have the same modulus, as required.

In fact $a$, $b$, $c$ are the three cube roots of $p$, which sit at the vertices of an equilateral triangle centred on the origin.




**Part (ii)**

Suppose $a+b+c=0$. Then $c=-(a+b)$ and

$$a^3+b^3+c^3=a^3+b^3-(a+b)^3=-3a^2b-3ab^2=-3ab(a+b)=3abc.$$

If also $a^3+b^3+c^3=0$, then $abc=0$, so at least one of the numbers is zero. Hence there are no three *non-zero* complex numbers with both properties.

**Part (iii)**

The useful identity this time is

$$(a+b+c)^3=a^3+b^3+c^3+3(a+b)(b+c)(c+a),$$

which you can verify by expanding: both sides contain each of $a^2b$, $ab^2$ and so on three times, and $abc$ six times.

Now $d=-(a+b+c)$, so $d^3=-(a+b+c)^3$ and

$$0=a^3+b^3+c^3+d^3=a^3+b^3+c^3-(a+b+c)^3=-3(a+b)(b+c)(c+a).$$

**?** **Ask yourself** Why does a product being zero tell us anything about moduli?

A product of complex numbers is zero only if one factor is zero. So one of $a+b$, $b+c$, $c+a$ vanishes; say $a+b=0$ (the other cases are the same with letters relabelled). Then $b=-a$, so $|b|=|a|$. Hence at least two of the four numbers have the same modulus.

(Nothing here used $d\neq 0$ or that the numbers are non-zero; the conclusion is simply that the four numbers split into two pairs of negatives, as in $a,-a,c,-c$.)

**Part (iv)**

With $c=1$, $d=-2$, $e=3$ we have

$$c+d+e=2, \qquad c^3+d^3+e^3=1-8+27=20,$$

so we need real $a$, $b$ with

$$a+b=-2, \qquad a^3+b^3=-20.$$

Using $a^3+b^3=(a+b)^3-3ab(a+b)$, the second condition becomes

$$-20=-8+6ab \quad\Longrightarrow\quad ab=-2.$$

So $a$ and $b$ are the roots of $t^2+2t-2=0$, namely

$$a=-1+\sqrt{3}, \qquad b=-1-\sqrt{3},$$

which are real since the discriminant $4+8=12$ is positive. Their magnitudes are $\sqrt{3}-1\approx 0.73$ and $\sqrt{3}+1\approx 2.73$, neither of which is 1, 2 or 3 (both are irrational). So

$$-1+\sqrt{3}, \quad -1-\sqrt{3}, \quad 1, \quad -2, \quad 3$$

are five real numbers with distinct magnitudes, summing to 0, whose cubes also sum to 0.

## Checks

- In (i), the cube roots of 1, namely $1$, $\omega$, $\omega^2$, satisfy $1+\omega+\omega^2=0$ and $1+\omega^2+\omega^4=1+\omega^2+\omega=0$: all modulus 1, not all real, as predicted.
- In (iii), try $a=2$, $b=-2$, $c=\mathrm{i}$, $d=-\mathrm{i}$: both sums vanish, and indeed the numbers pair off into negatives.
- In (iv), numerically $0.7320-2.7320+1-2+3=0$ and $0.3923-20.3923+1-8+27=0$, using $(\sqrt{3}-1)^3=6\sqrt{3}-10$ and $(-1-\sqrt{3})^3=-6\sqrt{3}-10$.

## Common slips

- In (i), concluding $|a|=|b|$ from $a^3=b^3$ by "taking cube roots": $a$ and $b$ need not be equal, but their moduli are, because moduli are non-negative reals.
- In (iii), forgetting to say *why* $(a+b)(b+c)(c+a)=0$ forces one factor to be zero (there are no zero divisors in $\mathbb{C}$).
- In (iv), not checking that the magnitudes of $a$ and $b$ differ from 1, 2, 3 and from each other.

## Beyond STEP

Every symmetric polynomial in several variables can be written in terms of the elementary symmetric polynomials, and Newton's identities give the link with the power sums $a^k+b^k+c^k+\dots$ explicitly; this is why the conditions here pin the numbers down so tightly. A close cousin is the Prouhet–Tarry–Escott problem: find two different sets of integers whose sums of first, second, …, $k$th powers all agree. It is still partly open. For the algebra of polynomials and their roots, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

# STEP III 2025, Question 7

Pure • Inverse Tangent Functions

**STEP III 2025, Question 7**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Everything turns on one fact: $f(x)\,f(-x) = 1$, because $(\sqrt{x^2+1} - x)(\sqrt{x^2+1} + x) = 1$. Together with $f(x) > 0$, this turns $g(x) + g(-x)$ into the familiar $\tan^{-1} t + \tan^{-1}(1/t)$. For (c), $\tan\frac{1}{2}\theta$ is the half-angle tangent, and with $\tan\theta = 1/x$ it simplifies to $f(x)$. Once g and k are known to agree for $x > 0$, the two integrals in (e) are tied together by symmetry.

## Solution

### Part (i)

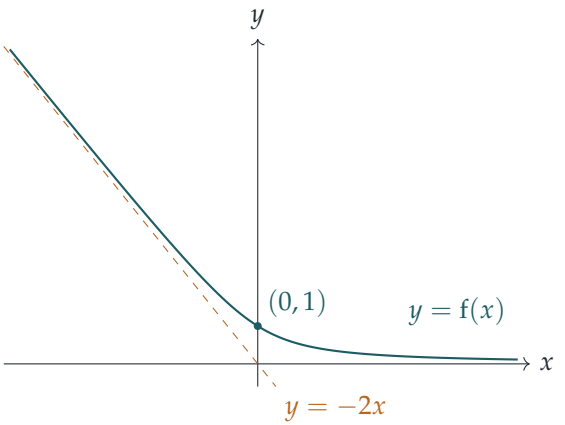
For $x \neq 0$ we have $x^2 = |x|^2$, so

$$\sqrt{x^2+1} = |x|\left(1 + \frac{1}{x^2}\right)^{1/2} = |x|\left(1 + \frac{1}{2x^2} - \frac{1}{8x^4} + \cdots\right) = |x| + \frac{1}{2|x|} - \frac{1}{8|x|^3} + \cdots,$$

using the binomial series, which is valid because $1/x^2 < 1$ when $|x|$ is large. Dropping the terms of order $|x|^{-3}$ gives $\sqrt{x^2+1} \approx |x| + \dfrac{1}{2|x|}$, as required.

*The sketch.* Gather the facts first.

- $f(x) > 0$ for every $x$, since $\sqrt{x^2+1} > \sqrt{x^2} = |x| \geq x$. Also $f(0) = 1$.
- $f'(x) = \dfrac{x}{\sqrt{x^2+1}} - 1 < 0$, since $x < \sqrt{x^2+1}$, so f is decreasing. Moreover $f''(x) = (x^2+1)^{-3/2} > 0$, so the curve is convex.
- As $x \to +\infty$: $f(x) \approx x + \frac{1}{2x} - x = \frac{1}{2x} \to 0^+$, so the $x$-axis is an asymptote.
- As $x \to -\infty$: $|x| = -x$, so $f(x) \approx -2x + \frac{1}{2|x|}$. The line $y = -2x$ is an oblique asymptote, and the curve lies just above it.




**Part (ii)(a)**

Multiplying out a difference of two squares gives

$$\mathrm{f}(x)\,\mathrm{f}(-x) = \left(\sqrt{x^2+1}-x\right)\left(\sqrt{x^2+1}+x\right) = (x^2+1)-x^2 = 1,$$

so $\mathrm{f}(-x) = 1/\mathrm{f}(x)$. Write $t = \mathrm{f}(x)$, which is positive by part (i). Then

$$\mathrm{g}(x)+\mathrm{g}(-x) = \tan^{-1} t + \tan^{-1}\frac{1}{t}.$$

? **Ask yourself** Why does $\tan^{-1} t + \tan^{-1}(1/t) = \frac{1}{2}\pi$ need $t > 0$?

Let $\alpha = \tan^{-1} t$. Since $t > 0$, $\alpha \in (0, \frac{1}{2}\pi)$, so $\frac{1}{2}\pi - \alpha$ is also in $(0, \frac{1}{2}\pi)$, which lies inside the range of $\tan^{-1}$. Its tangent is $\cot\alpha = 1/t$, so $\tan^{-1}(1/t) = \frac{1}{2}\pi - \alpha$. That gives $\mathrm{g}(x)+\mathrm{g}(-x) = \frac{1}{2}\pi$. (For $t < 0$ the sum would be $-\frac{1}{2}\pi$ instead, which is why the positivity of f matters.)

**Part (ii)(b)**

$\tan^{-1}$ is an odd function, so for $x \neq 0$,

$$\mathrm{k}(-x) = \tfrac{1}{2}\tan^{-1}\left(-\frac{1}{x}\right) = -\tfrac{1}{2}\tan^{-1}\frac{1}{x} = -\mathrm{k}(x),$$

which means $\mathrm{k}(x)+\mathrm{k}(-x) = 0$.

**Part (ii)(c)**

Let $x > 0$ and $\theta = \tan^{-1}(1/x)$, so $\theta \in (0, \frac{1}{2}\pi)$ and $\mathrm{k}(x) = \frac{1}{2}\theta$. A right-angled triangle with opposite side 1 and adjacent side $x$ gives $\sin\theta = \dfrac{1}{\sqrt{x^2+1}}$ and $\cos\theta = \dfrac{x}{\sqrt{x^2+1}}$. Using the half-angle identity $\tan\frac{1}{2}\theta = \dfrac{1-\cos\theta}{\sin\theta}$ (from $1-\cos\theta = 2\sin^2\frac{1}{2}\theta$ and $\sin\theta =$

$2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$),

$$\tan \mathrm{k}(x) = \frac{1 - \dfrac{x}{\sqrt{x^2+1}}}{\dfrac{1}{\sqrt{x^2+1}}} = \sqrt{x^2+1} - x = \mathrm{f}(x) = \tan \mathrm{g}(x).$$

This proves the result.

There is more to say. For $x > 0$, $\mathrm{k}(x) \in (0, \frac{1}{4}\pi)$. Also $0 < \mathrm{f}(x) < \mathrm{f}(0) = 1$, so $\mathrm{g}(x) \in (0, \frac{1}{4}\pi)$ too. Since tan is one-to-one on that interval,

$$\mathrm{k}(x) = \mathrm{g}(x) \quad \text{for } x > 0.$$

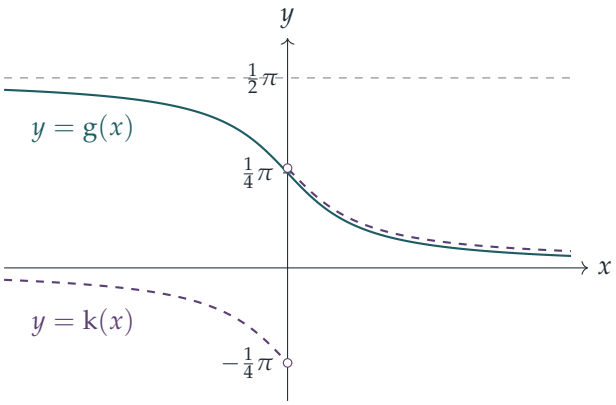
For $x < 0$, we have $-x > 0$, so by (b), then this, then (a):

$$\mathrm{k}(x) = -\mathrm{k}(-x) = -\mathrm{g}(-x) = \mathrm{g}(x) - \tfrac{1}{2}\pi \quad \text{for } x < 0.$$

**Part (ii)(d)**

$\mathrm{g} = \tan^{-1} \circ \mathrm{f}$ is decreasing, because both $\tan^{-1}$ (increasing) and f (decreasing) are monotonic. As $x \to -\infty$, $\mathrm{f} \to \infty$ and $\mathrm{g} \to \frac{1}{2}\pi$. As $x \to +\infty$, $\mathrm{f} \to 0^+$ and $\mathrm{g} \to 0$. Also $\mathrm{g}(0) = \tan^{-1} 1 = \frac{1}{4}\pi$, and by (a) the graph has rotational symmetry about $(0, \frac{1}{4}\pi)$.

From (c), the graph of k is the same as that of g for $x > 0$, falling from $\frac{1}{4}\pi$ (not attained) towards 0. For $x < 0$ it is the graph of g moved down by $\frac{1}{2}\pi$, falling from 0 (as $x \to -\infty$) towards $-\frac{1}{4}\pi$ as $x \to 0^-$. So k is odd and jumps by $\frac{1}{2}\pi$ at $x = 0$, where it is undefined.




(For $x > 0$ the two curves coincide. The dashed curve is drawn slightly above the solid one there only so that both can be seen.)

**Part (ii)(e)**

For $x > 0$, $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}\dfrac{1}{x} = \dfrac{-1/x^2}{1+1/x^2} = -\dfrac{1}{1+x^2}$. Integrating by parts, with $\tan^{-1}(1/x)$ differentiated and 1 integrated,

$$\begin{aligned}\int_0^1 \mathrm{k}(x)\,\mathrm{d}x &= \tfrac{1}{2}\Big[x\tan^{-1}\frac{1}{x}\Big]_0^1 + \tfrac{1}{2}\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x \\ &= \tfrac{1}{2}\cdot\tfrac{1}{4}\pi + \tfrac{1}{4}\Big[\ln(1+x^2)\Big]_0^1 = \frac{\pi}{8} + \frac{1}{4}\ln 2.\end{aligned}$$

At the lower limit, $x\tan^{-1}(1/x) \to 0$ as $x \to 0^+$, since $\tan^{-1}(1/x)$ stays below $\frac{1}{2}\pi$. (The integrand is bounded, so the integral is perfectly proper.)

*Hence.* By (c), $\mathrm{g}(x) = \mathrm{k}(x) + \frac{1}{2}\pi$ for $-1 \le x < 0$. By (b), k is odd, so $\int_{-1}^0 \mathrm{k}(x)\,\mathrm{d}x = -\int_0^1 \mathrm{k}(x)\,\mathrm{d}x$. Therefore

$$\int_{-1}^0 \mathrm{g}(x)\,\mathrm{d}x = \tfrac{1}{2}\pi - \Big(\frac{\pi}{8} + \frac{1}{4}\ln 2\Big) = \frac{3\pi}{8} - \frac{1}{4}\ln 2.$$

## Checks

- On $[-1, 0]$, g decreases from $\mathrm{g}(-1) = \tan^{-1}(\sqrt{2}+1) = \frac{3}{8}\pi \approx 1.178$ to $\frac{1}{4}\pi \approx 0.785$, so the integral must lie between these values. Our answer is $\frac{3}{8}\pi - \frac{1}{4}\ln 2 \approx 1.0048$, and numerical quadrature gives the same to ten places. Likewise $\int_0^1 \mathrm{k}\,\mathrm{d}x \approx 0.56599$.
- Differentiating gives $\mathrm{g}'(x) = -\dfrac{1}{2(1+x^2)}$, so in fact $\mathrm{g}(x) = \frac{1}{4}\pi - \frac{1}{2}\tan^{-1}x$. This formula makes (a) obvious, and integrating it directly reproduces the answer to (e).
- At $x = 1$: $\mathrm{f}(1) = \sqrt{2} - 1 = \tan\frac{1}{8}\pi$, and $\mathrm{k}(1) = \frac{1}{2}\cdot\frac{1}{4}\pi = \frac{1}{8}\pi$, as (c) predicts.

## Common slips

- Writing $\sqrt{x^2+1} \approx x + \frac{1}{2x}$ for all large $|x|$. For negative $x$ this has the wrong sign, and it hides the asymptote $y = -2x$.
- Concluding $\mathrm{k} = \mathrm{g}$ everywhere from (c). Equal tangents only give equal angles up to multiples of $\pi$, and for $x < 0$ the two functions differ by $\frac{1}{2}\pi$.
- Forgetting to justify the limit of $x\tan^{-1}(1/x)$ at 0 in the integration by parts.

## Beyond STEP

The function f is $\mathrm{e}^{-\sinh^{-1}x}$, because $\sinh^{-1}x = \ln\big(x + \sqrt{x^2+1}\big)$. Its reciprocal property $\mathrm{f}(-x) = 1/\mathrm{f}(x)$ is just the oddness of $\sinh^{-1}$. The trap in (c), where equal tangents do

not force equal angles, is the real-variable version of the branch problem for inverse functions. It reappears more seriously with complex logarithms and arguments. Needham's *Visual Complex Analysis* is a beautiful place to see this, and Abbott's *Understanding Analysis* treats the limits and improper integrals carefully.

WORKED SOLUTION

# STEP III 2025, Question 8

Pure • Trigonometric Sums via Complex Numbers

**STEP III 2025, Question 8**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

The identity in (i) is a recurrence: it writes $z^{m+1} - z^{-(m+1)}$ in terms of the same kind of expression two steps lower, so applying it repeatedly makes a sum telescope out. With $z = \mathrm{e}^{\mathrm{i}\theta}$ every $z^k \pm z^{-k}$ becomes $2\cos k\theta$ or $2\mathrm{i}\sin k\theta$, turning the algebra into a trigonometric identity. The numerical parts then come from choosing $\theta$ so that $\sin 2n\theta$ equals $\sin\theta$ (or $\cos\theta$) and cancels.

## Solution

### Part (i)

Expanding the product,

$$\left(z - \frac{1}{z}\right)\left(z^m + \frac{1}{z^m}\right) = z^{m+1} + z^{1-m} - z^{m-1} - z^{-(m+1)}.$$

Adding $z^{m-1} - z^{1-m}$ cancels the two middle terms and leaves $z^{m+1} - z^{-(m+1)}$, which is the required identity.

*Induction.* For $n = 1$ the claim is $z^2 - z^{-2} = (z - z^{-1})(z + z^{-1})$, the difference of two squares, so it holds. Suppose it holds for some $n \geq 1$. Using the identity with $m = 2n + 1$ (so $m + 1 = 2n + 2$ and $m - 1 = 2n$),

$$\begin{aligned} z^{2n+2} - \frac{1}{z^{2n+2}} &= \left(z - \frac{1}{z}\right)\left(z^{2n+1} + \frac{1}{z^{2n+1}}\right) + \left(z^{2n} - \frac{1}{z^{2n}}\right) \\ &= \left(z - \frac{1}{z}\right)\left(z^{2n+1} + \frac{1}{z^{2n+1}}\right) + \left(z - \frac{1}{z}\right)\sum_{r=1}^{n}\left(z^{2r-1} + \frac{1}{z^{2r-1}}\right) \\ &= \left(z - \frac{1}{z}\right)\sum_{r=1}^{n+1}\left(z^{2r-1} + \frac{1}{z^{2r-1}}\right), \end{aligned}$$

by the inductive hypothesis, since the extra term is the $r = n + 1$ term. So the result holds for $n + 1$, and by induction for all $n \geq 1$.

*The version with* $z+\frac{1}{z}$. In the same way,

$$\Big(z+\frac{1}{z}\Big)\Big(z^m-\frac{1}{z^m}\Big)=z^{m+1}-z^{1-m}+z^{m-1}-z^{-(m+1)},$$

so this time we must *subtract* $z^{m-1}-z^{1-m}$:

$$z^{m+1}-\frac{1}{z^{m+1}}=\Big(z+\frac{1}{z}\Big)\Big(z^m-\frac{1}{z^m}\Big)-\Big(z^{m-1}-\frac{1}{z^{m-1}}\Big).$$

The minus sign makes the signs alternate as we step down, and we claim

$$z^{2n}-\frac{1}{z^{2n}}=\Big(z+\frac{1}{z}\Big)\sum_{r=1}^{n}(-1)^{n-r}\Big(z^{2r-1}-\frac{1}{z^{2r-1}}\Big).$$

For $n=1$ this is again $(z+z^{-1})(z-z^{-1})$. If it holds for $n$, the identity with $m=2n+1$ gives

$$z^{2n+2}-\frac{1}{z^{2n+2}}=\Big(z+\frac{1}{z}\Big)\Big[\Big(z^{2n+1}-\frac{1}{z^{2n+1}}\Big)-\sum_{r=1}^{n}(-1)^{n-r}\Big(z^{2r-1}-\frac{1}{z^{2r-1}}\Big)\Big],$$

and since $-(-1)^{n-r}=(-1)^{n+1-r}$, while the first term is the $r=n+1$ term with sign $(-1)^0=1$, the bracket is exactly $\sum_{r=1}^{n+1}(-1)^{n+1-r}\big(z^{2r-1}-z^{-(2r-1)}\big)$. This completes the induction. Written out,

$$z^{2n}-\frac{1}{z^{2n}}=\Big(z+\frac{1}{z}\Big)\Big[\Big(z^{2n-1}-\frac{1}{z^{2n-1}}\Big)-\Big(z^{2n-3}-\frac{1}{z^{2n-3}}\Big)+\cdots\pm\Big(z-\frac{1}{z}\Big)\Big].$$

**Part (ii)**

**(a)** With $z=\mathrm{e}^{\mathrm{i}\theta}$ we have $z^k+z^{-k}=2\cos k\theta$ and $z^k-z^{-k}=2\mathrm{i}\sin k\theta$. The first result of (i) becomes

$$2\mathrm{i}\sin 2n\theta=2\mathrm{i}\sin\theta\sum_{r=1}^{n}2\cos(2r-1)\theta,$$

and dividing by 2i gives $\sin 2n\theta=2\sin\theta\sum_{r=1}^{n}\cos(2r-1)\theta$, as required.

**(b)** With $n=2$: $\sin 4\theta=2\sin\theta(\cos\theta+\cos 3\theta)$. We want $\sin 4\theta$ to equal $\sin\theta$, and $\theta=\frac{1}{5}\pi$ does this, because $\sin\frac{4}{5}\pi=\sin(\pi-\frac{1}{5}\pi)=\sin\frac{1}{5}\pi$. Since $\sin\frac{1}{5}\pi\neq 0$ we may divide by it:

$$1=2\big(\cos\tfrac{1}{5}\pi+\cos\tfrac{3}{5}\pi\big).$$

Now $\cos\frac{3}{5}\pi=-\cos(\pi-\frac{3}{5}\pi)=-\cos\frac{2}{5}\pi$, so $\cos\frac{1}{5}\pi-\cos\frac{2}{5}\pi=\frac{1}{2}$, that is, $\cos\frac{2}{5}\pi=\cos\frac{1}{5}\pi-\frac{1}{2}$.

**(c)** With $n=7$ and $\theta=\frac{1}{15}\pi$, again $\sin 14\theta=\sin(\pi-\theta)=\sin\theta\neq 0$, so

$$\sum_{r=1}^{7}\cos\frac{(2r-1)\pi}{15}=\frac{1}{2}.$$

**Ask yourself** The seven angles here are odd multiples of $\frac{1}{15}\pi$, but the target has even ones. How do we get from one to the other?

Use $\cos(\pi-\alpha) = -\cos\alpha$ on the angles beyond $\frac{1}{2}\pi$, and note $\cos\frac{5}{15}\pi = \cos\frac{1}{3}\pi = \frac{1}{2}$, and $\cos\frac{3}{15}\pi = \cos\frac{1}{5}\pi$:

$$\cos\tfrac{9}{15}\pi = -\cos\tfrac{2}{5}\pi, \qquad \cos\tfrac{11}{15}\pi = -\cos\tfrac{4}{15}\pi, \qquad \cos\tfrac{13}{15}\pi = -\cos\tfrac{2}{15}\pi.$$

The sum becomes

$$\cos\tfrac{1}{15}\pi + \cos\tfrac{7}{15}\pi - \cos\tfrac{2}{15}\pi - \cos\tfrac{4}{15}\pi + \tfrac{1}{2} + \left(\cos\tfrac{1}{5}\pi - \cos\tfrac{2}{5}\pi\right) = \tfrac{1}{2}.$$

By (b) the bracket is $\frac{1}{2}$, so

$$\cos\tfrac{2}{15}\pi + \cos\tfrac{4}{15}\pi - \cos\tfrac{7}{15}\pi - \cos\tfrac{1}{15}\pi = \tfrac{1}{2}.$$

Finally $-\cos\frac{7}{15}\pi = \cos(\pi - \frac{7}{15}\pi) = \cos\frac{8}{15}\pi$ and $-\cos\frac{1}{15}\pi = \cos(\pi + \frac{1}{15}\pi) = \cos\frac{16}{15}\pi$, giving

$$\cos\tfrac{2}{15}\pi + \cos\tfrac{4}{15}\pi + \cos\tfrac{8}{15}\pi + \cos\tfrac{16}{15}\pi = \tfrac{1}{2},$$

as required.

#### Part (iii)

**Ask yourself** The target is an alternating sum of sines of odd multiples of $\theta$. Which result from (i) produces exactly that?

The $z + \frac{1}{z}$ version. With $z = \mathrm{e}^{\mathrm{i}\theta}$ it becomes

$$2\mathrm{i}\sin 2n\theta = 2\cos\theta \sum_{r=1}^{n} (-1)^{n-r}\, 2\mathrm{i}\sin(2r-1)\theta,$$

so

$$\sin 2n\theta = 2\cos\theta \sum_{r=1}^{n} (-1)^{n-r} \sin(2r-1)\theta.$$

Take $n = 3$:

$$\sin 6\theta = 2\cos\theta\,(\sin 5\theta - \sin 3\theta + \sin\theta).$$

Now we want $\sin 6\theta = \cos\theta$, and $\theta = \frac{1}{14}\pi$ works: $\sin\frac{6}{14}\pi = \cos(\frac{1}{2}\pi - \frac{6}{14}\pi) = \cos\frac{1}{14}\pi$, which is non-zero. Dividing by it,

$$\sin\tfrac{1}{14}\pi - \sin\tfrac{3}{14}\pi + \sin\tfrac{5}{14}\pi = \tfrac{1}{2}.$$

## Checks

- Test the identities at a simple value: $z = 2$, $n = 2$ gives $16 - \frac{1}{16} = \frac{255}{16}$, while $(2-\frac{1}{2})(2+\frac{1}{2}+8+\frac{1}{8}) = \frac{3}{2}\cdot\frac{85}{8} = \frac{255}{16}$ and $(2+\frac{1}{2})(8-\frac{1}{8}-2+\frac{1}{2}) = \frac{5}{2}\cdot\frac{51}{8} = \frac{255}{16}$.

- Numerically, $\cos\frac{2}{5}\pi = 0.309017\ldots$ and $\cos\frac{1}{5}\pi - \frac{1}{2} = 0.809017\ldots - 0.5$ agree; this is the familiar $\cos\frac{1}{5}\pi = \frac{1+\sqrt{5}}{4}$, $\cos\frac{2}{5}\pi = \frac{\sqrt{5}-1}{4}$.
- The sums in (ii)(c) and (iii) both evaluate to 0.500000 on a calculator.

## Common slips

- Getting the sign wrong in the $z + \frac{1}{z}$ recurrence: the lower term is subtracted, so the sum alternates and must end with the $+$ sign on the top term $z^{2n-1} - z^{-(2n-1)}$.
- Dividing by $\sin\theta$ or $\cos\theta$ without noting it is non-zero at the chosen angle.
- In (ii)(c), forgetting that $\cos\frac{5}{15}\pi = \frac{1}{2}$ and that (b) is needed to clear the remaining pair of terms.

## Beyond STEP

The expressions $z^k + z^{-k}$ with $z = \mathrm{e}^{\mathrm{i}\theta}$ are $2\cos k\theta$, and recurrences like the one in (i) are exactly those satisfied by Chebyshev polynomials, which turn $\cos k\theta$ and $\sin k\theta / \sin\theta$ into polynomials in $\cos\theta$. The neat sums in (ii)(c) and (iii) come from roots of unity: the angles $\frac{2}{15}\pi$, $\frac{4}{15}\pi$, $\frac{8}{15}\pi$, $\frac{16}{15}\pi$ are generated by repeated doubling, a first glimpse of Gauss's periods and Galois theory. Needham's *Visual Complex Analysis* is a beautiful place to see this geometry.

WORKED SOLUTION

# STEP III 2025, Question 9

Mechanics • Centre of Mass

**STEP III 2025, Question 9**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.8, *Tilted Equilibria*: https://beyondstep.keithmcnulty.org/ch11/tilted-equilibria.html

## The key idea

A solid resting on a table touches it at one point, and only two forces act: the weight at the centre of mass $G$ and the reaction at the contact point. They can balance only if they act along the same vertical line, so the normal to the surface at the contact point must pass through $G$. Part (i) locates $G$, part (ii) tells us where normals to the curved surface meet the axis, and part (iii) puts the two together: we count the points on the surface whose normals pass through $G$.

## Solution

Throughout, write the curve as $y = \dfrac{r^n - x^n}{r^{n-1}}$ for $0 \le x \le r$. It runs from $(0, r)$ on the $y$-axis down to $(r, 0)$ on the $x$-axis, with $y \ge 0$ in between.

### Part (i)

We cut the solid into thin cylindrical shells about the $y$-axis. The shell at radius $x$, of thickness $\delta x$ and height $y$, has volume $2\pi x y\,\delta x$, and since the density is uniform its centre of mass is at height $\frac{1}{2}y$. So the volume and the moment about the $xz$-plane are

$$V = 2\pi \int_0^r xy \,\mathrm{d}x, \qquad V\bar{y} = 2\pi \int_0^r x \cdot \tfrac{1}{2}y^2 \,\mathrm{d}x.$$

The first integral is

$$\int_0^r x\,\frac{r^n - x^n}{r^{n-1}}\,\mathrm{d}x = \frac{1}{r^{n-1}}\left(\frac{r^{n+2}}{2} - \frac{r^{n+2}}{n+2}\right) = \frac{n\,r^3}{2(n+2)},$$

so $V = \dfrac{\pi n r^3}{n+2}$. For the second, expand $(r^n - x^n)^2$:

$$\int_0^r x\,\frac{(r^n - x^n)^2}{r^{2n-2}}\,\mathrm{d}x = \frac{r^{2n+2}}{r^{2n-2}}\left(\frac{1}{2} - \frac{2}{n+2} + \frac{1}{2n+2}\right)$$
$$= r^4 \cdot \frac{(n+2)^2 - 4(n+1)}{2(n+1)(n+2)} = \frac{n^2 r^4}{2(n+1)(n+2)},$$

so $V\bar{y} = \pi \cdot \dfrac{n^2 r^4}{2(n+1)(n+2)}$. Dividing,

$$\bar{y} = \frac{\pi n^2 r^4}{2(n+1)(n+2)} \cdot \frac{n+2}{\pi n r^3} = \frac{nr}{2(n+1)},$$

as required. By symmetry the centre of mass lies on the $y$-axis.

**Part (ii)**

Differentiating, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{nx^{n-1}}{r^{n-1}}$, which at $x = rp$ is $-np^{n-1}$. This is non-zero since $p > 0$, so the normal has gradient $\dfrac{1}{np^{n-1}}$ and equation

$$y - r(1 - p^n) = \frac{x - rp}{np^{n-1}}.$$

Putting $x = 0$,

$$Y = r(1 - p^n) - \frac{rp}{np^{n-1}} = r\left(1 - p^n - \frac{1}{np^{n-2}}\right),$$

which is the required result.

For $n = 4$, $Y(p) = r\left(1 - p^4 - \frac{1}{4}p^{-2}\right)$ and

$$\frac{\mathrm{d}Y}{\mathrm{d}p} = r\left(-4p^3 + \frac{1}{2p^3}\right) = \frac{r(1 - 8p^6)}{2p^3}.$$

**Ask yourself** How do we know the stationary point gives the greatest value, and not just a local one?

For $p > 0$ the sign of $\mathrm{d}Y/\mathrm{d}p$ is the sign of $1 - 8p^6$: positive for $p^6 < \frac{1}{8}$ and negative for $p^6 > \frac{1}{8}$. So $Y$ increases up to $p^6 = \frac{1}{8}$, that is $p^2 = \frac{1}{2}$, $p = 1/\sqrt{2}$ (which lies in $(0, 1]$), and decreases after it. That makes it the greatest value on the whole interval:

$$Y_{\max} = r\left(1 - \tfrac{1}{4} - \tfrac{1}{4} \cdot 2\right) = \tfrac{1}{4}r.$$

**Part (iii)**

**The two pieces.** The upper boundary $r^3 y = r^4 - x^4$ is the curve of part (i) with $n = 4$, so the top piece (above $y = 0$) is the solid of part (i) with $n = 4$. The lower boundary

$ry = -(r^2 - x^2)$ is the reflection in the $x$-axis of $ry = r^2 - x^2$, the $n = 2$ curve, so the bottom piece is the $n = 2$ solid turned upside down. The two meet in the circular rim of radius $r$ in the plane $y = 0$. Using part (i), with $V = \pi n r^3/(n+2)$:

| | volume | height of centre of mass |
|---|---|---|
| top ($n = 4$) | $\frac{2}{3}\pi r^3$ | $\frac{4r}{10} = \frac{2}{5}r$ |
| bottom ($n = 2$, reflected) | $\frac{1}{2}\pi r^3$ | $-\frac{2r}{6} = -\frac{1}{3}r$ |

Taking moments (assuming, as before, uniform density),

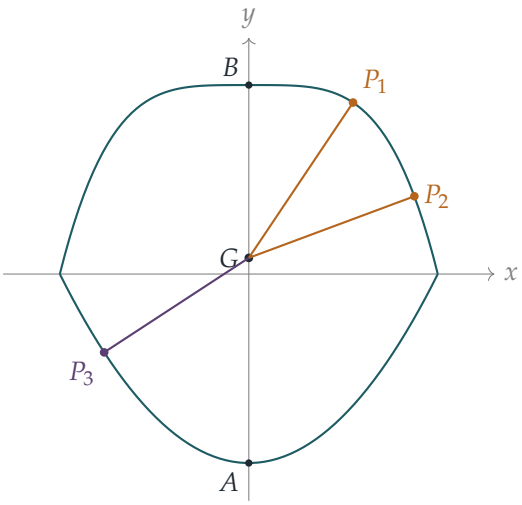
$$\bar{y} = \frac{\frac{2}{3}\cdot\frac{2}{5}r - \frac{1}{2}\cdot\frac{1}{3}r}{\frac{2}{3}+\frac{1}{2}} = \frac{\frac{1}{10}r}{\frac{7}{6}} = \frac{3r}{35},$$

so $G = \left(0, \frac{3}{35}r\right)$, a little above the rim.

**The equilibrium condition.**

? **Ask yourself** Why must the normal at the point of contact pass through $G$?

Resting on the surface at a single point $P$, the solid feels only its weight, acting vertically at $G$, and the reaction from the surface, acting at $P$. Two forces balance only if they are equal, opposite and along the same line, so $G$ must be vertically above $P$. The surface is the tangent plane at $P$, so the vertical through $P$ is the normal to the solid there: the normal at $P$ must pass through $G$. Conversely, if it does, the solid can rest there, because the solid is convex (the profile $y = r - x^4/r^3$ is concave and $y = (x^2 - r^2)/r$ is convex), so it lies entirely on one side of its tangent plane at $P$. The normal to a surface of revolution lies in a plane through the axis, so we only need to look at the cross-section in the $xy$-plane. Since $G$ is on the axis, we need normals that meet the $y$-axis at $Y = \frac{3}{35}r$.



**Contact on the top surface.** By part (ii) with $n = 4$ (and $r(1-p^4)$ the height of the point), the normal at $\big(rp, r(1-p^4)\big)$ meets the axis at $r\big(1-p^4-\frac{1}{4}p^{-2}\big)$. Setting this equal to $\frac{3}{35}r$ and multiplying by $140p^2$:

$$140p^2 - 140p^6 - 35 = 12p^2 \iff h(q) := 140q^3 - 128q + 35 = 0, \quad q = p^2.$$

Now $h(0) = 35 > 0$, $h(\frac{1}{2}) = 17.5 - 64 + 35 = -11.5 < 0$ and $h(1) = 47 > 0$. So $h$ has a root $q_1 \in (0, \frac{1}{2})$ and a root $q_2 \in (\frac{1}{2}, 1)$, giving two points $P_1, P_2$ on the top surface with $0 < p_1 < 1/\sqrt{2} < p_2 < 1$. (This is what part (ii) predicts: $\frac{3}{35}r$ is below the greatest value $\frac{1}{4}r$, so the level is reached once on each side of the maximum.)

**Contact on the bottom surface.** For the $n = 2$ curve the normal at parameter $p$ meets the axis at $r(1-p^2-\frac{1}{2}) = r(\frac{1}{2}-p^2)$; reflecting in the $x$-axis, the normal to the bottom surface at $\big(rp, -r(1-p^2)\big)$ meets the axis at $r(p^2-\frac{1}{2})$. Setting this equal to $\frac{3}{35}r$ gives

$$p^2 = \tfrac{1}{2} + \tfrac{3}{35} = \tfrac{41}{70}, \qquad p_3 = \sqrt{41/70} \in (0,1),$$

a third point $P_3$ (taking the positive root, since $p > 0$ on the half-section we are using; the figure shows its mirror image, which is equivalent by symmetry).

**The three angles are different and non-zero.** When the solid rests at $P$, the upward vertical points from $P$ towards $G$, that is, along the inward normal at $P$.

- At $P_3$ on the bottom, the inward normal $(-2p_3, 1)$ (the curve has gradient $2p_3$) points upwards, making the acute angle $\arctan(2p_3)$ with $\overrightarrow{AB}$, which points along the positive $y$-axis.
- At $P_1$ or $P_2$ on the top, the inward normal $(-4p^3, -1)$ points downwards, so the angle with $\overrightarrow{AB}$ is the obtuse angle $180^\circ - \arctan(4p^3)$.

Since $p_1 \neq p_2$ and $\arctan(4p^3)$ is strictly increasing, the two obtuse angles differ, and both differ from the acute one. None is zero, because $p_1, p_2, p_3 > 0$. So the solid can rest in equilibrium with $\overrightarrow{AB}$ at three different non-zero angles to the upward vertical, as required.

(There are in fact more: the solid can also balance on its rim, or on $B$ with $\overrightarrow{AB}$ pointing straight down. The three curved-surface positions are enough for the question.)

## Checks

- Part (i) with $n = 2$ gives $\frac{1}{3}r$ for a paraboloid of height $r$, the standard result that a solid paraboloid's centre of mass is two-thirds of the way from its vertex to its base. As $n \to \infty$ the solid becomes a cylinder of height $r$, and $\bar{y} \to \frac{1}{2}r$, as it should.
- In part (ii), at $p = 1$ (the rim) with $n = 4$ we get $Y = -\frac{1}{4}r$; the gradient there is $-4$, so the normal has gradient $\frac{1}{4}$ and drops $\frac{1}{4}r$ over a horizontal distance $r$. It agrees.

- With $r = 1$ the roots are $p_1 \approx 0.552$, $p_2 \approx 0.876$, $p_3 \approx 0.765$, and the angles are about $146.1^\circ$, $110.4^\circ$ and $56.8^\circ$. A computer search for the tilts at which the height of $G$ above the table is stationary found exactly these three strictly between $0^\circ$ and $180^\circ$, together with the rim position (about $85.1^\circ$).

## Common slips

- Placing the bottom piece's centre of mass at $+\frac{1}{3}r$: it is the $n = 2$ solid turned upside down, so its centre of mass is below the rim.
- Using disc volumes $\pi x^2\,\delta y$ with $x$ still in terms of $y$ to the power $\frac{2}{n}$: it works, but the shell method gives polynomial integrals.
- Finding the cubic in $q = p^2$ but not justifying that it has two roots with $0 < p < 1$; a sign check at $q = 0, \frac{1}{2}, 1$ settles it.

## Beyond STEP

The condition that the normal passes through $G$ makes the height of $G$ stationary as the body rolls, and that height is the potential energy. Stable positions are its minima and unstable ones its maxima. A smooth convex body must have at least two equilibria, and the "Gömböc", discovered in 2006, is a convex shape with only one stable and one unstable position. The general machinery of potential energy and stability is in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

# STEP III 2025, Question 10

Mechanics • Friction on Moving Plank

**STEP III 2025, Question 10**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Solve the differential equation for h first: it is simple harmonic motion, and it says that the plank's angle of inclination grows at the steady rate $\omega$. Once the angle is known, the particle's position is a fixed fraction of the way along the plank, so differentiating twice gives its acceleration. The forces then follow from Newton's second law, resolved perpendicular to and along the plank, which is much tidier than resolving horizontally and vertically.

## Solution

We model the plank as a rigid rod and $P$ as a particle. Let $\theta$ be the angle the plank makes with the horizontal.

### Part (i)

The equation $\ddot{\mathrm{h}} = -\omega^2 \mathrm{h}$ has general solution $\mathrm{h} = a\cos\omega t + b\sin\omega t$. The condition $\mathrm{h}(0) = 0$ gives $a = 0$, and $\dot{\mathrm{h}}(0) = b\omega = \omega L$ gives $b = L$, so

$$\mathrm{h}(t) = L\sin\omega t.$$

Since $A$ is at height h and the plank has length $L$, $\sin\theta = \mathrm{h}/L = \sin\omega t$, so $\theta = \omega t$ (for $0 \le \omega t \le \frac{\pi}{2}$, which is the whole of the motion: after that $A$ would have to rise above height $L$). Thus $A = (0, L\sin\omega t)$ and $B = (L\cos\omega t, 0)$.

The point $C$ is the fraction $s$ of the way from $A$ to $B$, so $C = A + s(B - A)$:

$$x = sL\cos\omega t, \qquad y = (1-s)L\sin\omega t.$$

The first of these is the required $x$-coordinate, and the second is the matching expression for the $y$-coordinate.

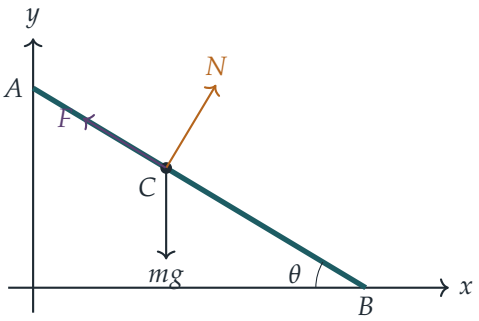
**Part (ii)**

Differentiating twice with respect to $t$,

$$\ddot{x} = -sL\omega^2 \cos \omega t, \qquad \ddot{y} = -(1-s)L\omega^2 \sin \omega t.$$

**Part (iii)**

The forces on $P$ are its weight $mg$ downwards, the normal reaction $N$ perpendicular to the plank (upwards), and friction $F$ along the plank, which we take as positive when it points up the slope, towards $A$.




The unit vector perpendicular to the plank (upwards) is $(\sin\theta, \cos\theta)$, and the unit vector along the plank towards $A$ is $(-\cos\theta, \sin\theta)$, with $\theta = \omega t$. Resolving the equation of motion $m(\ddot{x}, \ddot{y}) = N(\sin\theta, \cos\theta) + F(-\cos\theta, \sin\theta) + (0, -mg)$ in these two directions (that is, taking the dot product with each unit vector):

*Perpendicular to the plank.*

$$\begin{aligned} N - mg\cos\theta &= m(\ddot{x}\sin\theta + \ddot{y}\cos\theta) \\ &= -mL\omega^2\big(s + (1-s)\big)\sin\theta\cos\theta = -mL\omega^2 \sin\theta\cos\theta. \end{aligned}$$

With $L\omega^2 = kg$, this gives

$$N = mg\cos\theta - mgk\sin\theta\cos\theta = mg(1 - k\sin\omega t)\cos\omega t,$$

as required.

*Along the plank, towards $A$.*

$$\begin{aligned} F - mg\sin\theta &= m(-\ddot{x}\cos\theta + \ddot{y}\sin\theta) \\ &= mL\omega^2\big(s\cos^2\theta - (1-s)\sin^2\theta\big) = mgk\big(s - \sin^2\theta\big), \end{aligned}$$

using $\cos^2\theta + \sin^2\theta = 1$. Hence

$$F = mgsk + mg\sin\theta(1 - k\sin\theta) = mgsk + \underbrace{mg(1 - k\sin\theta)\cos\theta}_{N}\tan\theta,$$

that is, $F = mgsk + N\tan\omega t$, as required. Notice that $F$ comes out positive whenever $N > 0$, so friction always acts up the plank while $P$ is pressed against it.

**Part (iv)**

Write $\mu = \tan\alpha$. While $N > 0$, the particle stays at $C$ without slipping exactly when $|F| \leq \mu N$; since $F > 0$ there, this is $F/N \leq \tan\alpha$. Dividing the result of part (iii) by $N$,

$$\frac{F}{N} = \frac{sk}{(1 - k\sin\theta)\cos\theta} + \tan\theta.$$

*Initially*, $\theta = 0$, so $N = mg > 0$ and $F/N = sk$. So the particle does not slip at the start provided $sk < \tan\alpha$.

> **Ask yourself** Why must the ratio $F/N$ eventually reach $\tan\alpha$ before $N$ vanishes?

Now suppose $sk < \tan\alpha$. The factor $\cos\theta$ vanishes at $\theta = \frac{\pi}{2}$, and if $k > 1$ the factor $1 - k\sin\theta$ vanishes earlier, at $\sin\theta = 1/k$. Let $\theta_0$ be the first angle at which $N = 0$ (either $\frac{\pi}{2}$ or $\arcsin(1/k)$). On $0 \leq \theta < \theta_0$ we have $N > 0$, and $F/N$ is continuous. As $\theta \to \theta_0^-$, the denominator $(1 - k\sin\theta)\cos\theta$ tends to 0 through positive values while the numerator $sk$ is a fixed positive number, so the first term tends to $+\infty$ (and $\tan\theta$ is either bounded or also tends to $+\infty$). Hence $F/N \to +\infty$.

So $F/N$ starts at $sk < \tan\alpha$ and grows without bound before $N$ reaches zero. By the intermediate value theorem there is a first angle $\theta_1 < \theta_0$ at which $F/N = \tan\alpha$, and the particle begins to slip there. This happens at an angle where $N$ is still positive, which is the first claim.

At this angle, since $sk > 0$ and $(1 - k\sin\theta_1)\cos\theta_1 > 0$,

$$\tan\alpha = \frac{F}{N} = \frac{sk}{(1 - k\sin\theta_1)\cos\theta_1} + \tan\theta_1 > \tan\theta_1.$$

Both $\alpha$ and $\theta_1$ lie in $[0, \frac{\pi}{2})$, where tan is increasing, so $\theta_1 < \alpha$: the plank makes an angle less than $\alpha$ with the horizontal when slipping starts. This is the second claim.

The second claim is less surprising than it looks. A particle at rest on a fixed plank slips exactly when the plank's angle reaches $\alpha$, the angle of friction. Here the plank is accelerating, and the extra force needed to carry the particle along with it (the $mgsk$ term) means friction gives out sooner.

## Checks

- At $t = 0$ the plank is horizontal and the particle's acceleration is $(-sL\omega^2, 0)$, horizontal. Then $N = mg$ and $F = msL\omega^2 = mgsk$ should provide exactly that acceleration, and they do.
- If $\omega \to 0$ (so $k \to 0$), the plank is raised very slowly; then $N \to mg\cos\theta$ and $F \to mg\sin\theta$, the familiar equilibrium values on a fixed slope, and slipping happens at $\theta = \alpha$, the boundary of our inequality.
- Numerically, with $s = \frac{1}{2}$, $k = \frac{1}{2}$ and $\alpha = \frac{\pi}{4}$ (so $sk = \frac{1}{4} < 1$), solving $F/N = 1$ shows slipping at $\theta \approx 0.545$ rad, about $31°$, where $N \approx 0.63mg > 0$ and $\theta < 45°$.

A computer check over thousands of random choices of $s$, $k$ and $\alpha$ with $sk < \tan\alpha$ found slipping at an angle below $\alpha$ with $N > 0$ every time.

## Common slips

- Taking $\theta$ to be the angle at $A$ instead of at $B$: then the sines and cosines swap, and the given expressions will not come out.
- Resolving horizontally and vertically and then trying to solve two simultaneous equations with $\sin\theta$ and $\cos\theta$ everywhere. Resolving along and perpendicular to the plank separates $N$ and $F$ immediately.
- In part (iv), arguing only from $\theta$ approaching $\frac{\pi}{2}$ and forgetting that when $k > 1$ the normal reaction vanishes earlier, at $\sin\theta = 1/k$.

## Beyond STEP

This question is about forces in a moving frame: from the plank's point of view, the particle feels an extra "fictitious" force equal to $-m$ times the acceleration of the point $C$, and the term $mgsk$ is exactly that inertial force at the start. At university, this idea is developed into the full theory of accelerating and rotating reference frames, including the centrifugal and Coriolis forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames with many worked examples.

WORKED SOLUTION

# STEP III 2025, Question 11

Probability and Statistics • Distribution of Maximum

**STEP III 2025, Question 11**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.25, *The Largest of Many*: https://beyondstep.keithmcnulty.org/ch12/extremes.html

## The key idea

The largest of the $X_i$ is at most $y$ exactly when *every* $X_i$ is at most $y$, and independence turns that into a product. After that the question is all about reading quantiles off $G$ and then seeing what happens to them for large $n$. The single approximation that does the work is $\alpha^{1/n} = \mathrm{e}^{(\ln\alpha)/n} \approx 1 + \frac{\ln\alpha}{n}$, because $(\ln\alpha)/n$ is small when $n$ is large.

## Solution

### Part (i)

For $x \geq 0$,

$$F(x) = \int_0^x \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1 - \mathrm{e}^{-\lambda x}.$$

Since $Y = \max(X_1, \ldots, X_n)$, the event $\{Y \leq y\}$ is the same as the event that $X_i \leq y$ for every $i$. The $X_i$ are independent, so for $y \geq 0$

$$G(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(X_1 \leq y) \cdots \mathrm{P}(X_n \leq y) = F(y)^n = \left(1 - \mathrm{e}^{-\lambda y}\right)^n,$$

as required.

**Part (ii)**

$Y$ is a continuous random variable, so $\mathrm{P}(Y < y) = \mathrm{P}(Y \leq y) = G(y)$. The condition on $L(\alpha)$ is therefore $G(L(\alpha)) = \alpha$:

$$\begin{aligned}\left(1 - \mathrm{e}^{-\lambda L}\right)^n = \alpha &\iff 1 - \mathrm{e}^{-\lambda L} = \alpha^{1/n} \qquad \text{(taking the positive } n\text{th root)} \\ &\iff \mathrm{e}^{-\lambda L} = 1 - \alpha^{1/n} \\ &\iff L(\alpha) = -\frac{1}{\lambda}\ln\left(1 - \alpha^{1/n}\right).\end{aligned}$$

The logarithm is defined because $0 < \alpha < 1$ gives $0 < \alpha^{1/n} < 1$.

For $U(\alpha)$, the condition $\mathrm{P}(Y > U) = \alpha$ means $G(U) = 1 - \alpha$, so the same steps with $\alpha$ replaced by $1 - \alpha$ give

$$U(\alpha) = -\frac{1}{\lambda}\ln\left(1 - (1-\alpha)^{1/n}\right).$$

Since $\alpha \leq \frac{1}{2} \leq 1 - \alpha$ and $G$ is increasing, $L(\alpha) \leq U(\alpha)$, with equality only when $\alpha = \frac{1}{2}$.

**Part (iii)**

Write $\alpha^{1/n} = \mathrm{e}^{(\ln\alpha)/n}$. For fixed $\alpha$ and large $n$, $t = (\ln\alpha)/n$ is small, so $\mathrm{e}^t \approx 1 + t$ gives

$$1 - \alpha^{1/n} \approx -\frac{\ln\alpha}{n} = \frac{1}{n}\ln\left(\frac{1}{\alpha}\right).$$

Note that this is positive, since $\alpha < 1$. Substituting into part (ii),

$$\lambda L(\alpha) = -\ln\left(1 - \alpha^{1/n}\right) \approx -\ln\left(\frac{1}{n}\ln\frac{1}{\alpha}\right) = \ln(n) - \ln\left(\ln\frac{1}{\alpha}\right).$$

**Ask yourself** Is it safe to take the logarithm of an approximation? An error in $1 - \alpha^{1/n}$ could be large compared with $1 - \alpha^{1/n}$ itself, which is tiny.

What matters is the *relative* error. With $t = (\ln\alpha)/n$ we have $1 - \mathrm{e}^t = -t\,(1 + \frac{t}{2} + \cdots)$, so the ratio of the exact value to the approximation is $1 + O(1/n)$. Taking logarithms, the error in $\lambda L(\alpha)$ is $\ln(1 + O(1/n))$, which tends to 0. So the approximation is not only good for large $n$; the difference between the two sides genuinely tends to zero.

**Part (iv)**

The median $m$ of $Y$ satisfies $G(m) = \frac{1}{2}$, so $m = L(\frac{1}{2})$ (which is also $U(\frac{1}{2})$). By part (iii),

$$\lambda m \approx \ln n - \ln(\ln 2).$$

Since $\ln(\ln 2)$ is a fixed number (about $-0.37$) and $\ln n \to \infty$, the median tends to infinity as $n$ increases.

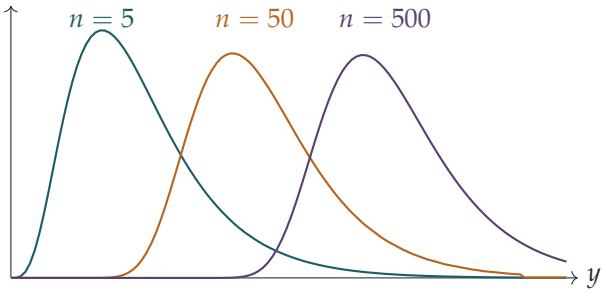
For $U(\alpha)$, replace $\alpha$ by $1-\alpha$ in part (iii):

$$\lambda U(\alpha) \approx \ln n - \ln\left(\ln \frac{1}{1-\alpha}\right).$$

Subtracting, the $\ln n$ terms cancel:

$$U(\alpha) - L(\alpha) \approx \frac{1}{\lambda}\left[\ln\left(\ln \frac{1}{\alpha}\right) - \ln\left(\ln \frac{1}{1-\alpha}\right)\right].$$

Since the errors in both approximations tend to zero (as in the box above), the width tends to this value, which depends on $\alpha$ and $\lambda$ but not on $n$. So as $n$ grows, the distribution of $Y$ drifts off to infinity like $\frac{1}{\lambda}\ln n$ without spreading out.




The sketch shows the density of $Y$ for $\lambda = 1$: each tenfold increase in $n$ moves the curve right by about $\ln 10 \approx 2.3$, and its shape settles down.

**Part (v)**

Take $\alpha = 0.05$. Then

$$\mathrm{P}\big(L(\alpha) \le Y \le U(\alpha)\big) = 1 - \mathrm{P}(Y < L) - \mathrm{P}(Y > U) = 1 - 2(0.05) = 0.9,$$

so $Y$ lies in the interval $[L(0.05), U(0.05)]$ with probability exactly 0.9, whatever $n$ is. For large $n$, part (iv) gives its width as approximately

$$\frac{1}{\lambda}\left[\ln(\ln 20) - \ln\left(\ln \frac{1}{0.95}\right)\right].$$

Now we use the two facts we were given:

- $\mathrm{e}^3 \approx 20$, so $\ln 20 \approx 3$ and $\ln(\ln 20) \approx \ln 3 \approx 1.1$;
- $\ln \frac{1}{0.95} = -\ln(1-0.05) \approx 0.05 = \frac{1}{20}$, using $\ln(1+t) \approx t$ with $t = -0.05$; so $\ln\left(\ln \frac{1}{0.95}\right) \approx \ln \frac{1}{20} = -\ln 20 \approx -3$.

Hence the width is approximately

$$\frac{1}{\lambda}(\ln 3 + 3) \approx \frac{4.1}{\lambda} \approx 4\lambda^{-1},$$

as required.

## Checks

- In part (ii), substituting $L(\alpha)$ back into $G$ gives $\left(1-(1-\alpha^{1/n})\right)^n = \alpha$ exactly.
- With $\alpha = 0.05$ and $n = 10^4$, the exact value of $\lambda L(\alpha)$ is 8.11330 and the approximation $\ln n - \ln\ln 20$ gives 8.11315; for $n = 10$ they are 1.35 and 1.21, so "sufficiently large" really matters.
- The exact limiting width in part (v) is $4.067/\lambda$, comfortably close to $4/\lambda$. A computer simulation of the maximum of five exponentials also matched $G(y)$ to three decimal places.

## Common slips

- Writing $G(y) = 1-(\mathrm{e}^{-\lambda y})^n$: that is the cumulative distribution function of the *minimum*, not the maximum.
- Setting $\mathrm{P}(Y > U) = \alpha$ as $G(U) = \alpha$ instead of $G(U) = 1-\alpha$.
- In part (v), choosing $\alpha = 0.1$: the interval must leave 0.05 in each tail, not 0.1.

## Beyond STEP

This is a first glimpse of *extreme value theory*. Part (iii) is really showing that $\lambda Y - \ln n$ has a limiting distribution, with $\mathrm{P}(\lambda Y - \ln n \le x) \to \exp(-\mathrm{e}^{-x})$: the Gumbel distribution, which models record floods, the strongest gusts in a year and the largest insurance claims. A remarkable theorem says that suitably rescaled maxima can only ever converge to one of three families of distributions. Blitzstein and Hwang, *Introduction to Probability*, treat order statistics and are a good place to start.

WORKED SOLUTION

# STEP III 2025, Question 12

Probability and Statistics • Recursive Random Variables

**STEP III 2025, Question 12**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## The key idea

Part (i) is just a double sum written two ways: both sides add up $\mathrm{f}(r)\,\mathrm{g}(s)$ over the same set of pairs $(r,s)$, one counting row by row and the other column by column. In part (ii), conditioning on the previous value $X_{n-1}$ gives each probability $\mathrm{P}(X_n = r)$ as exactly the kind of sum that appears on the left of (i). Swapping the order of summation then turns $\mathrm{E}(X_n)$ into something we can recognise in terms of $\mathrm{E}(X_{n-1})$.

## Solution

### Part (i)

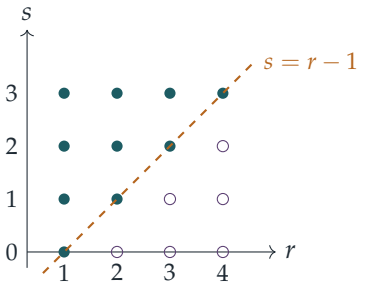
Expanding the brackets, the left-hand side is

$$\sum_{r=1}^{m+1}\sum_{s=r-1}^{m}\mathrm{f}(r)\,\mathrm{g}(s),$$

a sum of $\mathrm{f}(r)\,\mathrm{g}(s)$ over all integer pairs $(r,s)$ with

$$1\le r\le m+1 \quad\text{and}\quad r-1\le s\le m.$$

We describe the same set of pairs by fixing $s$ first. The conditions force $0\le s\le m$ (since $s\ge r-1\ge 0$), and for a given such $s$ the allowed $r$ are those with $1\le r$ and $r\le s+1$; note $s+1\le m+1$, so the bound $r\le m+1$ is then automatic. Conversely any pair with $0\le s\le m$ and $1\le r\le s+1$ satisfies the original conditions. The picture below shows the triangle of pairs for $m=3$.




Summing by columns (fixed $r$) gives the left-hand side; summing by rows (fixed $s$) gives

$$\sum_{s=0}^{m}\sum_{r=1}^{s+1} f(r)\,g(s) = \sum_{s=0}^{m}\Big(g(s)\sum_{r=1}^{s+1} f(r)\Big),$$

which is the right-hand side, as required. Since the sums are finite, reordering them is always allowed.

**Part (ii)(a)**

$X_1$ is equally likely to be 0 or 1, so $\mathrm{E}(X_1) = \frac{1}{2}$.

For $X_2$, condition on $X_1$. If $X_1 = 0$, then $X_2$ is uniform on $\{0,1\}$; if $X_1 = 1$, then $X_2$ is uniform on $\{0,1,2\}$. By the law of total probability,

$$\begin{aligned}
\mathrm{P}(X_2 = 0) &= \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{5}{12},\\
\mathrm{P}(X_2 = 1) &= \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{5}{12},\\
\mathrm{P}(X_2 = 2) &= \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{6},
\end{aligned}$$

as required (and the three add to 1, a useful check). Hence

$$\mathrm{E}(X_2) = 0\cdot\tfrac{5}{12} + 1\cdot\tfrac{5}{12} + 2\cdot\tfrac{1}{6} = \tfrac{3}{4}.$$

**Part (ii)(b)**

By induction $X_{n-1}$ takes values in $\{0,1,\ldots,n-1\}$ (each step raises the maximum possible value by one). Given $X_{n-1} = s$, $X_n$ is uniform on the $s+2$ values $0,1,\ldots,s+1$, so

$$\mathrm{P}(X_n = r \mid X_{n-1} = s) = \begin{cases} \dfrac{1}{s+2}, & 0 \le r \le s+1,\\ 0, & r > s+1. \end{cases}$$

The value $r = 0$ is always possible, so the law of total probability gives

$$\mathrm{P}(X_n = 0) = \sum_{s=0}^{n-1} \frac{\mathrm{P}(X_{n-1} = s)}{s+2}.$$

For $1 \le r \le n$, only those $s$ with $s+1 \ge r$, that is $s \ge r-1$, contribute:

$$\mathrm{P}(X_n = r) = \sum_{s=r-1}^{n-1} \frac{\mathrm{P}(X_{n-1} = s)}{s+2}, \qquad r = 1,2,\ldots,n.$$

(For $r = 1$ this is the same sum as for $r = 0$, so $\mathrm{P}(X_n = 0) = \mathrm{P}(X_n = 1)$ for every $n \geq 1$, matching the two values $\frac{5}{12}$ in (a).)

#### Part (ii)(c)

Write $p_s = \mathrm{P}(X_{n-1} = s)$. The $r = 0$ term contributes nothing to the expectation, so by (b)

$$\mathrm{E}(X_n) = \sum_{r=1}^{n} r\,\mathrm{P}(X_n = r) = \sum_{r=1}^{n} \Big( r \sum_{s=r-1}^{n-1} \frac{p_s}{s+2} \Big).$$

**?** **Ask yourself** How does part (i) apply here?

This is the left-hand side of (i) with $m = n - 1$ (allowed, as $n \geq 1$), $\mathrm{f}(r) = r$ and $\mathrm{g}(s) = p_s/(s+2)$. Swapping the order of summation,

$$\begin{aligned} \mathrm{E}(X_n) &= \sum_{s=0}^{n-1} \Big( \frac{p_s}{s+2} \sum_{r=1}^{s+1} r \Big) = \sum_{s=0}^{n-1} \frac{p_s}{s+2} \cdot \frac{(s+1)(s+2)}{2} \\ &= \frac{1}{2} \sum_{s=0}^{n-1} (s+1)\, p_s = \frac{1}{2} \Big( \sum_{s=0}^{n-1} p_s + \sum_{s=0}^{n-1} s\, p_s \Big) \\ &= \tfrac{1}{2} \big( 1 + \mathrm{E}(X_{n-1}) \big), \end{aligned}$$

using $\sum p_s = 1$. This is the required result.

Intuitively: given $X_{n-1} = s$, the mean of a uniform variable on $\{0, \ldots, s+1\}$ is $\frac{1}{2}(s+1)$, so this is the tower law in disguise.

To solve the recurrence, write $e_n = \mathrm{E}(X_n)$. Subtracting 1 from both sides,

$$e_n - 1 = \tfrac{1}{2}\,(e_{n-1} - 1),$$

so $e_n - 1$ halves at each step. Since $e_0 = 0$, we get $e_n - 1 = -\big(\frac{1}{2}\big)^n$, that is

$$\mathrm{E}(X_n) = 1 - \frac{1}{2^n}, \qquad n = 1, 2, \ldots.$$

## Checks

The formula gives $\mathrm{E}(X_1) = \frac{1}{2}$ and $\mathrm{E}(X_2) = \frac{3}{4}$, agreeing with (a). Computing the exact distribution of $X_n$ step by step up to $n = 15$ confirms both the expressions in (b) and $\mathrm{E}(X_n) = 1 - 2^{-n}$; for example $\mathrm{E}(X_5) = \frac{31}{32} \approx 0.969$, and a computer simulation of 200 000 runs gave 0.967. As $n \to \infty$ the mean tends to 1, which is plausible: each step pulls the mean halfway towards 1.

## Common slips

- Starting the sum for $\mathrm{P}(X_n = r)$ at $s = r$ rather than $s = r - 1$: the value $r$ is reachable from $s = r - 1$, since $X_n$ can go up to $X_{n-1} + 1$.

- Getting the limits wrong when swapping the order in (c): after the swap, $r$ runs from 1 to $s+1$, giving $\frac{1}{2}(s+1)(s+2)$, and the factor $s+2$ cancels neatly.
- Solving the recurrence with $e_1$ as starting value but writing $1-2^{-(n-1)}$; check against $e_1 = \frac{1}{2}$.

## Beyond STEP

The process here is a Markov chain: the distribution of $X_n$ depends only on $X_{n-1}$, and (b) is one step of multiplying a probability vector by a transition matrix. The shortcut in (c) is the law of total expectation, $\mathrm{E}(X_n) = \mathrm{E}\big(\mathrm{E}(X_n \mid X_{n-1})\big)$, which reduces many such problems to a linear recurrence. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*.

# Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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